# From Brackets to Amplitudes: Quantum Mechanics

## Quantizing the canonical structure

Lecture notes

July 23, 2026


**Abstract**

This document is the quantum layer of the series: it takes the canonical structures the previous four documents built — conjugate pairs, Poisson brackets, symmetries traded for conserved charges, the harmonic oscillator, coordinates adapted to preserved structure — and quantizes them. The route is structural rather than historical. Part I begins where the mechanics notes ended: the Poisson bracket's algebra admits a second realization, by commutators of operators, and Dirac's rule $\{\,,\,\} \to [\,,\,]/\mathrm{i}\hbar$ is the replacement of realization with the algebra kept; the canonical commutator forces an infinite-dimensional stage (a three-line trace argument), the stage is Hilbert space, and the measurement postulates are installed on it, with the uncertainty relations derived from them. Position and momentum representations are constructed, with $\hat{p} = -\mathrm{i}\hbar\,\partial_x$ derived from the translation group. Part II is dynamics: the evolution operator, the Heisenberg picture *first* — quantum bracket flow, the verbatim mirror of the mechanics notes — with the Schrödinger picture as a change of bookkeeping; the harmonic oscillator solved algebraically, the series' opening object now with a discrete spectrum; and the propagator computed twice, canonically and by path integral, making explicit at the one level where both routes are fully tractable the fork that the field-theory notes could only mark. Part III is symmetry: rotations represented unitarily, angular momentum as their Noether charges, the $SU(2)$ double cover with its measurable $2\pi \neq \mathbf{1}$, the angular momentum spectrum from pure algebra, and the hydrogen atom as the capstone worked example. The outlook quantizes the field modes of the companion notes — every mode one quantum oscillator, occupation numbers, particles — closing the series' longest arc for the second and final time. Functional-analytic fine print (domains, self-adjointness, the Stone–von Neumann theorem) is flagged where it lives and deferred to Hall; every statement used is either proved here or explicitly tagged.

# Contents

# 1 Conventions and Notation

Operators wear hats ($\hat{q}$, $\hat{p}$, $\hat{H}$) until §3 declares them safe to remove in unambiguous contexts; classical quantities are bare. Dots are time derivatives. Three-vectors carry arrows. Cross-references to the companion documents: [Osc. §$n$] for the oscillator reference, [Mech. §$n$] for the variational mechanics course, and [Fld. §$n$] for the spacetime-and-fields notes.

**Convention 1.1** ($\hbar$ stays visible)**.** This document keeps $\hbar = 1.055 \times 10^{-34}\,\mathrm{J\,s}$ explicit throughout — a deliberate reversal of the companion notes' $c = 1$ policy [Fld. §3.9], and the contrast is the reason. There, $c$ was a conversion factor between units of length and time, carrying no dynamics; hiding it lost nothing. Here, $\hbar$ is the *subject*: it is the deformation parameter by which quantum mechanics differs from classical mechanics, the knob whose $\hbar \to 0$ limit must recover everything the series built, and the scale that decides when quantum effects matter. Setting it to 1 would hide the very parameter under study. Note its dimensions: $[\hbar] = [q][p] = \textit{action}$ — the very quantity the series has been extremizing since [Mech. §3].

**Convention 1.2** (Nonrelativistic throughout)**.** The speed of light does not appear: this document's mechanics is Galilean, its Hamiltonians are $\hat{p}^2/2m + V$, and the marriage of quantum mechanics with special relativity is exactly the handoff of the outlook (§17) — quantum *field* theory, for the reasons [Fld. §4.4] previewed.

Recurring symbols, each re-introduced where it first appears:

| Symbol | Definition | First appears | Meaning |
|---|---|---|---|
| $[\hat{A}, \hat{B}]$ | $\hat{A}\hat{B} - \hat{B}\hat{A}$ | §2 | commutator |
| $\hbar$ | $h/2\pi$ | §2 | quantum of action |
| $\mathcal{H}$ | — | §3 | Hilbert space |
| $\lvert\alpha\rangle$, $\langle\alpha\rvert$ | — | §3 | ket, bra |
| $\hat{U}(t)$ | $\mathrm{e}^{-\mathrm{i}\hat{H}t/\hbar}$ | §6 | evolution operator |
| $\hat{a}$, $\hat{a}^\dagger$ | — | §9 | ladder operators |
| $K(x_b, t_b; x_a, t_a)$ | — | §10 | propagator |
| $\hat{J}_i$ | — | §11 | angular momentum |

(The $\lvert\,\rangle/\langle\,\rvert$ notation is defined in §3; the table lists it for reference only. Mathematical fine print — operator domains, self-adjointness versus Hermiticity, spectral theory of unbounded operators — is flagged with [stated; Hall] where it arises, pointing to Hall, *Quantum Theory for Mathematicians*; this document works at the rigor of a careful physicist, with the potholes marked rather than paved.)

# Part I
# Kinematics

## 2 From Brackets to Commutators

The mechanics notes ended with dynamics as bracket flow: observables are functions on phase space, the Poisson bracket organizes them into an algebra, and $\dot{F} = \{F, H\}$ runs the world [Mech. §8.2]. This section makes the single structural observation on which the whole document stands: *that algebra admits a second realization*, by operators, and nature uses it. Everything else — Hilbert space, wavefunctions, spectra — is the working-out of where the second realization is forced to live.

### 2.1 What classical mechanics could not do

Three experimental facts, each fatal on its own, name the problem. [All three: experimental input.]

- **Atoms are stable and discrete.** A classical electron orbiting a nucleus is an accelerating charge, and accelerating charges radiate (Larmor's formula — one classical result this series did *not* derive [stated; Griffiths, *Introduction to Electrodynamics*, Ch. 11]); losing energy to radiation, the electron must spiral in, in about $10^{-11}$ s. It does not — and when atoms do radiate, they emit *lines*, not continua: something restricts the energies available to the electron to a discrete set, and no classical Hamiltonian $H(q, p)$ with a continuum of orbits does that.
- **The oscillator's energy comes in lumps.** Blackbody radiation and low-temperature specific heats both fail classically in the same way, and both are repaired by one assumption: an oscillator of frequency $\omega$ can hold energy only in units of $\hbar\omega$. The failing system is the series' own mascot [Osc. §2].
- **Measurements interfere with each other.** A Stern–Gerlach apparatus sorts silver atoms by a two-valued quantity ($S_z = \pm\hbar/2$). Sort by $S_z$, keep the + beam, sort it by $S_x$, keep the + beam, and sort *that* by $S_z$ again: both $S_z$ values reappear. Measuring $S_x$ *destroyed* the previously sharp value of $S_z$. Classical observables — mere functions on phase space — can all be read simultaneously; these two demonstrably cannot. The order of observations matters: whatever mathematical objects $S_z$ and $S_x$ are, *they do not commute.*

The third fact is the structural one, and it points directly at the bracket.

### 2.2 The algebra the bracket built

Recall from [Mech. §8.1] what makes the Poisson bracket more than a formula — its axioms. For observables $A, B, C$ and numbers $\lambda$:

$$\begin{aligned} &\text{antisymmetry:} && \{A, B\} = -\{B, A\}, \\ &\text{linearity:} && \{A, \lambda B + C\} = \lambda\{A, B\} + \{A, C\}, \\ &\text{Leibniz:} && \{A, BC\} = \{A, B\}\, C + B\, \{A, C\}, \\ &\text{Jacobi:} && \{A, \{B, C\}\} + \{B, \{C, A\}\} + \{C, \{A, B\}\} = 0. \end{aligned} \tag{1}$$

Everything the mechanics notes did with brackets — conserved quantities, canonical transformations, the flow $\dot{F} = \{F, H\}$ — used only (1) plus the fundamental bracket $\{q, p\} = 1$; the specific realization as $\sum(\partial_q A\, \partial_p B - \partial_p A\, \partial_q B)$ mattered only for computing.

Now the observation. Let observables instead be *operators* — linear maps on some vector space, composed by application, so that $\hat{A}\hat{B}$ and $\hat{B}\hat{A}$ need not agree — and define

$$[\hat{A}, \hat{B}] \;\equiv\; \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{2}$$

Antisymmetry and linearity are immediate. Leibniz is one line of adding and subtracting:

$$[\hat{A}, \hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \big(\hat{A}\hat{B} - \hat{B}\hat{A}\big)\hat{C} + \hat{B}\big(\hat{A}\hat{C} - \hat{C}\hat{A}\big) = [\hat{A}, \hat{B}]\,\hat{C} + \hat{B}\,[\hat{A}, \hat{C}], \tag{3}$$

and Jacobi is a patient expansion — twelve terms, canceling in pairs (write it out once; the cancellation needs no cleverness). So the commutator satisfies *every axiom* in (1): **the bracket algebra has a second realization**, in which observables are operators, "bracket" means commutator, and — this is the license the Stern–Gerlach fact demanded — observables need not commute.

**Definition 2.1** (Dirac's quantization rule)**.** Quantum mechanics keeps classical mechanics' algebra and replaces its realization:

$$\boxed{\{A, B\} \quad\longrightarrow\quad \frac{1}{\mathrm{i}\hbar}\,[\hat{A}, \hat{B}],} \tag{4}$$

and in particular the fundamental bracket $\{q, p\} = 1$ becomes the *canonical commutation relation*,

$$\boxed{[\hat{q}, \hat{p}] = \mathrm{i}\hbar.} \tag{5}$$

Three notes on the rule before it is put to work. *The* i: observables should be represented by operators whose measured values are real — Hermitian operators, §3 — and the commutator of two Hermitian operators is *anti*-Hermitian ($[\hat{A}, \hat{B}]^\dagger = [\hat{B}^\dagger, \hat{A}^\dagger] = -[\hat{A}, \hat{B}]$); the i repairs the mismatch, making $[\hat{A}, \hat{B}]/\mathrm{i}\hbar$ Hermitian again, eligible to be an observable as $\{A, B\}$ was. *The* $\hbar$: each term of the Poisson bracket carries one $\partial_q$ and one $\partial_p$, so the bracket operation costs a factor $1/([q][p]) = 1/[\text{action}]$ relative to the plain product — $\{A, B\}$ has dimensions $[A][B]/[\text{action}]$, and $\{q, p\} = 1$ is dimensionless. The commutator involves no derivatives: $[\hat{q}, \hat{p}]$ has dimensions $[q][p] = [\text{action}]$, plain. The two realizations differ by exactly one unit of action, so the dictionary between them needs a constant with those dimensions — and nature supplies exactly one. And it is small in a precisely quantifiable sense: a desk pendulum ($m = 0.1\,\mathrm{kg}$, energy $\sim 10^{-2}\,\mathrm{J}$, period $\sim 1\,\mathrm{s}$) carries an action per cycle of order $10^{-2}\,\mathrm{J\,s} \sim 10^{32}\,\hbar$, while one vibrational quantum of a molecule carries, per cycle, exactly $ET = \hbar\omega \cdot 2\pi/\omega = h \approx 6\hbar$. The dimensionless ratio $S/\hbar$ is the switch: where it is astronomical, the deformation (4) is invisible and the mechanics notes rule; where it is of order one, this document takes over. (§10 converts this slogan into a theorem — the stationary-phase limit of the path integral.) *The status*: (4) is a *construction principle*, not a theorem — for polynomials in $q, p$ beyond quadratic it suffers ordering ambiguities (which operator does $q^2p^2$ become?), and the precise statement is that (5) plus the algebra is the postulate, with (4) its guiding reading. [The obstruction to a perfect $\{\,,\,\} \to [\,,\,]$ dictionary is a real theorem — Groenewold's no-go theorem — stated here, proved nowhere near here; Hall §13.4.]

## 2.3 First consequences from the algebra alone

Remarkably much follows from (5) before any stage for the operators is chosen — three results, in increasing order of consequence.

**Derivative identities from the algebra.** By induction with the Leibniz rule (3): $[\hat{q}, \hat{p}^n] = [\hat{q}, \hat{p}]\hat{p}^{n-1} + \hat{p}\,[\hat{q}, \hat{p}^{n-1}]$, and unwinding,

$$[\hat{q}, \hat{p}^n] = \mathrm{i}\hbar\, n\, \hat{p}^{n-1} \qquad \Longrightarrow \qquad \boxed{[\hat{q}, F(\hat{p})] = \mathrm{i}\hbar\, F'(\hat{p}), \qquad [\hat{p}, G(\hat{q})] = -\mathrm{i}\hbar\, G'(\hat{q})} \tag{6}$$

for functions defined by power series — the commutator with $\hat{q}$ *differentiates in* $\hat{p}$, exactly as the classical $\{q, F(p)\} = F'(p)$ does (one line from the bracket's definition [Mech. §8.1]: $\{q, F\} = \partial F/\partial p$): the new realization reproduces the old one's derivative structure operator-by-operator, which is the algebra-keeping promise of (4) kept.

**No common eigenvectors.** Suppose $|\psi\rangle$ had sharp values of both: $\hat{q}|\psi\rangle = q_0|\psi\rangle$ and $\hat{p}|\psi\rangle = p_0|\psi\rangle$ (notation borrowed from §3, content clear already). Then $[\hat{q}, \hat{p}]|\psi\rangle = (q_0p_0 - p_0q_0)|\psi\rangle = 0$, contradicting $\mathrm{i}\hbar|\psi\rangle \neq 0$. *No state has sharp position and sharp momentum* — the uncertainty principle's skeleton, three lines from the algebra; §4 adds the flesh (the quantitative bound $\Delta q\, \Delta p \geq \hbar/2$).

**The stage must be infinite-dimensional.** Here is the theorem that decides where the subject must live.

**Theorem 2.2** (No finite-dimensional realization)**.** *No pair of finite-dimensional matrices satisfies* $[\hat{q}, \hat{p}] = \mathrm{i}\hbar\,\mathbf{1}$.

*Proof.* The trace is cyclic, $\mathrm{tr}(\hat{A}\hat{B}) = \mathrm{tr}(\hat{B}\hat{A})$, so the trace of any commutator vanishes: $\mathrm{tr}[\hat{q}, \hat{p}] = 0$. But $\mathrm{tr}(\mathrm{i}\hbar\,\mathbf{1}) = \mathrm{i}\hbar\, d \neq 0$ in dimension $d$. No $d < \infty$ survives. □

So the canonical commutation relation cannot live where the mechanics of the series lived — on finite lists of coordinates — nor even on finite-dimensional vector spaces: it *forces* an infinite-dimensional stage, on which the trace argument dissolves (the trace of $\mathbf{1}$ is no longer finite, and indeed $\hat{q}, \hat{p}$ turn out to be unbounded operators, for which the cyclic property fails — the first pothole, marked: [stated; Hall Ch. 9, esp. §9.8 for these two operators]). The spin observables of Stern–Gerlach, by contrast, close their commutators without any $\mathbf{1}$ on the right (§11: $[\hat{S}_x, \hat{S}_y] = \mathrm{i}\hbar\,\hat{S}_z$), which is why *they* fit in two dimensions — for both algebras here, the size of the smallest stage is visible in the right-hand sides.

## 2.4 The representation question

An algebra is not yet a physical theory; it needs a stage — a vector space for the operators to act on, an inner product to make predictions with, and a dictionary between vectors and physical states. Building that stage is §3's job, and one theorem, quoted now and leaned on throughout, guarantees the job has essentially one answer:

*Remark* 2.3 (Stone–von Neumann and why the founders agreed). Up to unitary equivalence, the canonical commutation relation (5) has exactly *one* irreducible representation (in the technically careful, exponentiated form). [stated; Hall Ch. 14 — the theorem needs the Weyl form of the CCR (Hall §14.2) precisely because of the unboundedness pothole above.] The physical content is uniqueness of quantum mechanics itself: Heisenberg's matrix mechanics and Schrödinger's wave mechanics, born separately in 1925–26, are the same representation in different coordinates — the position representation of §5 is not *a* choice among physically distinct options but a convenient basis in the unique theory, exactly as normal coordinates were a basis in the unique chain [Osc. §8]. When §7 finds two descriptions of dynamics and §10 finds two routes to the propagator, the same moral will hold: one theory, many coordinates.

## 3 The Stage: Hilbert Space and Dirac Notation

Theorem 2.2 ordered an infinite-dimensional complex vector space with an inner product; this section builds it, installs Dirac's notation on it, and stocks it with the operator vocabulary the rest of the document speaks. One deliberate exception to the infinite-dimensionality runs through the section and the document: the *two-state system*, spin-$\frac{1}{2}$, small enough to compute everything in and rich enough to exhibit almost every quantum phenomenon — the standing laboratory, introduced at the end and never dismissed.

### 3.1 States: kets, superposition, and rays

**Postulate 3.1** (States)**.** The states of a quantum system are the nonzero vectors of a complex Hilbert space $\mathcal{H}$ — a complex vector space with an inner product, complete in the induced norm [the completeness clause is the first pothole: it matters only in infinite dimensions, and only for the analysis this document defers; stated here, leaned on never — Hall App. A.4 does it right] — with two vectors describing the same physical state when they differ by a nonzero complex factor.

A state is written as a *ket*, $|\alpha\rangle$, the label $\alpha$ naming whatever physics identifies it. Two pieces of physics hide in "vector space" and one in "complex":

- **Superposition.** If $|\alpha\rangle$ and $|\beta\rangle$ are states, so is $c_1|\alpha\rangle+c_2|\beta\rangle$ — not a mixture, not an either-or, but a genuinely new state. This is the vector-space axiom promoted to physics, and it is the structural fact behind interference: amplitudes add before probabilities are taken.
- **Rays.** Since $|\alpha\rangle$ and $c|\alpha\rangle$ are the same physics, states are really *rays*; in practice one normalizes, $\langle\alpha|\alpha\rangle = 1$, leaving a physically empty overall phase $\mathrm{e}^{\mathrm{i}\gamma}$. (Empty for a lone state — *relative* phases inside superpositions are physical, and §3.5 measures one.)
- **Complex.** The i in $[\hat{q},\hat{p}] = \mathrm{i}\hbar$ already forced complex scalars: a real vector space has no operator algebra in which (5) can hold with Hermitian $\hat{q},\hat{p}$ (§3.2's Hermiticity computation shows the commutator of Hermitians is anti-Hermitian — on a real space, where anti-Hermitian means antisymmetric, $\mathrm{i}\hbar\,\mathbf{1}$ has no home).

*Bras.* Every ket $|\beta\rangle$ defines a linear machine that eats kets and returns numbers — "take the inner product with $|\beta\rangle$" — written as the *bra* $\langle\beta|$, so that the inner product is the bracket

$$\langle\beta|\alpha\rangle \;=\; \big(\langle\beta|\big)\big(|\alpha\rangle\big), \qquad \langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*, \qquad \langle\alpha|\alpha\rangle \geq 0, \tag{7}$$

conjugate-linear in the bra's label and linear in the ket's. (That every bounded linear machine is of this form — that the bras are exactly the duals — is the Riesz representation theorem [stated; Hall App. A.4.3]; in finite dimensions it is row-vectors-meet-column-vectors, and that mental model is safe throughout.) The notation's design intent will pay off in §3.3: bars and brackets are engineered so that the move made on nearly every page below — inserting a complete set of states — is typographically frictionless.

### 3.2 Operators: Hermitian and unitary

Observables need operators whose measurement outcomes are real; time evolution and symmetries need operators that preserve probability. Two definitions cover both needs.

The *adjoint* $\hat{A}^\dagger$ of an operator $\hat{A}$ is defined by

$$\langle\beta|\big(\hat{A}^\dagger|\alpha\rangle\big) = \big(\langle\alpha|\hat{A}|\beta\rangle\big)^* \qquad \text{for all } |\alpha\rangle, |\beta\rangle, \tag{8}$$

i.e. $\hat{A}^\dagger$ acting on kets matches $\hat{A}$ acting inside bras; in a basis it is the conjugate transpose. An operator is *Hermitian* if $\hat{A}^\dagger = \hat{A}$, and *unitary* if $\hat{U}^\dagger\hat{U} = \hat{U}\hat{U}^\dagger = \mathbf{1}$ — the latter exactly the condition $\langle \hat{U}\beta\,|\hat{U}\alpha\rangle = \langle\beta|\alpha\rangle$: unitaries are the inner product's isometries, the Hilbert-space entry in the series' table of structure-preserving maps (rotations for $\delta_{ij}$, Lorentz for $\eta_{\mu\nu}$ [Fld. §3.1] — and now unitaries for $\langle\cdot\,|\cdot\rangle$; same template, third preserved pairing).

Two properties of Hermitian operators carry the probabilistic interpretation, and both proofs take two lines:

**Proposition 3.2.** *A Hermitian operator's eigenvalues are real, and eigenvectors with distinct eigenvalues are orthogonal.*

*Proof.* Let $\hat{A}|a\rangle = a|a\rangle$ and $\hat{A}|a'\rangle = a'|a'\rangle$, eigenkets normalized. Sandwich once: $\langle a|\hat{A}|a\rangle = a\langle a|a\rangle = a$, while Hermiticity says the same number equals its own conjugate, $a = a^*$. Sandwich twice, crosswise: $\langle a'|\hat{A}|a\rangle$ evaluated right-to-left gives $a\langle a'|a\rangle$, and left-to-right (using $\hat{A}^\dagger = \hat{A}$ and $a'$ real) gives $a'\langle a'|a\rangle$; subtracting, $(a - a')\langle a'|a\rangle = 0$, so $a \neq a'$ forces $\langle a'|a\rangle = 0$. ☐

*Remark* 3.3 (The pothole made concrete). In infinite dimensions, "Hermitian" as used above (symmetric on its domain) is strictly weaker than *self-adjoint* — equality of $\hat{A}$ and $\hat{A}^\dagger$ *including their domains* — and the spectral machinery of the next subsection belongs to the self-adjoint. The gap is not pedantic. Take the momentum-type operator $\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ on $[0, 1]$ with hard-wall conditions $\psi(0) = \psi(1) = 0$: integration by parts kills the boundary terms, so it is symmetric — and it has *no eigenvectors whatsoever* (the candidates $\mathrm{e}^{-\mathrm{i}\lambda x/\hbar}$ refuse to vanish at the endpoints), no spectral decomposition, no basis of its own. Self-adjointness is restored only by *choosing* boundary conditions of the periodic type $\psi(1) = \mathrm{e}^{\mathrm{i}\theta}\psi(0)$, one legitimate operator per choice of $\theta$. [stated, with the example verified against Hall §9.6; the systematic theory of such choices — deficiency indices — is Hall Ch. 9.] The working policy of this document: "Hermitian" everywhere, potholes flagged where a domain actually bites (it bites once more in §4).

*Hats, henceforth.* With the operator vocabulary in place, the promise of §1 is kept: hats are dropped whenever context distinguishes operator from number ($H$, $p$, $a$), and retained where confusion threatens ($\hat{q}$ versus an eigenvalue $q$).

### 3.3 Eigenbases and the resolution of the identity

**Theorem 3.4** (Spectral theorem, working version)**.** *A Hermitian operator on a finite-dimensional Hilbert space possesses an orthonormal basis of eigenvectors, $A|a\rangle = a|a\rangle$, $\langle a'|a\rangle = \delta_{a'a}$. [For genuinely self-adjoint operators in infinite dimensions the statement survives with sums traded for integrals where the spectrum is continuous — §5 uses exactly that trade — and its full formulation and proof are Hall Ch. 10; stated.]*

The basis being complete, every ket expands in it, and the expansion is worth writing as an operator identity:

$$\boxed{\mathbf{1} \;=\; \sum_a |a\rangle\langle a|} \qquad \text{(the } \textit{resolution of the identity}\text{)}, \tag{9}$$

for then every expansion, every change of basis, and every matrix element is the *same* typographical move — insert (9) where needed:

$$|\psi\rangle = \sum_a |a\rangle \underbrace{\langle a|\psi\rangle}_{\text{coefficients}}\,, \qquad \langle\beta|B|\alpha\rangle = \sum_{a,a'} \langle\beta|a\rangle \underbrace{\langle a|B|a'\rangle}_{\text{matrix of } B} \langle a'|\alpha\rangle. \tag{10}$$

An operator, in its own eigenbasis, is diagonal: $A = \sum_a a\,|a\rangle\langle a|$ — the spectral theorem restated as a formula.

*Remark* 3.5 (Adapted coordinates: third appearance in the series). This is the series' standing pattern. Normal coordinates diagonalized the coupled pair and the chain [Osc. §§6, 8]; null coordinates diagonalized the boost [Fld. §2.4, Remark 2.7]; and here, the eigenbasis of whatever operator matters diagonalizes it — energy eigenstates for dynamics (§6), momentum eigenstates for free motion, angular momentum eigenstates for rotations. What is new is not the strategy but its *cost*: on the chain, finding normal coordinates was work; here, Dirac notation makes changing to adapted coordinates a one-symbol operation, (9) inserted at will. The notation is not a convenience layered on the theory — it is a systematic, zero-cost form of the series' oldest strategy.

### 3.4 What the notation computes

The notation so far is grammar; here is its semantics. Bras, kets, and operators combine into exactly four composite objects, and each carries a physical reading. (The readings involving probability lean on the measurement postulate, stated properly in §4; they are previewed here because the notation is unintelligible without them — and because Example 3.7 below already needs them.)

**(a) The inner product is an amplitude.** The number $\langle\beta|\alpha\rangle$ is the *overlap* of the two states — operationally, the *probability amplitude* that a system prepared in $|\alpha\rangle$ answers "yes" when asked "are you $|\beta\rangle$?", with

$$P(\alpha \to \beta) = \left|\langle\beta|\alpha\rangle\right|^2 \tag{11}$$

the probability itself (Born's rule, installed as a postulate in §4). The two extremes calibrate the reading: $\langle\alpha|\alpha\rangle = 1$ is certainty of self, and orthogonality $\langle\beta|\alpha\rangle = 0$ is *perfect distinguishability* — a measurement can tell the two states apart without error, which is why eigenstates of an observable (orthogonal, by Proposition 3.2) are the states a measurement of it can cleanly report.

**(b) The outer product is an operator.** The object $|\alpha\rangle\langle\beta|$ — ket first, bra second — eats a ket and returns a ket:

$$\big(|\alpha\rangle\langle\beta|\big)|\gamma\rangle = |\alpha\rangle\,\langle\beta|\gamma\rangle, \tag{12}$$

"detect $\beta$, output $\alpha$, weighted by the overlap." Its adjoint swaps the factors, $(|\alpha\rangle\langle\beta|)^\dagger = |\beta\rangle\langle\alpha|$ (one line from the adjoint's definition: take matrix elements of both sides). The special case $\beta = \alpha$ is the workhorse: the *projector*

$$P_\alpha = |\alpha\rangle\langle\alpha|, \qquad P_\alpha^2 = |\alpha\rangle\,\underbrace{\langle\alpha|\alpha\rangle}_{1}\langle\alpha| = P_\alpha, \qquad P_\alpha^\dagger = P_\alpha: \tag{13}$$

Hermitian and idempotent — asking the question twice is the same as asking it once. The resolution of the identity (9) now reads as physics: the projectors of a complete eigenbasis exhaust the space, $\mathbf{1} = \sum_a P_a$ — "the answer is *always* one of the $a$'s" — and they are the mathematical atoms out of which §4 builds measurement.

**(c) The closed sandwich is an average.** For a normalized state, define the *expectation value* $\langle A\rangle_\psi \equiv \langle\psi|A|\psi\rangle$. Its statistical meaning is two insertions of (9) away:

$$\langle\psi|A|\psi\rangle = \sum_a \langle\psi|A|a\rangle\langle a|\psi\rangle = \sum_a a\,\langle\psi|a\rangle\langle a|\psi\rangle = \boxed{\sum_a a\,\left|\langle a|\psi\rangle\right|^2} \tag{14}$$

— the eigenvalues, weighted by exactly the Born probabilities (11): $\langle A\rangle$ is the *mean of the measurement statistics*, not the outcome of any single measurement. The distinction has teeth: a spin-$\frac{1}{2}$ state can have $\langle S_z\rangle = 0.3\,\hbar$ while every individual measurement returns $\pm\hbar/2$ and nothing else. The spread about the mean,

$$(\Delta A)^2 \equiv \big\langle (A - \langle A\rangle)^2 \big\rangle = \langle A^2\rangle - \langle A\rangle^2, \tag{15}$$

is the quantity the uncertainty relations of §4 will bound.

**(d) The open sandwich is a transition amplitude.** The general matrix element $\langle\beta|A|\alpha\rangle$ is the amplitude for $\alpha$ to be found as $\beta$ *with the process $A$ in between* — the object dynamics computes: §6's central quantity is $\langle x|\,U(t)|\psi\rangle$, "the amplitude to be found at $x$ after evolving for time $t$," and §10 is one long evaluation of a single such sandwich. In its own eigenbasis an operator's sandwich structure is laid bare by the spectral formula $A = \sum_a a\,P_a$: every operator this document meets is a weighted sum of "are you $a$?"-questions.

In summary — the four moves, which the laboratory below runs in miniature:

| object | type | reading |
|---|---|---|
| $\langle\beta\|\alpha\rangle$ | number | amplitude to find $\alpha$ as $\beta$ |
| $\|\alpha\rangle\langle\beta\|$ | operator | detect $\beta$, output $\alpha$; $P_\alpha = \|\alpha\rangle\langle\alpha\|$ asks "are you $\alpha$?" |
| $\langle\psi\|A\|\psi\rangle$ | number | mean of $A$'s measurement statistics |
| $\langle\beta\|A\|\alpha\rangle$ | number | amplitude for $\alpha \to \beta$ through the process $A$ |

### 3.5 The standing laboratory: spin-$\frac{1}{2}$

Theorem 2.2 exempted observables whose commutators close without an $\mathbf{1}$, and the exemption's smallest instance is the two-dimensional Hilbert space of a spin-$\frac{1}{2}$ system — the silver atom of §2.1, finally given its mathematics. Take the $S_z$ eigenbasis as computational home,

$$S_z|\pm\rangle = \pm\tfrac{\hbar}{2}|\pm\rangle, \qquad \langle +|-\rangle = 0, \quad \langle +|+\rangle = \langle -|-\rangle = 1, \tag{16}$$

and write the three spin components in it. In matrix form $S_i = \frac{\hbar}{2}\,\sigma_i$ with the *Pauli matrices*

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{17}$$

Hermitian, traceless, and obeying the identity that packages their entire algebra (verify once by direct multiplication — nine products, each a one-liner):

$$\boxed{\sigma_i\,\sigma_j = \delta_{ij}\,\mathbf{1} + \mathrm{i}\,\epsilon_{ijk}\,\sigma_k,} \tag{18}$$

whose antisymmetric part is the commutation relation $[S_i, S_j] = \mathrm{i}\hbar\,\epsilon_{ijk}S_k$ — a promise of §2.3 that this verifies and that §11 will derive from rotations — and whose symmetric part, $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1}$, is a structure the Klein–Gordon-to-Dirac story will one day need [stated; used in the QFT sequel].

The general normalized state, with the empty overall phase fixed to make the $|+\rangle$ coefficient real:

$$|\theta,\varphi\rangle = \cos\tfrac{\theta}{2}\,|+\rangle \;+\; \mathrm{e}^{\mathrm{i}\varphi}\sin\tfrac{\theta}{2}\,|-\rangle, \tag{19}$$

two real parameters — and they are exactly the polar coordinates of a unit vector: a direct computation of the closed sandwich (14) with the matrices (17) gives

$$\langle\theta,\varphi|\vec{S}|\theta,\varphi\rangle = \tfrac{\hbar}{2}\left(\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta\right) : \tag{20}$$

the state space of a spin-$\frac{1}{2}$ is a sphere (Fig. 1), each state "pointing" somewhere, with $|\pm\rangle$ at the poles and the relative phase $\varphi$ — physical, as promised — as the longitude.

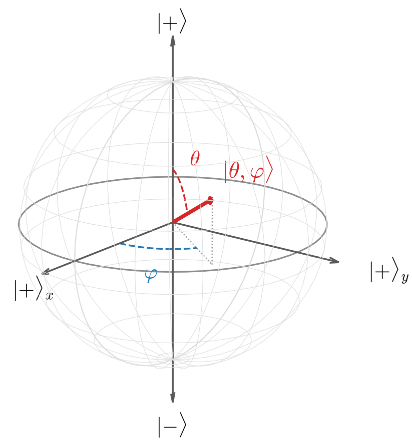


Figure 1: The *pure-state* Bloch sphere: the states (19) of a spin-$\frac{1}{2}$ system, mapped to unit vectors by the exact correspondence $\langle\vec{S}\rangle = \frac{\hbar}{2}(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. Poles: the $S_z$ eigenstates; equator at $\varphi = 0$: the $S_x$ eigenstate $|+\rangle_x$. Antipodal points — and only antipodal points — are orthogonal: the sphere's geometry encodes the Hilbert space's. (This surface is the boundary of a ball; Remark 3.6.)

*Remark* 3.6 (The sphere is the boundary of a ball). The name "Bloch sphere" refers to a larger object than Postulate 3.1 describes. The general state concept of quantum mechanics — covering ensembles and subsystems, not just maximally-known preparations — is the *density matrix* $\rho$, and for two levels every such object is

$$\rho = \tfrac{1}{2}\big(\mathbf{1} + \vec{r}\cdot\vec{\sigma}\big), \qquad |\vec{r}\,| \le 1, \tag{21}$$

one point of the solid *Bloch ball* per state. Purity is the boundary condition: $\rho^2 = \rho \iff |\vec{r}\,| = 1$, and there $\vec{r} = \langle\vec{\sigma}\rangle$ reproduces exactly the map of Fig. 1 — what this document calls a state is the ball's surface, with the interior (mixed states, the center $\frac{1}{2}\mathbf{1}$ the "knows-nothing" state) describing situations where the preparation itself is uncertain. This document works with pure states throughout and does not develop $\rho$; the machinery is stated here so the figure is not mistaken for the whole story. [stated; see Sakurai's treatment of density operators and ensembles — density matrices are where quantum mechanics meets statistical mechanics and decoherence.]

**Example 3.7** (Stern–Gerlach made quantitative)**.** The sequence of §2.1 can now be computed. The $S_x$ eigenstates sit on the equator, at $(\theta, \varphi) = (\frac{\pi}{2}, 0)$ and $(\frac{\pi}{2}, \pi)$ in (19):

$$|\pm\rangle_x = \tfrac{1}{\sqrt{2}}\big(|+\rangle \pm |-\rangle\big), \qquad \big|{}_x\langle +|+\rangle\big|^2 = \left|\tfrac{1}{\sqrt{2}}\right|^2 = \tfrac{1}{2}. \tag{22}$$

So: the beam filtered to $|+\rangle$ splits 50:50 at the $S_x$ magnet; the $|+\rangle_x$ beam, expressed in the $z$ basis, is an equal superposition again, and the final $S_z$ magnet splits it 50:50 — one quarter of the twice-filtered atoms emerge with $S_z = -\hbar/2$, the value the first filter had *removed*. The numbers say what §2.1 could only gesture at: measuring $S_x$ prepared an $S_x$ eigenstate, which is not an $S_z$ eigenstate but a superposition of both — filtering is *preparation*, and incompatible preparations overwrite each other. (What "measurement prepares eigenstates" means as a postulate is §4's opening business.)

### 3.6 Composite systems: the tensor product

One construction completes the kit. Two systems with spaces $\mathcal{H}_1, \mathcal{H}_2$ compose into one with the *tensor product* $\mathcal{H}_1 \otimes \mathcal{H}_2$: basis kets are pairs $|a\rangle \otimes |b\rangle \equiv |a, b\rangle$, dimensions *multiply*, operators of system 1 act as $A \otimes \mathbf{1}$ (each subsystem's operators ignore the other), and inner products factorize on product kets. The physically loaded fact is what the construction contains beyond products: a general vector of $\mathcal{H}_1 \otimes \mathcal{H}_2$ is a superposition of product kets that need not *be* a product ket — for two spins,

$$\tfrac{1}{\sqrt{2}}\big(|+,-\rangle - |-,+\rangle\big) \tag{23}$$

assigns neither spin a state of its own. Such states are called *entangled*, they are the generic case (product states are a measure-zero subset), and their consequences — correlations no classical joint distribution reproduces — are flagged here and delivered as a physical specimen in §15 (Remark 15.2); the outlook's identical particles are where nature makes the choice of symmetric versus antisymmetric combinations for us. [stated at flag level; the Bell-inequality story is real, quantitative, and outside this document's arc.]

The stage is built. What has not been said is what happens on it when someone *looks* — the postulates connecting the formalism to laboratory statistics, and the inequality that §2.3 promised to make quantitative.

## 4 Measurement and Uncertainty

The stage of §3 connects to laboratory statistics through two postulates — what the outcomes are, and what remains afterward. Both were used, unproved, in Example 3.7; this section supplies them, extracts the algebraic criterion for when two observables tolerate each other, and converts §2.3's qualitative "no common eigenvectors" into the quantitative uncertainty relation — derived, then *stress-tested*: the section ends by exhibiting, with Hall, a state that seems to violate it, and locating the loophole.

### 4.1 The measurement postulates

**Postulate 4.1** (Born rule)**.** Measuring an observable $A = \sum_a a\, P_a$ (spectral form, §3.4) on a normalized state $|\psi\rangle$ returns one of the eigenvalues $a$, with probability

$$\boxed{P(a) = \langle\psi|P_a|\psi\rangle} \qquad \big(= |\langle a|\psi\rangle|^2 \text{ when } a \text{ is nondegenerate}\big). \tag{24}$$

**Postulate 4.2** (Projection)**.** If the outcome is $a$, the state immediately after the measurement is

$$|\psi\rangle \;\longrightarrow\; \frac{P_a|\psi\rangle}{\sqrt{P(a)}}. \tag{25}$$

Three consistency checks, each one line, each delivering a piece of §3.4:

- **Probabilities sum to one** because the projectors do: $\sum_a P(a) = \langle\psi|\big(\sum_a P_a\big)|\psi\rangle = \langle\psi|\psi\rangle = 1$ — the resolution of the identity (9) *is* the conservation of probability, which is the physical reason completeness sat in the postulates at all.

- **Measurement is repeatable**: measuring $A$ again on the post-measurement state (25) returns $a$ with probability $\langle\psi|P_a^3|\psi\rangle/P(a) = 1$ by idempotence — "asking twice equals asking once," now a law rather than a slogan, and exactly the "filtering is preparation" that Example 3.7 read off the beams.

- **Statistics match**: $\sum_a a\,P(a) = \langle\psi|A|\psi\rangle$ — the expectation value (14) is the mean of the Born distribution, as §3.4 promised.

*Remark* 4.3 (What the postulates do not say). Equation (25) is a *rule*, not a mechanism: it does not say what physical process implements the jump, when exactly "a measurement" has occurred, or how the linear, deterministic dynamics of Part II coexists with a nonlinear, stochastic update. These questions — the measurement problem, decoherence, the classical limit of measuring devices — are genuine, quantitative, and open in parts; this document uses the postulates as every working calculation does, and flags the foundations rather than litigating them. [stated; decoherence theory is where the modern discussion lives.]

## 4.2 Compatible and incompatible observables

When can two observables have sharp values together? Exactly when their operators share an eigenbasis — and the criterion is the commutator, closing the loop opened by Stern–Gerlach in §2.1.

**Proposition 4.4** (Compatibility criterion)**.** *Two Hermitian operators with* $[A, B] = 0$ *possess a common orthonormal eigenbasis, and conversely.*

*Proof (nondegenerate case, displayed; general case stated).* Let $A|a\rangle = a|a\rangle$ with $a$ nondegenerate. Then

$$A\left(B|a\rangle\right) = B\,A|a\rangle = a\left(B|a\rangle\right): \tag{26}$$

$B|a\rangle$ is again an $a$-eigenvector of $A$, hence (nondegeneracy) proportional to $|a\rangle$ — which says $B|a\rangle = b|a\rangle$: the $A$-eigenbasis already diagonalizes $B$. The converse is immediate (diagonal matrices commute). [Degenerate case: diagonalize $B$ within each $A$-eigenspace; Sakurai's compatible-observables discussion runs the induction.] □

Compatible observables can be measured in either order with the same statistics, and their eigenvalues can label states jointly — the practice, formalized as a *complete set of commuting observables* (CSCO), of naming a state by the full list of sharp quantum numbers it carries. The hydrogen atom's $|n, \ell, m\rangle$ (§14) is the series' destination example. Incompatible observables are the interesting ones: $[S_z, S_x] = i\hbar\, S_y \neq 0$, and the Stern–Gerlach overwriting of §2.1 now has its one-line algebraic diagnosis. How *much* incompatibility costs is the next subsection's theorem.

### 4.3 Derivation of the uncertainty relation

**Theorem 4.5** (Robertson)**.** *For Hermitian $A, B$ and a normalized state $|\psi\rangle$ (in the domains of the operators involved — a clause that Remark 4.8 will show is load-bearing),*

$$\boxed{\Delta A\,\Delta B \;\geq\; \tfrac{1}{2}\left|\langle\psi|\,[A,B]\,|\psi\rangle\right|.} \tag{27}$$

*Proof.* Center the operators: $\bar{A} \equiv A - \langle A\rangle$, $\bar{B} \equiv B - \langle B\rangle$, so that $(\Delta A)^2 = \|\bar{A}|\psi\rangle\|^2$ by (15), and note the shifts cancel inside the commutator, $[\bar{A}, \bar{B}] = [A, B]$. Cauchy–Schwarz on the two vectors $\bar{A}|\psi\rangle$, $\bar{B}|\psi\rangle$:

$$(\Delta A)^2(\Delta B)^2 = \|\bar{A}|\psi\rangle\|^2\,\|\bar{B}|\psi\rangle\|^2 \;\geq\; \left|\langle\psi|\bar{A}\,\bar{B}|\psi\rangle\right|^2. \tag{28}$$

Split the product into its Hermitian and anti-Hermitian parts,

$$\bar{A}\bar{B} = \tfrac{1}{2}\{\bar{A}, \bar{B}\} + \tfrac{1}{2}[\bar{A}, \bar{B}], \tag{29}$$

whose expectation values are respectively real and purely imaginary (the anticommutator of Hermitians is Hermitian; the commutator, anti-Hermitian — the same computation as in §2's "the i" note). A complex number's modulus dominates its imaginary part:

$$\left|\langle\psi|\bar{A}\bar{B}|\psi\rangle\right|^2 = \tfrac{1}{4}\left\langle\{\bar{A}, \bar{B}\}\right\rangle^2 + \tfrac{1}{4}\left|\left\langle[A,B]\right\rangle\right|^2 \;\geq\; \tfrac{1}{4}\left|\left\langle[A,B]\right\rangle\right|^2, \tag{30}$$

and chaining (28)–(30) and taking square roots gives (27). (Keeping the anticommutator term instead of discarding it strengthens the bound to the *Schrödinger* inequality; the discarded term measures the $A$–$B$ correlations in the state.) □

Feed the canonical pair to the theorem: $[\hat{q}, \hat{p}] = \mathrm{i}\hbar\,\mathbf{1}$ has a *state-independent* expectation, so

$$\boxed{\Delta q\;\Delta p \;\geq\; \frac{\hbar}{2}} \tag{31}$$

for every state in the operators' domains (the theorem's parenthetical clause rides along) — the quantitative flesh on §2.3's skeleton, with the classical limit visible in the formula: the floor under the statistics is set by exactly the deformation parameter of Convention 1.1. For non-canonical pairs the right-hand side is an expectation value and therefore *state-dependent*:

$$\Delta S_x\,\Delta S_y \;\geq\; \tfrac{\hbar}{2}\left|\langle S_z\rangle\right|, \tag{32}$$

a weaker-looking statement with a physical point — there exist states ($\langle S_z\rangle = 0$, anywhere on the Bloch equator) where nothing prevents sharp $S_x$ *and* $S_y$ statistics from coexisting in the bound, and the bound correctly goes slack.

**Example 4.6** (The pole saturates)**.** Run (32) on the laboratory. In the state $|+\rangle$ (north pole of Fig. 1): the Pauli identity (18) gives $S_x^2 = S_y^2 = \frac{\hbar^2}{4}\mathbf{1}$, while $\langle S_x\rangle = \langle S_y\rangle = 0$ (the matrices (17) are purely off-diagonal). So

$$\Delta S_x = \Delta S_y = \frac{\hbar}{2}, \qquad \Delta S_x\,\Delta S_y = \frac{\hbar^2}{4} = \frac{\hbar}{2}\left|\langle S_z\rangle\right|: \tag{33}$$

the inequality is *saturated* — the spin-up state is a minimum-uncertainty state of the transverse pair, its unavoidable transverse spread exactly the floor the algebra charges for a sharp $S_z$. (Geometrically: a Bloch vector pinned to the pole has its transverse components maximally undetermined — the sphere picture and the algebra agree.)

**What saturation requires in general.** Equality in (27) needs equality at both inequalities of the proof: Cauchy–Schwarz collapses only when the two vectors are parallel, $\bar{B}|\psi\rangle = c\,\bar{A}|\psi\rangle$, and the discarded anticommutator term vanishes only when $\mathrm{Re}\,c = 0$ (substitute the parallelism into $\langle\{\bar{A},\bar{B}\}\rangle = 2\,\mathrm{Re}(c)\,(\Delta A)^2$). Hence the *minimum-uncertainty condition*:

$$\big(\bar{B} - \mathrm{i}\lambda\,\bar{A}\big)|\psi\rangle = 0, \qquad \lambda \in \mathbb{R}. \tag{34}$$

For the canonical pair this is a first-order equation whose solution requires a representation to solve in — an IOU: §5 builds the position representation, solves (34) there in two lines, and finds the Gaussian wave packet: *the* state, unique up to translations, that sits exactly on the floor (31).

*Remark* 4.7 (Interpretive care: preparation statistics versus disturbance). Everything in (27) refers to *preparation statistics*: $\Delta A$ and $\Delta B$ are standard deviations of outcome distributions over many runs on identically prepared systems, each run measuring *one* observable. The theorem says incompatible statistics cannot both be sharp *in the same preparation* — it does not, by itself, quantify how measuring $A$ disturbs a subsequent measurement of $B$ on the same system (Heisenberg's original microscope heuristic). The disturbance question is real physics with its own modern formulations, and conflating the two readings is a standard misstatement of the uncertainty principle. [stated; error–disturbance relations are a living literature.]

*Remark* 4.8 (The counterexample and where the proof leans). Now the stress test, following Hall §12.2. Work on $L^2[-1,1]$ with the bounded "position" $A\psi(x) = x\,\psi(x)$ and the "momentum" $B = -\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ on continuously differentiable functions with *periodic* boundary values $\psi(-1) = \psi(1)$ — a legitimate, essentially self-adjoint operator (its eigenvectors $\psi_n = \mathrm{e}^{\mathrm{i}\pi n x}/\sqrt{2}$, $n \in \mathbb{Z}$, form an orthonormal basis). On a dense domain one checks $AB - BA = \mathrm{i}\hbar$, exactly the canonical value. Yet take $|\psi\rangle = \psi_n$: it is an eigenvector of $B$, so $\Delta B = 0$, while $\Delta A$ is finite — and

$$\Delta A\,\Delta B = 0 \;<\; \frac{\hbar}{2}\,, \tag{35}$$

an apparent violation. The resolution is the parenthetical clause of Theorem 4.5: the proof applies $B$ after $A$, and $A\psi_n = x\,\psi_n$ *violates the periodic boundary condition* ($x\,\psi_n$ takes different values at $\pm 1$), so $\psi_n \notin \mathrm{Dom}(BA)$ — the state sits exactly outside the hypotheses, and the inequality never applied to it. Two morals, one local, one global. Local: the physical system here is a particle on a *ring*, $A$ is (essentially) an angle and $B$ an angular momentum, and the lesson is that a bounded, periodic coordinate admits no naive uncertainty relation with its conjugate momentum — a fact that returns, constructively, when §13 builds angular momentum. Global: this is the promised second bite of Remark 3.3 — domains are not decoration, and the hypotheses of a theorem are part of its physics. [Example verified against Hall §12.2.]

Kinematics still lacks its most-used representation: the one in which states are *functions*, $\psi(x)$, and (34) becomes a differential equation. Building it — and deriving $\hat{p} = -\mathrm{i}\hbar\,\partial_x$ within it — is the next section's business.

# 5 Position, Momentum, and Translation

Kinematics closes with its most-used representation: states as functions. Building it means confronting an operator whose spectrum is a continuum — which imports the delta apparatus of the field-theory notes — and the payoff is structural: the canonical commutator, *postulated* in §2 as a deformation of the classical bracket, will here be *derived* a second, independent way, from the geometry of translations. Two roads, one commutator; their convergence is the first entry of the quantum Noether dictionary.

### 5.1 The position basis and its fine print

A particle on a line has a position operator $\hat{q}$ whose possible sharp values fill $\mathbb{R}$: a continuous spectrum. Write formal eigenkets $\hat{q}|x\rangle = x|x\rangle$ and trade Theorem 3.4's sums for integrals, replacing the Kronecker delta of orthonormality by its continuum original:

$$\langle x'|x\rangle = \delta(x - x'), \qquad \boxed{\mathbf{1} = \int \mathrm{d}x\, |x\rangle\langle x|.} \tag{36}$$

The delta function arrives with the license already issued in [Fld. §10.2, App. B]: not a function but a distribution, every manipulation below the shadow of a legitimate operation on smeared objects — the identical fiction, imported wholesale. The fine print: $|x\rangle$ itself is *not* a vector of the Hilbert space (its "norm" is $\delta(0)$); it is a computational scaffold, and the physically meaningful objects are the coefficients it produces,

$$\psi(x) \equiv \langle x|\psi\rangle, \qquad \langle\psi|\psi\rangle = \int \mathrm{d}x\, \left|\psi(x)\right|^2 = 1, \tag{37}$$

— the *wavefunction*: the state's components in the position basis, exactly as $\langle a|\psi\rangle$ were its components in a discrete one. Born's rule (24) reads through the trade: with no normalizable eigenvector, $|\psi(x)|^2$ is a probability *density*, its integral over a region the probability of finding the particle there. [The mathematically clean route is the spectral theorem without eigenvectors, Hall Ch. 10; rigged Hilbert spaces legitimizing $|x\rangle$ directly are a second route, outside Hall; stated, and the working notation is safe for everything this document does.]

### 5.2 Translation and the operator it generates

Now the geometry. Define the *translation operator* by its action on the position basis,

$$T(a)\,|x\rangle = |x + a\rangle, \tag{38}$$

extended by linearity. It is unitary (it maps the orthonormal continuum (36) onto itself), composes as the line's additive group, $T(a)\,T(b) = T(a+b)$, $T(0) = \mathbf{1}$, and carries one commutator on its face: on any basis ket,

$$\hat{q}\,T(a)|x\rangle = (x+a)|x+a\rangle, \qquad T(a)\,\hat{q}|x\rangle = x|x+a\rangle \qquad \Longrightarrow \qquad [\hat{q},\, T(a)] = a\,T(a). \tag{39}$$

Look at an infinitesimal translation. Unitarity and smoothness force the form

$$T(\mathrm{d}a) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\,\hat{p}\,\mathrm{d}a + O(\mathrm{d}a^2), \tag{40}$$

which *defines* an operator $\hat{p}$ — the *generator* of translations — Hermitian because $T$ is unitary (expand $T^\dagger T = \mathbf{1}$ to first order: $\hat{p}^\dagger = \hat{p}$), with the $\hbar$ inserted so that $[\hat{p}] = [\text{action}]/[\text{length}] = [\text{momentum}]$: the name is a promise, about to be kept. Insert (40) into (39) and keep first order:

$$-\frac{\mathrm{i}}{\hbar}\,[\hat{q}, \hat{p}]\,\mathrm{d}a = \mathrm{d}a\,\mathbf{1} \qquad \Longrightarrow \qquad \boxed{[\hat{q}, \hat{p}] = \mathrm{i}\hbar\,\mathbf{1}.} \tag{41}$$

*Remark* 5.1 (Two roads to one commutator: the quantum Noether dictionary). Equation (41) is the canonical commutation relation (5) again — reached by a completely different road. Section 2 *deformed* the classical bracket ($\{q, p\} = 1$, Dirac's rule); this section never mentioned a bracket, only the statement *"$\hat{p}$ is what translates."* That the two roads meet is not a coincidence to admire but a dictionary to use: in mechanics, Noether traded each symmetry for a conserved quantity, and momentum *was* the translation entry [Mech. §3.3], as it was again for fields [Fld. §9.4]. Quantum mechanics keeps the trade and sharpens it: *the conserved quantity is the symmetry's generator*, as an operator —

| symmetry | generator | where |
|---|---|---|
| spatial translation | $\hat{p}$ | here |
| time translation | $\hat{H}$ | §6 |
| rotation | $\hat{J}_i$ | §11 |

— and the commutators among generators are the transcription of the classical bracket algebra, one theorem per row ahead.

### 5.3 $\hat{p}$ in the position representation

What does $\hat{p}$ do to wavefunctions? Compute $\langle x|T(\mathrm{d}a)|\psi\rangle$ twice. Geometrically: acting leftward, $\langle x|T(\mathrm{d}a) = \langle x - \mathrm{d}a|$ (the adjoint of $T(\mathrm{d}a)|x\rangle = |x + \mathrm{d}a\rangle$, since $T^\dagger(\mathrm{d}a) = T(-\mathrm{d}a)$), so

$$\langle x|T(\mathrm{d}a)|\psi\rangle = \psi(x - \mathrm{d}a) = \psi(x) - \mathrm{d}a\,\frac{\partial\psi}{\partial x} + O(\mathrm{d}a^2). \tag{42}$$

Algebraically, via (40): $\langle x|T(\mathrm{d}a)|\psi\rangle = \psi(x) - \frac{\mathrm{i}}{\hbar}\,\mathrm{d}a\,\langle x|\hat{p}|\psi\rangle$. Equate the first orders:

$$\boxed{\langle x|\,\hat{p}\,|\psi\rangle = -\,\mathrm{i}\hbar\,\frac{\partial}{\partial x}\,\psi(x)} \tag{43}$$

— the textbook formula for momentum: here not a postulate but the unique answer to "what must the translation generator look like on functions that get dragged." Hermiticity in this representation is one integration by parts, $\int \psi_1^*(-\mathrm{i}\hbar\,\partial_x\psi_2)\,\mathrm{d}x = \int(-\mathrm{i}\hbar\,\partial_x\psi_1)^*\psi_2\,\mathrm{d}x$ + boundary terms — and the boundary terms are precisely where the potholes of Remarks 3.3 and 4.8 lived: on the full line with decay they vanish and all is well; on intervals, they were the whole story.

The eigenvalue problem of (43) is immediate: $-\mathrm{i}\hbar\,\partial_x\langle x|p\rangle = p\langle x|p\rangle$ gives

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\,\mathrm{e}^{\mathrm{i}px/\hbar}, \qquad \langle p'|p\rangle = \frac{1}{2\pi\hbar}\int \mathrm{d}x\,\mathrm{e}^{\mathrm{i}(p-p')x/\hbar} = \delta(p - p') \tag{44}$$

(the normalization check is the Fourier representation of the delta: the $\sigma \to \infty$ limit of the Gaussian pair (178), read distributionally under the license of [Fld. App. B]). A momentum eigenstate is a plane wave of wavelength

$$\lambda = \frac{2\pi\hbar}{p} = \frac{h}{p} \tag{45}$$

— de Broglie's relation, here neither postulate nor analogy but the shape of the translation generator's eigenfunctions.

### 5.4 The momentum representation and the Fourier bridge

The momentum eigenkets are delta-normalized and complete in their own right, so a state has momentum-basis components $\varphi(p) \equiv \langle p|\psi\rangle$, and the bridge between the two descriptions is one insertion of (36):

$$\varphi(p) = \int \mathrm{d}x\,\langle p|x\rangle\,\langle x|\psi\rangle = \int \frac{\mathrm{d}x}{\sqrt{2\pi\hbar}}\,\mathrm{e}^{-\mathrm{i}px/\hbar}\,\psi(x) \tag{46}$$

— the Fourier transform, with its inverse running the other way. This is Remark 3.5 delivered in full: position and momentum representations are the two adapted coordinate systems ($\hat{q}$ diagonal versus $\hat{p}$ diagonal), the change between them is a basis change like any other, and the celebrated transform is nothing but $\langle p|x\rangle$ used as a transition matrix. In the momentum representation the roles swap symmetrically: $\hat{p}$ multiplies and $\langle p|\hat{q}|\psi\rangle = +\mathrm{i}\hbar\,\partial_p\varphi(p)$ (same derivation, sign from conjugating (44)).

## 5.5 The Gaussian minimum-uncertainty packet

Section 4.3 left one item open: the minimum-uncertainty condition (34), waiting for a representation to be solved in. In the position representation, with $\bar{p} = \hat{p} - p_0$ and $\bar{q} = \hat{q} - x_0$ (expectations $p_0, x_0$), the condition $(\bar{p} - \mathrm{i}\lambda\bar{q})|\psi\rangle = 0$ reads

$$\Big(-\mathrm{i}\hbar\,\frac{\partial}{\partial x} - p_0 - \mathrm{i}\lambda\,(x - x_0)\Big)\,\psi(x) = 0 \qquad \Longrightarrow \qquad \frac{\partial\psi}{\partial x} = \Big[\frac{\mathrm{i}p_0}{\hbar} - \frac{\lambda}{\hbar}(x - x_0)\Big]\psi, \tag{47}$$

a first-order linear equation with a two-line solution:

$$\boxed{\psi(x) = N\,\exp\Big[\frac{\mathrm{i}p_0 x}{\hbar} - \frac{(x-x_0)^2}{4(\Delta x)^2}\Big], \qquad (\Delta x)^2 = \frac{\hbar}{2\lambda},} \tag{48}$$

normalizable precisely when $\lambda > 0$, with the variance read off the profile $|\psi|^2 \propto \exp[-(x - x_0)^2/2(\Delta x)^2]$ (integrals: Appendix A). No further computation is needed for the momentum spread: the state was *constructed* to saturate, so $\Delta p = \hbar/(2\,\Delta x)$ by Theorem 4.5's equality case — and the direct check confirms it: the Fourier bridge (46) of a Gaussian is again a Gaussian (Appendix A), with exactly the reciprocal width. *The Gaussian wave packet is the unique family of states sitting on the floor* $\Delta x\,\Delta p = \hbar/2$ (Fig. 2): squeeze the left panel and the right panel broadens in exact compensation, the deformation parameter $\hbar$ fixing the product.

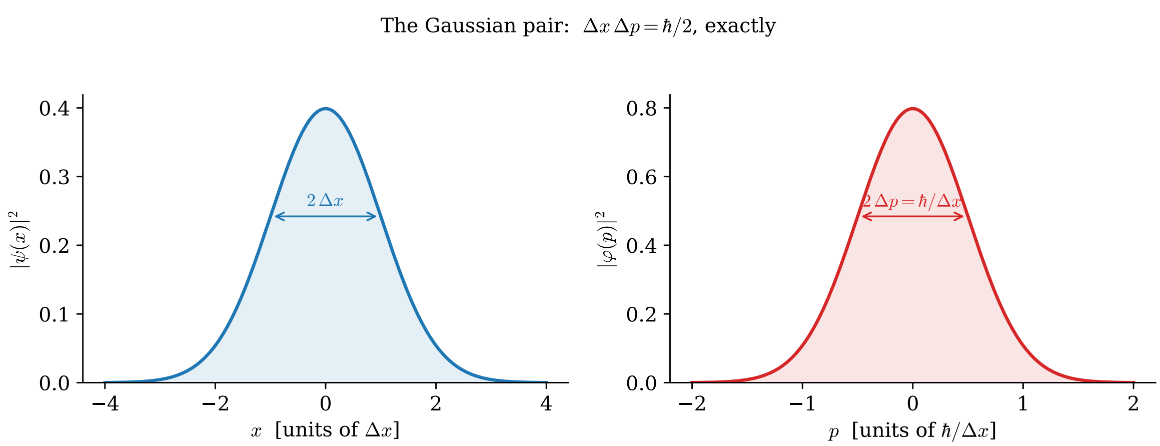


Figure 2: The minimum-uncertainty state (48) in both adapted coordinate systems, drawn from the exact formulas ($\hbar = 1$, $\Delta x = 1$, $x_0 = p_0 = 0$). The widths are reciprocal, $\Delta p = \hbar/(2\Delta x)$: narrowing the position distribution forces a wider momentum distribution, the product fixed at $\Delta x\,\Delta p = \hbar/2$.

**Example 5.2** (Why atoms are the size they are)**.** The uncertainty floor is not philosophy; it is a pressure, and it holds atoms up — answering the first puzzle of §2.1. Confine an electron to an atomic length, $\Delta x \approx 0.5\,\text{Å} = 5 \times 10^{-11}\,\text{m}$. The floor (31) forces

$$p \gtrsim \frac{\hbar}{2\,\Delta x} \approx 1.1 \times 10^{-24}\,\text{kg m/s} \qquad \Longrightarrow \qquad E_{\text{kin}} \sim \frac{p^2}{2m_e} \approx 6.1 \times 10^{-19}\,\text{J} \approx 3.8\,\text{eV} : \tag{49}$$

electron-volt-scale kinetic energy, forced by localization alone — the atomic energy scale, from nothing but $\hbar$, $m_e$, and a length. Squeezing the electron further toward the nucleus raises this cost as $1/(\Delta x)^2$, faster than the Coulomb energy $\sim -1/\Delta x$ can compensate; the balance point *is* the atom, and the classical spiral-in of §2.1 is arrested by the same inequality drawn in Fig. 2. (The balance is made exact, with the numerical constant earned, at the hydrogen capstone of §14.)

Part I is complete: an algebra (§2), a stage (§3), measurement rules (§4), and coordinates (§5). Nothing yet *moves*. Part II turns on time — and the first thing it will do is run the quantum Noether dictionary's second row.

# Part II
# Dynamics

## 6 Time Evolution

Nothing in Part I moved. This section turns on time, and the structure is deliberately a rerun: exactly as §5.2 derived the momentum operator from "something must implement spatial shifts," here the Hamiltonian is derived from "something must implement waiting" — which fills the second row of the quantum Noether dictionary (Remark 5.1). The section ends where quantum dynamics always ends: in the energy eigenbasis, where the whole of time evolution is a set of rotating phases — and with the series' laboratory precessing on the Bloch sphere — the working principle of MRI.

### 6.1 The evolution operator and where $H$ comes from

Let $|\psi(t)\rangle$ be the state at time $t$ of a system prepared as $|\psi(0)\rangle$. Two physical requirements pin down the map between them:

- **Probability is conserved**: $\langle\psi(t)|\psi(t)\rangle = 1$ at all times, for every initial state — the map $U(t)$ with $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ must be *unitary* (§3.2: unitaries are exactly the inner product's isometries).
- **Waiting composes**: evolving for $t_1$ and then $t_2$ is evolving for $t_1 + t_2$, and no moment is special (a closed system; explicit time dependence would encode an outside agent): $U(t_2)\,U(t_1) = U(t_1 + t_2)$, $U(0) = \mathbf{1}$.

These are verbatim the properties that §5.2 used for spatial shifts — a one-parameter unitary group, now in time — so the same move applies: expand the infinitesimal element,

$$U(\mathrm{d}t) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\,\hat{H}\,\mathrm{d}t + O(\mathrm{d}t^2), \tag{50}$$

which *defines* a Hermitian operator $\hat{H}$ — the generator of time translations — with $\hbar$ inserted so that $[\hat{H}] = [\text{action}]/[\text{time}] = [\text{energy}]$. The name *Hamiltonian* is the dictionary's second row: in mechanics, the conserved quantity guaranteed by time-translation invariance *was* the Hamiltonian [Mech. §§3.3, 6.2], and quantum mechanics promotes that conserved quantity to the generator itself. [That every strongly continuous one-parameter unitary group has such a self-adjoint generator is Stone's theorem — Hall §10.2; stated, and it is the precise justification for (50).]

Differentiate the composition law at finite $t$ — or simply apply (50) to $U(t + \mathrm{d}t) = U(\mathrm{d}t)\,U(t)$ — to get the operator equation and its state-level shadow:

$$\boxed{\mathrm{i}\hbar\,\frac{\partial}{\partial t}\,U(t) = \hat{H}\,U(t), \qquad \mathrm{i}\hbar\,\frac{\partial}{\partial t}\,|\psi(t)\rangle = \hat{H}\,|\psi(t)\rangle} \tag{51}$$

— the *Schrödinger equation*, obtained here as the differential form of "waiting is unitary and composes," with all its famous features inherited rather than postulated: linear (because $U$ is), probability-conserving (because $U$ is), and first order in time (because groups have one-parameter tangents). For an $\hat{H}$ with no explicit time dependence the solution closes:

$$U(t) = \exp\!\Big[-\frac{\mathrm{i}\,\hat{H}\,t}{\hbar}\Big]. \tag{52}$$

(Sandwiching (51) in the position basis, with the Hamiltonian $\hat{p}^2/2m + V(\hat{q})$ and the representation (43), produces the wave equation $\mathrm{i}\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m}\partial_x^2\psi + V\psi$ — the form §8 will live in.)

## 6.2 The energy eigenbasis: dynamics is phases

Now the adapted-coordinates discipline (Remark 3.5), fourth appearance — and its most consequential. Diagonalize the operator that generates the dynamics: $\hat{H}|n\rangle = E_n|n\rangle$. In this basis (52) is diagonal, and every initial state evolves by

$$\boxed{|\psi(t)\rangle = \sum_n c_n \, \mathrm{e}^{-\mathrm{i}E_n t/\hbar}\,|n\rangle, \qquad c_n = \langle n|\psi(0)\rangle :} \tag{53}$$

*the entire dynamics of quantum mechanics is a set of rotating phases*, one per energy level, at rates set by the spectrum. Solving a quantum system *means* finding the spectrum; everything after is (53). Two immediate consequences, one per extreme:

- **One term: stationary states.** An energy eigenstate evolves as $\mathrm{e}^{-\mathrm{i}E_n t/\hbar}|n\rangle$ — an overall phase, physically empty (§3.1): every probability, every expectation value is frozen. Here, at last, is the full answer to §2.1's stability crisis: an atom in its ground state is not an electron *orbiting* — not a charge accelerating, not a system doing anything — it is a stationary state, and stationary states have no dynamics to radiate away.

- **Two terms: beats at the Bohr frequency.** A superposition $c_1|1\rangle + c_2|2\rangle$ has one *relative* phase — physical (§3.1) — rotating at

$$\omega_{21} = \frac{E_2 - E_1}{\hbar}, \tag{54}$$

  so every observable connecting the two levels oscillates at $\omega_{21}$: the quantum system *beats*, exactly as two coupled classical modes did [Osc. §6.6], with energy differences playing the role of frequency differences. This is why atoms emit *lines* and why the lines sit at differences of a fixed set of terms (the Rydberg–Ritz combination principle, mysterious in 1908, one line in (54)): this answers the second experimental puzzle of §2.1 in structure; the numbers arrive at the hydrogen capstone. [One gap, flagged: an atom *prepared* in an excited stationary state also radiates — spontaneous emission — which no isolated-atom stationary state can explain; the mechanism is the coupling to the quantized field of the outlook, and the beat picture above is its semiclassical face. Stated.]

## 6.3 The laboratory precesses: spin in a magnetic field

Run the machinery end to end on the two-state laboratory. A spin-$\frac{1}{2}$ magnetic moment in a uniform field $\vec{B} = B\hat{z}$ has

$$\hat{H} = -\gamma B\,\hat{S}_z, \qquad E_\pm = \mp\frac{\gamma\hbar B}{2}, \qquad \omega \equiv \frac{E_- - E_+}{\hbar} = \gamma B \tag{55}$$

($\gamma$ the gyromagnetic ratio; the $S_z$ eigenstates are the stationary states, and $\omega$ is the Bohr frequency of the only pair there is). Prepare the state at the equator, $|\psi(0)\rangle = |+\rangle_x = (|+\rangle + |-\rangle)/\sqrt{2}$, and apply (53):

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\Big(\mathrm{e}^{\mathrm{i}\omega t/2}\,|+\rangle + \mathrm{e}^{-\mathrm{i}\omega t/2}\,|-\rangle\Big), \tag{56}$$

which is the Bloch state (19) with $\theta = \frac{\pi}{2}$, $\varphi = -\omega t$: the expectation vector,

$$\langle S_x\rangle = \tfrac{\hbar}{2}\cos\omega t, \qquad \langle S_y\rangle = -\tfrac{\hbar}{2}\sin\omega t, \qquad \langle S_z\rangle = 0, \tag{57}$$

*precesses* around the field at the *Larmor frequency* $\omega = \gamma B$ — uniform circular motion of the Bloch vector on the equator of Fig. 1: the figure of §3.5, now a movie.

**Example 6.1** (Larmor precession in the clinic: MRI)**.** For a proton, $\gamma_p/2\pi = 42.58\,\text{MHz/T}$ [experimental input; CODATA]. In the 3 T field of a clinical MRI magnet, the hydrogen nuclei of the body's water precess at

$$f = \frac{\gamma_p B}{2\pi} = 42.58\,\text{MHz/T} \times 3\,\text{T} \approx 128\,\text{MHz} \tag{58}$$

— a radio frequency. An MRI machine is a Larmor-precession transceiver: it tips the protons' Bloch vectors off axis, listens for the 128 MHz signal of their collective precession, and maps tissue by how the precession decays. A clinical scan is Eq. (53) run on $\sim 10^{27}$ copies of the two-state laboratory.

*Remark* 6.2 (The $2\pi$ seed)*.* Watch the state, not just its expectation vector, through one full turn. At $t = 2\pi/\omega$ the Bloch vector has returned — but the ket has not:

$$|\psi(2\pi/\omega)\rangle = \frac{1}{\sqrt{2}}\big(\mathrm{e}^{\mathrm{i}\pi}|+\rangle + \mathrm{e}^{-\mathrm{i}\pi}|-\rangle\big) = -\,|\psi(0)\rangle\ : \tag{59}$$

a full $2\pi$ of precession multiplies the state by $-1$, and only $4\pi$ restores it. For an isolated spin the sign is an overall phase, invisible (§3.1) — but a phase becomes *relative*, and hence measurable, the moment the spin is one arm of an interferometer. That experiment exists, with neutrons, and §12 owes the reader its number; the present remark is the seed: the $-1$ is not an artifact of this Hamiltonian but the signature of what kind of representation of the rotation group a spin-$\frac{1}{2}$ carries.

*Remark* 6.3 (Energy–time is not Robertson)*.* A caution the two-level beat invites. There is no time operator to feed Theorem 4.5 — in this formalism $t$ is a parameter, not an observable [stated; the obstruction to a self-adjoint time conjugate to a bounded-below $H$ is Pauli's observation], so "$\Delta E\,\Delta t \geq \hbar/2$" cannot mean what $\Delta q\,\Delta p$ means. The precise version is one line away. Differentiate an expectation value with (51):

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle A\rangle = \frac{\mathrm{i}}{\hbar}\left\langle\, [\hat{H}, \hat{A}]\,\right\rangle \qquad (A \text{ with no explicit } t\text{-dependence}), \tag{60}$$

then run Robertson on the pair $(\hat{A}, \hat{H})$ and substitute: $\Delta A\,\Delta E \geq \frac{\hbar}{2}\,|\mathrm{d}\langle A\rangle/\mathrm{d}t|$, i.e.

$$\Delta E\;\tau_A \;\geq\; \frac{\hbar}{2}, \qquad \tau_A \equiv \frac{\Delta A}{|\mathrm{d}\langle A\rangle/\mathrm{d}t|} \tag{61}$$

(Mandelstam–Tamm): $\tau_A$ is the time for the statistics of $A$ to shift by one standard deviation, and the theorem says *states with sharp energy change slowly* — the stationary state ($\Delta E = 0$, nothing moves) is its extreme case, and a spectral line's width–lifetime relation its laboratory face. Equation (60) is more than this remark's tool: it is the quantum mirror of $\dot{F} = \{F, H\}$ [Mech. §8.2], about to be promoted from expectation values to the operators themselves — which is precisely the next section.

# 7 Two Pictures of One Physics

Remark 6.3 ended mid-promotion: the flow equation (60) moved *expectation values*, and the promise was to move the operators themselves. This section keeps it — and the payoff, for a reader of the mechanics notes: the quantum equation of motion *is* the classical bracket flow, transcribed by Dirac's rule with nothing lost in translation. The section then locates the classical limit (Ehrenfest), and closes by dissolving a historical rivalry into a choice of coordinates.

### 7.1 The Heisenberg picture: operators flow

Everything measurable is a sandwich, and a sandwich has two slices:

$$\langle\psi(t)|\,\hat{A}\,|\psi(t)\rangle = \langle\psi(0)|\,\underbrace{U^\dagger(t)\,\hat{A}\,U(t)}_{\equiv\ \hat{A}_H(t)}\,|\psi(0)\rangle. \tag{62}$$

Section 6 attached the evolution to the state (the *Schrödinger picture*); attach it instead to the operator — states frozen at $|\psi(0)\rangle$, observables carrying the time — and nothing measurable changes, because (62) is an identity. This is the *Heisenberg picture*, and its equation of motion is one derivative away. With $\hat{H}$ time-independent, $U(t) = \mathrm{e}^{-\mathrm{i}\hat{H}t/\hbar}$ commutes with $\hat{H}$, and differentiating $\hat{A}_H = U^\dagger \hat{A}\, U$ by the product rule (using (51) and its adjoint):

$$\frac{\mathrm{d}\hat{A}_H}{\mathrm{d}t} = \frac{\mathrm{i}}{\hbar}\,U^\dagger\,[\hat{H},\hat{A}]\,U = \frac{\mathrm{i}}{\hbar}\,[\hat{H},\hat{A}_H], \tag{63}$$

i.e., written to make the point unmissable:

$$\boxed{\frac{\mathrm{d}\hat{A}_H}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\,[\hat{A}_H,\,\hat{H}]} \qquad \text{versus} \qquad \frac{\mathrm{d}F}{\mathrm{d}t} = \{F,H\} \quad \text{[Mech. §8.2]}. \tag{64}$$

Dirac's rule (4) was installed in §2 as a statement about *kinematics* — replace the bracket's realization, keep its algebra. Equation (64) says the replacement extends to *dynamics* verbatim: the law of motion is the same sentence in the new realization. Three corollaries fall out at the speed of substitution:

- **Conservation is commutation.** $[\hat{A},\hat{H}] = 0$ freezes $\hat{A}_H$ for all time — the quantum Noether loop closes at the operator level: the generators of Remark 5.1's table are conserved exactly when their symmetry is a symmetry *of the dynamics*, and by §4.2 the conserved quantities are precisely the observables sharable with energy. A CSCO containing $\hat{H}$ is therefore the natural naming scheme for stationary states — the $(n,\ell,m)$ of the hydrogen capstone are exactly such a list.

- **The canonical structure rides along.** Unitary conjugation preserves commutators, so $[\hat{q}_H(t),\hat{p}_H(t)] = \mathrm{i}\hbar$ at every (equal) time — the quantum echo of the classical fact that time evolution preserves the fundamental brackets [time evolution is the flow generated by $H$, Mech. §8.3; that Hamiltonian flows preserve the fundamental brackets is part of the canonical-transformation theory the mechanics notes' outlook only points to — stated there, proved neither there nor here].

- **The algebra of §2.3 was built for this.** The derivative formulas (6) turn (64) into operator-valued classical mechanics, next.

### 7.2 Operator Newton: two flows run explicitly

*Free particle*, $\hat{H} = \hat{p}^2/2m$: the flow equations

$$\frac{\mathrm{d}\hat{p}_H}{\mathrm{d}t} = 0, \qquad \frac{\mathrm{d}\hat{q}_H}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\Big[\hat{q}_H,\frac{\hat{p}_H^2}{2m}\Big] = \frac{\hat{p}_H}{m} \tag{65}$$

(the second by (6)) integrate to

$$\hat{q}_H(t) = \hat{q}(0) + \frac{\hat{p}(0)}{m}\,t \tag{66}$$

— the classical trajectory, *as an operator identity*. The quantum content hides in what the classical formula cannot say:

$$\big[\hat{q}_H(t),\, \hat{q}_H(0)\big] = \frac{t}{m}\,[\hat{p}(0), \hat{q}(0)] = -\,\frac{\mathrm{i}\hbar\, t}{m}\ \neq\ 0\,: \tag{67}$$

*the position now and the position later are incompatible observables.* By Theorem 4.5, a sharp position at $t = 0$ forces growing position spread later — $\Delta q(t)\,\Delta q(0) \geq \hbar t/2m$ — which is wave-packet spreading, obtained with no wavefunction in sight; the wave-mechanical version, with the full time profile, is §8's opening act.

*General potential*, $\hat{H} = \hat{p}^2/2m + V(\hat{q})$: the same two lines give

$$\frac{\mathrm{d}\hat{q}_H}{\mathrm{d}t} = \frac{\hat{p}_H}{m}, \qquad \frac{\mathrm{d}\hat{p}_H}{\mathrm{d}t} = -\,V'(\hat{q}_H) \qquad \Longrightarrow \qquad m\,\frac{\mathrm{d}^2\hat{q}_H}{\mathrm{d}t^2} = -\,V'(\hat{q}_H) \tag{68}$$

— Newton's second law, wearing hats.

## 7.3 Ehrenfest and where the classical world sits

Sandwich (68) in the (frozen) state:

$$m\,\frac{\mathrm{d}^2}{\mathrm{d}t^2}\,\langle\hat{q}\rangle = -\,\big\langle V'(\hat{q})\big\rangle \qquad \text{(Ehrenfest)}, \tag{69}$$

and resist the tempting misreading: the right side is $-\langle V'(\hat{q})\rangle$, *not* $-V'(\langle\hat{q}\rangle)$, and the two differ whenever the packet is wide enough to sample the potential's curvature — Taylor-expand about the center,

$$\big\langle V'(\hat{q})\big\rangle = V'(\langle q\rangle) + \tfrac{1}{2}\,V'''(\langle q\rangle)\,(\Delta q)^2 + \cdots\,, \tag{70}$$

so the center of a quantum state obeys classical mechanics exactly when (i) the potential is at most quadratic — $V''' = 0$: the free particle, the uniform field, and, decisively for §9, *the harmonic oscillator*, whose packet centers are perfectly classical — or (ii) the spread is negligible on the scale over which the force varies: the $S \gg \hbar$ regime of §2, now with its precise small parameter, $(\Delta q)^2 V'''/V'$. Classical mechanics is the dynamics of centers, valid where centers are all one can resolve.

*Remark* 7.1 (Matrix mechanics recovered). In the energy eigenbasis, the Heisenberg operator's matrix elements evolve by (53) applied twice:

$$\big(\hat{A}_H(t)\big)_{mn} = A_{mn}\,\mathrm{e}^{\mathrm{i}\omega_{mn} t}, \qquad \omega_{mn} = \frac{E_m - E_n}{\hbar}\ : \tag{71}$$

every observable is a table of numbers oscillating at the Bohr frequencies (54) — which is literally what Heisenberg wrote down in 1925, abstracted from the frequencies atoms were observed to emit, before wavefunctions existed. Matrix mechanics *is* the Heisenberg picture in the energy basis; the document's structural route has arrived at the historical starting point, from the other side.

## 7.4 One physics, a choice of bookkeeping

The formal content of the section fits in a table — where the time dependence is attached, everything else being forced by (62):

| | Schrödinger picture | Heisenberg picture |
|---|---|---|
| states | $\lvert\psi(t)\rangle = U(t)\lvert\psi(0)\rangle$ | frozen |
| operators | frozen | $\hat{A}_H(t) = U^\dagger \hat{A}\, U$ |
| equation of motion | (51) on kets | (64) on operators |
| natural kin | wave mechanics (§5) | matrix mechanics (Remark 7.1) |

*Remark* 7.2 (One theory, many coordinates — the dynamical instance). Remark 2.3 promised this moment: Heisenberg's matrix mechanics (1925) and Schrödinger's wave mechanics (1926) were briefly rival theories, with partisans; they are the two columns of the table, related by the unitary bookkeeping shift (62) — one theory, coordinatized twice, the rivalry dissolved by a change of variables. The pattern is the series' oldest (Remark 3.5), and it has one more consequence worth planting: when quantum mechanics meets relativity, it is the *Heisenberg* picture that survives gracefully — operators carrying the spacetime label, states carrying none, is exactly the form the quantization program of [Fld. §14.2] leads to once run in full, and the outlook carries this out. (A third attachment — splitting $\hat{H}$ and letting each piece carry part of the time — is the *interaction picture*, the natural home of time-dependent perturbation theory; it belongs to the sequel course and is flagged, not built. [stated.])

# 8 Stationary States in One Dimension

The energy eigenbasis is where dynamics lives (§6.2); this section actually finds it, for the potentials a line can hold — and it opens with a conservation law the document has met before, in relativistic dress. Everything here runs in the position representation: the Schrödinger equation of §6.1,

$$\mathrm{i}\hbar\,\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\,\frac{\partial^2\psi}{\partial x^2} + V(x)\,\psi, \tag{72}$$

with stationary states $\psi(x,t) = \mathrm{e}^{-\mathrm{i}Et/\hbar}\,u(x)$ and the time-independent problem $-\frac{\hbar^2}{2m}u'' + Vu = Eu$.

## 8.1 The probability current

Track the probability density $\rho = |\psi|^2$ in time, using (72) and its conjugate — the potential terms cancel (real $V$), the kinetic terms assemble into a total derivative:

$$\frac{\partial\rho}{\partial t} = \frac{\mathrm{i}\hbar}{2m}\Big(\psi^*\,\partial_x^2\psi - \psi\,\partial_x^2\psi^*\Big) = -\,\partial_x \underbrace{\Big[\frac{\hbar}{m}\,\mathrm{Im}\big(\psi^*\partial_x\psi\big)\Big]}_{\equiv\; j(x,t)}, \tag{73}$$

$$\boxed{\frac{\partial\rho}{\partial t} + \frac{\partial j}{\partial x} = 0, \qquad j = \frac{\hbar}{m}\,\mathrm{Im}\big(\psi^*\partial_x\psi\big).} \tag{74}$$

A continuity equation — and the reader of the field-theory notes has seen this exact machine: (74) is [Fld. §9.2]'s local conservation law, and its pedigree is the same too. The Schrödinger equation inherits the global phase symmetry $\psi \to \mathrm{e}^{\mathrm{i}\alpha}\psi$ of the complex scalar, the conserved current of that symmetry is exactly $j$, and probability conservation is *Noether's theorem for the phase* — the nonrelativistic shadow of the $U(1)$ current of [Fld. §9.3], with one instructive difference: there, $j^0$ was not positive definite and had to be read as charge; here $\rho = |\psi|^2 \geq 0$, and the probability reading Born installed is consistent — the very consistency that failed relativistically and forced the reinterpretation. (Global conservation, $\int \rho\,\mathrm{d}x = 1$ forever, was already guaranteed by unitarity in §6.1; (74) is the *local* refinement — probability does not teleport, it flows.)

Two immediate readings calibrate $j$. A plane wave $\mathrm{e}^{\mathrm{i}(px-Et)/\hbar}$ carries $j = \rho\, p/m = \rho v$ — density times velocity, as a current should. And in a *stationary* state $\partial_t \rho = 0$, so $j$ is constant in $x$: for a bound state ($u$ real up to a global phase, via the nondegeneracy of one-dimensional bound states[1]) that constant is zero, while for scattering states it is the nonzero number out of which the transmission coefficients of §8.5 are built.

## 8.2 The spreading of the free packet

Section 7.2 proved positions at different times incompatible, (67); here is the full time profile, and the Heisenberg picture delivers it without a single integral. From the operator trajectory (66), the variance at time $t$ is

$$\big(\Delta q(t)\big)^2 = \Big\langle \big(\bar{q} + \tfrac{t}{m}\,\bar{p}\big)^2 \Big\rangle = (\Delta q_0)^2 + \frac{t^2}{m^2}(\Delta p_0)^2 + \frac{t}{m}\,\big\langle \{\bar{q}, \bar{p}\} \big\rangle, \tag{75}$$

all expectations in the frozen initial state. For the Gaussian packet (48) the cross term vanishes — elegantly: the minimum-uncertainty condition (34) makes $\langle \bar{q}\,\bar{p}\rangle = \mathrm{i}\hbar/2$ *purely imaginary*, so the anticommutator's (real) expectation is zero — and saturation fixes $\Delta p_0 = \hbar/2\Delta q_0$:

$$\boxed{\Delta q(t) = \Delta q_0 \sqrt{1 + \Big(\frac{\hbar\, t}{2m\,(\Delta q_0)^2}\Big)^2}} \tag{76}$$

— ballistic spreading at late times, at speed $\Delta p_0/m$: the packet's own momentum spread, in motion. The timescale $\tau = 2m(\Delta q_0)^2/\hbar$ is the $S/\hbar$ switch of §2 wearing dynamics: an electron localized to $1\,\text{Å}$ doubles its width in $\tau\sqrt{3} \approx 3 \times 10^{-16}\,\mathrm{s}$ — quantum delocalization is, for electrons, effectively instantaneous — while a microgram dust grain localized to a micron needs $\sim 10^6$ *years*: the same formula, thirty orders of magnitude apart, and the reason tables do not diffuse.

## 8.3 The infinite well: an old set of modes

Hard walls at $x = 0, L$ impose $u(0) = u(L) = 0$ (the boundary-condition discipline of Remark 3.3, here doing constructive work), and inside, $u'' = -k^2 u$ with $k = \sqrt{2mE}/\hbar$. Requiring the sine to fit both walls quantizes it:

$$\boxed{u_n(x) = \sqrt{\frac{2}{L}}\,\sin\frac{n\pi x}{L}, \qquad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots} \tag{77}$$

(Fig. 3, left). Three readings, in ascending order of reach:

- **The floor is the uncertainty floor.** $E_1 > 0$: a confined particle cannot sit still, and the scaling is Example 5.2's — $\Delta x \lesssim L$ forces $E \gtrsim \hbar^2/8mL^2$, and the exact $\pi^2\hbar^2/2mL^2$ is that estimate with its constant earned.

[1] Three lines, so [verified here]: two solutions $u_1, u_2$ of $-\frac{\hbar^2}{2m}u'' + Vu = Eu$ at the same $E$ have Wronskian $W = u_1 u_2' - u_2 u_1'$ with $W' = 0$, since both satisfy the same equation; a bound state and its derivative vanish at infinity, so $W \to 0$, hence $W \equiv 0$ — which integrates to $u_2 \propto u_1$.

- **Discreteness comes from confinement.** The continuum of free momenta became a ladder because the walls admit only fitting waves — the same mechanism, and in fact the same *modes*, as the pinned string: $\sin(n\pi x/L)$ is verbatim [Osc. §9.3]'s pinned-endpoint mode set. One eigenvalue problem, two theories sharing it: the string attaches a frequency $\omega_n \propto n$ (linear dispersion), the box an energy $E_n \propto n^2$ (quadratic) — the difference between the two columns of physics built on one column of mathematics.

- **Nodes count.** $u_n$ has $n-1$ interior nodes — the ground state none, each excitation adding one — an instance of the general one-dimensional oscillation theorem [stated; Sturm–Liouville theory], useful as a free consistency check on any computed spectrum.

## 8.4 The finite well: counting bound states

*Remark* 8.1 (Parity). For a symmetric potential, $V(-x) = V(x)$, define the *parity operator* $(\hat{P}\psi)(x) \equiv \psi(-x)$: Hermitian, $\hat{P}^2 = \mathbf{1}$ (so eigenvalues $\pm 1$), and $[\hat{H}, \hat{P}] = 0$ (the kinetic term is even in $\partial_x$, and $V$ is even by hypothesis). A *nondegenerate* bound state $u$ is therefore an eigenstate of $\hat{P}$ — two lines: $\hat{P}u$ solves the same eigenvalue problem at the same $E$, nondegeneracy forces $\hat{P}u = c\,u$, and $\hat{P}^2 = \mathbf{1}$ forces $c = \pm 1$. Every bound state of the well below is *even or odd*; this is why its catalogue splits into two families, and the same operator, promoted to $L^2(\mathbb{R}^3)$, returns as the $(-1)^\ell$ of the dipole selection rules (§16.5).

Soften the walls: $V = -V_0$ on $|x| < a/2$, zero outside. Bound states ($-V_0 < E < 0$) oscillate inside and decay outside, $u \sim \mathrm{e}^{-\kappa|x|}$ with $\kappa = \sqrt{2m|E|}/\hbar$, and matching $u$ and $u'$ at the edges turns the spectrum into a transcendental intersection problem (the even and odd families of Remark 8.1, $\kappa = k\tan(ka/2)$ and $\kappa = -k\cot(ka/2)$). Two facts survive the details and are worth owning:

- **A one-dimensional well always binds.** However shallow, the symmetric well holds at least one (even) bound state — the even matching curve always intersects once [stated; the graphical argument is two lines, and the fact fails in three dimensions, where shallow wells can be empty — a genuine dimension-dependence with physical consequences, e.g. for the deuteron].

- **The count is the well's "size in quanta."** The number of bound states is $N = 1 + \lfloor z_0/\pi \rfloor$ with $z_0 = \sqrt{2mV_0}\,a/\hbar$ [stated; read off the intersection graph]: the dimensionless depth $z_0$ — an action in units of $\hbar$, once again — measures how many fitting waves the well can hold.

## 8.5 Tunneling and the microscope built on it

Now the inverted problem: a barrier, $V = V_0 > 0$ on $[0, a]$, and a wave incident with $0 < E < V_0$ — classically, certain reflection. Quantum mechanically the interior solution does not vanish; it decays, $u \sim \mathrm{e}^{\pm\kappa x}$ with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$, and a decaying tail that reaches the far edge leaks. Matching $u, u'$ at both edges (the algebra is mechanical and not displayed) and defining the transmission coefficient as the ratio of §8.1's currents, $T = j_{\mathrm{trans}}/j_{\mathrm{inc}}$:

$$\boxed{T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E\,(V_0 - E)}\right]^{-1} \xrightarrow[\kappa a \gg 1]{} \frac{16\,E(V_0 - E)}{V_0^2}\,\mathrm{e}^{-2\kappa a}\ :} \tag{78}$$

through a thick barrier, transmission is *exponential* in the width, with decay rate $2\kappa$ set by the deficit $V_0 - E$ (Fig. 3, right — which also shows the above-barrier resonances, $T = 1$ whenever a whole number of half-wavelengths fits the barrier: the same fitting-waves mechanism as the box, working in transmission). Lowest checks: $a \to 0$ gives $T \to 1$ — no barrier, no reflection — and $E \to V_0$ gives the finite $T = [1 + z_0^2/4]^{-1}$ from both branches, the sinh and sin forms meeting continuously at the top.

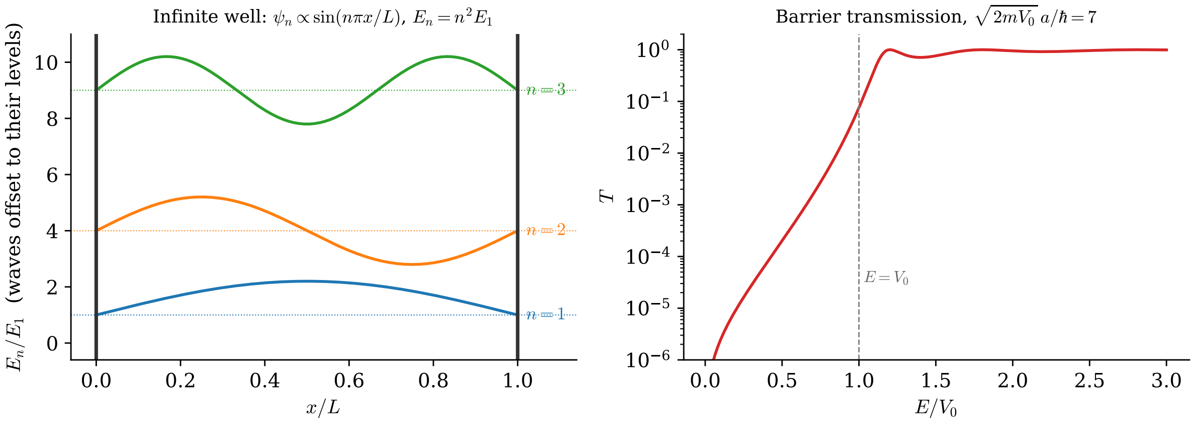


Figure 3: One dimension, both regimes, exact formulas. *Left*: the infinite well's first three stationary states (77), each drawn at its energy — the pinned string's mode shapes, now carrying quantum energies ($E_n \propto n^2$). *Right*: barrier transmission (78) (log scale) for $z_0 = \sqrt{2mV_0}\,a/\hbar = 7$: exponential tunneling below $V_0$, unit-transmission resonances above — fitting waves again.

**Example 8.2** (The scanning tunneling microscope)**.** The exponential in (78) is an instrument. Between an STM tip and a metal surface, the vacuum gap is a barrier of height set by the work function, $\phi \approx 4.5\,\mathrm{eV}$ for typical metals, so

$$\kappa = \frac{\sqrt{2m_e\phi}}{\hbar} \approx 1.09\ \text{Å}^{-1}, \qquad \frac{T(a)}{T(a+1\,\text{Å})} = \mathrm{e}^{2\kappa\cdot 1\,\text{Å}} \approx 9: \tag{79}$$

the tunneling current changes by nearly an *order of magnitude per ångström* of tip height. That extreme sensitivity is the whole instrument: a feedback loop holding the current constant traces the surface to sub-atomic precision, and every STM image of individual atoms is Eq. (78)'s exponential read out as topography.

*Remark* 8.3 (Slow potentials: WKB in brief). Between the exactly solvable and the numerical lies the *slowly-varying* potential, and the ansatz that owns it is worth one display even unproved:

$$u(x) \;\approx\; \frac{C}{\sqrt{p(x)}}\,\exp\Big[\pm\frac{\mathrm{i}}{\hbar}\int^x p(x')\,\mathrm{d}x'\Big], \qquad p(x) = \sqrt{2m\big(E - V(x)\big)} \tag{80}$$

— the phase is the *classical action* accumulating along the path, the amplitude $p^{-1/2}$ is flux conservation ((74): constant $j = \rho\,p/m$ forces $\rho \propto 1/p$), and the patching across turning points (where $p \to 0$ and the ansatz fails; the exact linear-potential solution, an Airy function, supplies the splice [stated; Hall Ch. 15]) yields the Bohr–Sommerfeld quantization $\oint p\,\mathrm{d}x = (n + \frac{1}{2})\,h$ of the old quantum theory [stated; Hall Ch. 15 gives the full treatment]. Why the action sits in the phase is not explicable at this level — but it is two sections ahead: §10 will show every amplitude is a sum of $\mathrm{e}^{\mathrm{i}S/\hbar}$ over paths, and (80) is what that sum looks like when one path dominates. The remark is a promise; §10 keeps it.

# 9 Quantizing the Harmonic Oscillator

The object that opened the series — one mass, one spring [Osc. §2] — makes its fifth appearance, and its most consequential: quantized here, it becomes in the outlook the atom of every quantum field. The solution method is pure algebra — no differential equation is solved until the wavefunctions are wanted — and the algebraic engine, a ladder built from the commutator, will be reused, unchanged, for angular momentum in §13.

## 9.1 Factorizing the Hamiltonian

The Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}\, m\omega^2 \hat{q}^2 \tag{81}$$

calls for factoring as a sum of squares, $u^2 + v^2 = (u - \mathrm{i}v)(u + \mathrm{i}v)$ — and the obstruction to doing so, $[\hat{q}, \hat{p}] \neq 0$, turns out to carry all of the problem's quantum content, as the derivation below makes explicit. Define the dimensionless combination (and its adjoint)

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\Big(\hat{q} + \frac{\mathrm{i}\,\hat{p}}{m\omega}\Big), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\Big(\hat{q} - \frac{\mathrm{i}\,\hat{p}}{m\omega}\Big), \tag{82}$$

and multiply out, letting the cross terms manufacture the commutator:

$$\hat{a}^\dagger \hat{a} = \frac{m\omega}{2\hbar}\Big(\hat{q}^2 + \frac{\hat{p}^2}{m^2\omega^2}\Big) + \frac{\mathrm{i}}{2\hbar}\,[\hat{q}, \hat{p}] = \frac{\hat{H}}{\hbar\omega} - \frac{1}{2}, \qquad [\hat{a}, \hat{a}^\dagger] = 1 \tag{83}$$

(the second relation by the same two lines with the sign flipped). Hence

$$\boxed{\hat{H} = \hbar\omega\left(\hat{N} + \tfrac{1}{2}\right), \qquad \hat{N} \equiv \hat{a}^\dagger \hat{a}, \qquad [\hat{a}, \hat{a}^\dagger] = 1.} \tag{84}$$

*Remark* 9.1 (The classical amplitude promoted to an operator). Equation (82) is not a clever guess; it is the series' oldest move. From the classical oscillator's amplitude–phase solution $q = A\cos(\omega t - \phi)$ [Osc. §2.3], form the combination $z \equiv q + \mathrm{i}p/m\omega = q + \mathrm{i}\dot{q}/\omega$: one line gives $z = A\,\mathrm{e}^{-\mathrm{i}(\omega t - \phi)}$ [verified here] — a complex amplitude that rotates rigidly, $z(t) = z(0)\,\mathrm{e}^{-\mathrm{i}\omega t}$. The operator $\hat{a}$ *is* the promotion of that combination — and the Heisenberg flow confirms it inherits the rigid rotation: $\mathrm{d}\hat{a}_H/\mathrm{d}t = [\hat{a}_H, \hat{H}]/\mathrm{i}\hbar = -\mathrm{i}\omega\,\hat{a}_H$, so $\hat{a}_H(t) = \hat{a}\,\mathrm{e}^{-\mathrm{i}\omega t}$, verbatim the classical solution with a hat.

## 9.2 The spectrum from the algebra alone

Everything now follows from (84) — three steps, no analysis.

*Step 1: the ladder.* From $[\hat{N}, \hat{a}] = [\hat{a}^\dagger \hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}]\,\hat{a} = -\,\hat{a}$ (Leibniz, (3)), and its adjoint $[\hat{N}, \hat{a}^\dagger] = +\,\hat{a}^\dagger$: if $\hat{N}|\nu\rangle = \nu|\nu\rangle$, then

$$\hat{N}\,\big(\hat{a}|\nu\rangle\big) = (\nu - 1)\,\big(\hat{a}|\nu\rangle\big), \qquad \hat{N}\,\big(\hat{a}^\dagger|\nu\rangle\big) = (\nu + 1)\,\big(\hat{a}^\dagger|\nu\rangle\big)\,: \tag{85}$$

$\hat{a}$ steps down the spectrum by one, $\hat{a}^\dagger$ up.

*Step 2: the floor.* Norms bound the ladder:

$$\big\|\hat{a}|\nu\rangle\big\|^2 = \langle\nu|\hat{a}^\dagger \hat{a}|\nu\rangle = \nu \;\geq\; 0. \tag{86}$$

Descending from any $\nu$ produces eigenvalues $\nu-1, \nu-2, \ldots$, which (86) forbids from passing below zero — so the descent must *terminate*: some rung is annihilated, $\hat{a}|\nu_{\min}\rangle = 0$, and then (86) says $\nu_{\min} = 0$. Termination is only possible if the starting $\nu$ was a whole number of steps above zero:

$$\nu = n \in \{0, 1, 2, \ldots\}. \tag{87}$$

*Step 3: the bookkeeping.* Fixing phases,

$$\boxed{E_n = \hbar\omega\Big(n + \tfrac{1}{2}\Big), \qquad \hat{a}|n\rangle = \sqrt{n}\,|n{-}1\rangle, \qquad \hat{a}^\dagger|n\rangle = \sqrt{n{+}1}\,|n{+}1\rangle, \qquad |n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle.} \tag{88}$$

The spectrum §2.1 demanded — energy in uniform lumps of $\hbar\omega$ — is now a theorem: the ladder has constant rungs because $[\hat{a}, \hat{a}^\dagger]$ is a *number*. [Completeness of the $|n\rangle$ — that the ladder built from $|0\rangle$ exhausts the Hilbert space — is true and quietly nontrivial; Hall §11.4. Stated.]

*Remark* 9.2 (The floor is the Gaussian again)*.* The termination condition $\hat{a}|0\rangle = 0$ is, written out via (82), $(\hat{p} - \mathrm{i}\, m\omega\, \hat{q})|0\rangle = 0$ — which is *verbatim* the minimum-uncertainty condition (34) with $\lambda = m\omega$ and centers at the origin. The oscillator's ground state *is* the Gaussian of §5.5, width $(\Delta q)^2 = \hbar/2m\omega$, sitting exactly on the floor $\Delta q\, \Delta p = \hbar/2$; and the zero-point energy is that floor expressed as an energy — minimizing $\langle H\rangle = (\Delta p)^2/2m + \frac{1}{2}m\omega^2(\Delta q)^2$ subject to the uncertainty bound gives exactly $\hbar\omega/2$, the algebra's $\frac{1}{2}$ re-derived as a variational fact. The particle cannot sit at the bottom of the well for the same reason the atom of Example 5.2 cannot collapse: stillness is not an available state.

**Example 9.3** (One molecule's quantum and frozen heat)**.** Carbon monoxide's vibrational frequency, spectroscopically $\tilde{\nu} = 2143\,\mathrm{cm}^{-1}$, puts its quantum at

$$\hbar\omega = hc\tilde{\nu} \approx 4.26 \times 10^{-20}\,\mathrm{J} \approx 0.266\,\mathrm{eV}, \qquad \left.\frac{\hbar\omega}{k_B T}\right|_{300\,\mathrm{K}} \approx 10.3: \tag{89}$$

ten times the thermal energy, so the Boltzmann weight of even the first excited rung is $\mathrm{e}^{-10.3} \approx 3 \times 10^{-5}$ — at room temperature the vibration is *frozen out*, the molecule locked to $n = 0$, and the vibrational degree of freedom silently withdraws from the specific heat. Classical equipartition's room-temperature failure — half of §2.1's second bullet — is this number; the first rung sits an order of magnitude above the thermal energy.

## 9.3 Wavefunctions from the ladder

When shapes are wanted, the algebra dictates them. The ground state solves $\hat{a}\, u_0 = 0$ in the position representation — a first-order equation, $(m\omega x + \hbar\, \partial_x)\, u_0 = 0$, whose solution is the Gaussian Remark 9.2 promised — and every excited state is $(\hat{a}^\dagger)^n$ applied to it: each application of $\hat{a}^\dagger \propto (m\omega x - \hbar\partial_x)$ multiplies by a polynomial of one higher degree, generating the Hermite family,

$$u_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}}\, H_n(\xi)\, \mathrm{e}^{-\xi^2/2}, \qquad \xi = x\sqrt{\frac{m\omega}{\hbar}} \tag{90}$$

[normalizations stated; the generation from $\hat{a}^\dagger$ is the proof strategy, and it also delivers $n$ nodes for $u_n$ — the oscillation theorem of §8.3 passing its check]. Figure 4 shows the family at its rungs, tails penetrating the classically forbidden region — tunneling's kinship (§8.5) — and, at $n = 20$, the correspondence principle in the flesh: the quantum density's ripples average to the classical dwell-time distribution $1/\pi\sqrt{A^2 - x^2}$, largest at the turning points where the classical particle lingers.

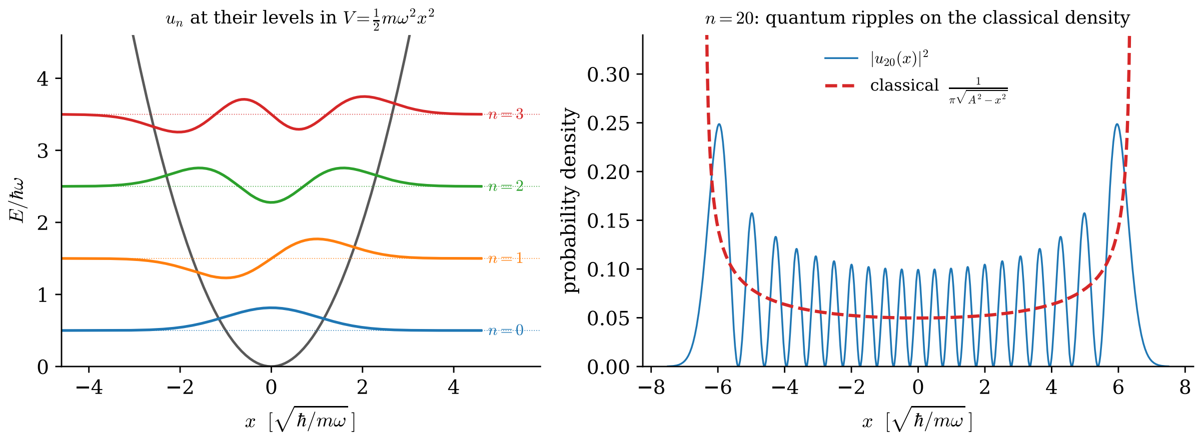


Figure 4: The quantized oscillator, exact formulas ($\hbar = m = \omega = 1$). *Left*: eigenfunctions (90), $n = 0$–$3$, at their levels $E_n = n + \frac{1}{2}$ in the potential — equal rungs, $n$ nodes, Gaussian floor, forbidden-region tails. *Right*: at $n = 20$ the quantum density ripples about the classical dwell-time distribution (dashed), which diverges at the turning points $\pm A$, $A = \sqrt{2E_n}$ — the correspondence principle, drawn.

### 9.4 Coherent states: where the classical oscillator lives

Ehrenfest (§7.3) promised that quadratic potentials evolve packet centers exactly classically; here are the packets that take fullest advantage. Define *coherent states* as eigenstates of the (non-Hermitian — eigenkets are still legal) lowering operator,

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle, \qquad \alpha \in \mathbb{C}, \qquad |\alpha\rangle = \mathrm{e}^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}}\,|n\rangle \tag{91}$$

(the expansion follows by feeding $\sum c_n|n\rangle$ to the eigenvalue equation, whence $c_n = \alpha\, c_{n-1}/\sqrt{n}$ by (88)). Three properties make them the classical limit's residents:

- **They stay coherent, and $\alpha$ rotates classically.** Apply (53) to (91): the phases $\mathrm{e}^{-\mathrm{i}E_n t/\hbar}$ reassemble the same sum with $\alpha \to \alpha\,\mathrm{e}^{-\mathrm{i}\omega t}$ (times a global phase) — the label traces exactly the classical complex amplitude of Remark 9.1, so $\langle q\rangle(t)$ and $\langle p\rangle(t)$ run the classical trajectory *exactly*, at any amplitude.

- **They ride the floor.** A coherent state is the ground -state Gaussian rigidly displaced (in $q$ and $p$ by Re, Im of $\alpha$), so it saturates $\Delta q\,\Delta p = \hbar/2$ at every instant [one-line check: $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ *is* the minimum-uncertainty condition with shifted centers] — a wave packet that oscillates forever *without spreading*, the loophole to §8.2 that only the quadratic potential's rigidity permits.

- **They are what "classical amplitude $A$" means.** The excitation statistics are Poisson, $P(n) = \mathrm{e}^{-|\alpha|^2}|\alpha|^{2n}/n!$, with mean $\langle N\rangle = |\alpha|^2$: a macroscopic oscillation is a coherent state with $|\alpha|^2 \sim S_{\mathrm{cl}}/\hbar \gg 1$ quanta — the $S/\hbar$ switch of §2 counting rungs — and laser light is, to excellent approximation, this state of the electromagnetic field's modes. [stated; quantum optics' founding fact.]

One bookkeeping note closes the section, aimed at the outlook: the pair $(\hat{a}, \hat{a}^\dagger)$ with $[\hat{a}, \hat{a}^\dagger] = 1$ is the entire quantum content of *one* oscillator — and [Fld. §14.1] showed a free field is a countable family of them, one per wavevector. Quantizing the field will mean writing one ladder per mode, $[\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = \delta_{\vec{k}\vec{k}'}$, and reading $\hat{a}^\dagger_{\vec{k}}|0\rangle$ as *a particle*. Every technical prerequisite for that sentence is now on the table.

# 10 The Propagator by Two Routes

The field-theory notes could only *mark* the fork between the two quantizations — canonical, with its time slicing, and the path integral, built on the scalar action [Fld. §10.3]. This section walks both roads in full, at the one level where both are completely tractable, and verifies they arrive at the same place. Along the way it keeps the promise of Remark 8.3, answers the series' oldest open question — *why* nature extremizes actions — and returns the central object of the mechanics notes, the action, as a phase.

## 10.1 The object: the propagator and its composition law

Define the *propagator* as the open sandwich of §3.4(d) with the evolution inside:

$$\boxed{K(x_b, T;\, x_a, 0) \;\equiv\; \langle x_b|\, U(T)\, |x_a\rangle} \tag{92}$$

— the amplitude to be found at $x_b$ after time $T$, having started at $x_a$. It carries the whole of dynamics: inserting $\mathbf{1} = \int \mathrm{d}x_a |x_a\rangle\langle x_a|$ into $|\psi(T)\rangle = U(T)|\psi(0)\rangle$,

$$\psi(x_b, T) = \int \mathrm{d}x_a\; K(x_b, T; x_a, 0)\; \psi(x_a, 0), \tag{93}$$

so solving for $K$ solves every initial-value problem at once. And it obeys one structural identity, obtained by splitting $U(T) = U(T - t_c)\, U(t_c)$ and inserting a position resolution between the factors:

$$K(x_b, T; x_a, 0) = \int \mathrm{d}x_c\; K(x_b, T; x_c, t_c)\; K(x_c, t_c; x_a, 0)\; : \tag{94}$$

*amplitudes for successive legs multiply; amplitudes for alternative waypoints add.* Equation (94) looks like bookkeeping. Iterated, it is a theory (§10.3).

## 10.2 Route 1: the spectral representation

Canonically, $K$ is computed by diagonalizing: insert the energy eigenbasis into (92),

$$K(x_b, T; x_a, 0) = \sum_n u_n(x_b)\, u_n^*(x_a)\; \mathrm{e}^{-\mathrm{i}E_n T/\hbar} \tag{95}$$

— the whole kernel, from the spectrum. For the free particle the sum is an integral over momentum eigenstates (44), and it is a Gaussian (Appendix A, in its Fresnel form (179)):

$$\begin{aligned} K_{\text{free}} &= \int \frac{\mathrm{d}p}{2\pi\hbar}\; \exp\Big[\frac{\mathrm{i}}{\hbar}\Big(p\,(x_b - x_a) - \frac{p^2 T}{2m}\Big)\Big] \\ &= \sqrt{\frac{m}{2\pi\mathrm{i}\hbar T}}\; \exp\Big[\frac{\mathrm{i}}{\hbar}\,\frac{m\,(x_b - x_a)^2}{2T}\Big] \end{aligned} \tag{96}$$

(complete the square in $p$; the shifted Gaussian is (177)–(179)). Now look hard at the exponent. The quantity $m(x_b - x_a)^2/2T$ is $\frac{1}{2}mv^2\,T$ for the uniform motion connecting the endpoints — it is *the classical action of the classical path*,

$$K_{\text{free}} = \sqrt{\frac{m}{2\pi \mathrm{i}\hbar T}}\; \mathrm{e}^{\,\mathrm{i}S_{\text{cl}}/\hbar}. \tag{97}$$

Nothing in Route 1 asked for the action; no Lagrangian was mentioned; and yet the action of [Mech.] stands in the exponent. Route 1 cannot explain this. Route 2 is the explanation.

### 10.3 Route 2: time slicing and the sum over paths

Iterate the composition law (94) $N-1$ times with uniform slices $\varepsilon = T/N$:

$$K(x_b, T; x_a, 0) = \int \mathrm{d}x_1 \cdots \mathrm{d}x_{N-1} \prod_{j=0}^{N-1} \langle x_{j+1}|\, U(\varepsilon)\, |x_j\rangle, \qquad x_0 = x_a,\ x_N = x_b. \tag{98}$$

Only the *short-time* amplitude is needed, and for $\hat{H} = \hat{p}^2/2m + V(\hat{q})$ the two terms can be exponentiated separately at cost $O(\varepsilon^2)$ per slice [stated; the Trotter product formula is the precise justification]. The kinetic factor is computed by inserting momentum states — the identical Gaussian as (96) with $T \to \varepsilon$ — and the potential factor is diagonal:

$$\langle x_{j+1}|U(\varepsilon)|x_j\rangle \approx \sqrt{\frac{m}{2\pi \mathrm{i}\hbar\varepsilon}}\; \exp\!\left[\frac{\mathrm{i}\varepsilon}{\hbar}\left(\frac{m}{2}\Big(\frac{x_{j+1}-x_j}{\varepsilon}\Big)^2 - V(x_j)\right)\right]. \tag{99}$$

Multiply the slices: the exponents *add*, and what they add up to is a Riemann sum —

$$\sum_j \varepsilon\left[\frac{m}{2}\Big(\frac{\Delta x_j}{\varepsilon}\Big)^2 - V(x_j)\right] \;\xrightarrow[N\to\infty]{}\; \int_0^T \mathrm{d}t\,\left[\frac{m}{2}\,\dot{x}^2 - V(x)\right] \;=\; S[x], \tag{100}$$

the *action* of [Mech. §3], kinetic minus potential, evaluated along the broken-line path through the waypoints. In the limit:

$$\boxed{K(x_b, T; x_a, 0) \;=\; \int \mathcal{D}x(t)\;\; \mathrm{e}^{\,\mathrm{i}S[x]/\hbar}} \tag{101}$$

— the *path integral*, with $\mathcal{D}x$ shorthand for the sliced measure and its prefactors [the continuum measure's existence is a genuine mathematical subtlety; stated, with the sliced definition as the operational meaning]. The reading (Fig. 5): *every* path from $(x_a, 0)$ to $(x_b, T)$ contributes — jagged, backtracking, absurd ones included — all with the *same magnitude*, distinguished only by a phase, and the phase is the classical action in units of $\hbar$. Democracy of paths; the classical trajectory holds no privilege in the integrand.

### 10.4 Stationary phase: the origin of Hamilton's principle

Where does the classical world sit inside (101)? In the interference. When $S \gg \hbar$ — the switch of §2, one last time — the phase $S[x]/\hbar$ spins wildly under any deformation of the path, and contributions cancel in pairs *except* where the phase is stationary:

$$\delta S[x] = 0 \qquad \text{— Hamilton's principle.} \tag{102}$$

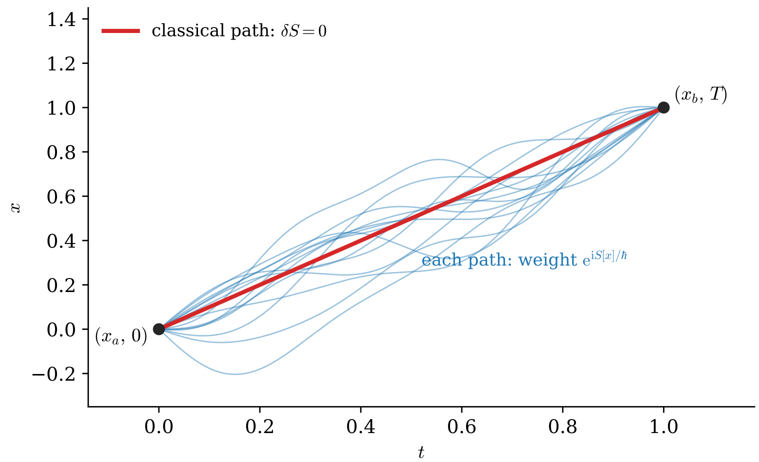


Figure 5: The sum over paths (101): every history from $(x_a, 0)$ to $(x_b, T)$ carries the same weight in magnitude, phase $S[x]/\hbar$. The classical path (bold) is distinguished not by the integrand but by *interference*: it is where the phase is stationary, so neighboring paths add coherently instead of canceling.

The variational principle that the mechanics notes installed as their axiom [Mech. §3] is hereby demoted and thereby explained: *it is the stationary-phase approximation to the sum over histories* — nature does not prefer the extremal path; it takes all of them, and only near the extremum do the amplitudes survive their own interference. This answers the question the mechanics notes could only postpone — why extremizing $S$ was ever the right principle — and it keeps Remark 8.3's promise: the WKB wavefunction (80), action in the phase and flux in the amplitude, is exactly what (101) looks like when one stationary path dominates and Gaussian fluctuations around it supply the prefactor.

For *quadratic* Lagrangians the story sharpens from approximation to identity: expanding $x(t) = x_{\mathrm{cl}}(t) + \eta(t)$, the action splits exactly — $S[x] = S_{\mathrm{cl}} + S_2[\eta]$ with no higher terms, the linear term absent by $\delta S = 0$ — and the fluctuation integral factors off as an endpoint-independent function of time:

$$K = F(T)\, e^{iS_{\mathrm{cl}}/\hbar} \qquad \text{(exactly, for quadratic } L\text{).} \tag{103}$$

That is why (97)'s exponent was *exactly* the classical action: the free particle is quadratic. So is the oscillator — the payoff:

**Route 2 applied to the oscillator, with Route 1 as the check.** The classical action of the harmonic path between fixed endpoints is a classical-mechanics exercise in [Mech.]'s toolkit:

$$S_{\mathrm{cl}} = \frac{m\omega}{2\sin\omega T}\Big[(x_a^2 + x_b^2)\cos\omega T - 2\,x_a x_b\Big], \tag{104}$$

and the fluctuation factor [computable as a product over the modes of $S_2$; stated — or fixed, more cheaply, by matching the $\omega \to 0$ free limit (96)] completes the kernel:

$$K_{\mathrm{SHO}} = \sqrt{\frac{m\omega}{2\pi i\hbar\,\sin\omega T}}\;\exp\Big[\frac{i}{\hbar}\,S_{\mathrm{cl}}\Big]. \tag{105}$$

Route 1 now checks Route 2. Set $x_a = x_b = x$ and integrate — the *trace* of the propagator, which by the spectral representation (95) must equal $\sum_n e^{-iE_nT/\hbar}$. The integral is one more Gaussian (exponent $\mathrm{i}m\omega\, x^2(\cos\omega T - 1)/\hbar\sin\omega T$, Appendix A), and with $1-\cos\omega T = 2\sin^2(\omega T/2)$:

$$\int \mathrm{d}x\; K_{\mathrm{SHO}}(x,T;x,0) = \frac{1}{2\mathrm{i}\,\sin(\omega T/2)} = \frac{\mathrm{e}^{-\mathrm{i}\omega T/2}}{1-\mathrm{e}^{-\mathrm{i}\omega T}} = \sum_{n=0}^{\infty} \mathrm{e}^{-\mathrm{i}\omega\left(n+\frac{1}{2}\right)T} \tag{106}$$

[the geometric series converges under the standard $T \to T(1-\mathrm{i}\epsilon)$ prescription; stated] — a geometric series whose exponents read off $E_n = \hbar\omega(n+\frac{1}{2})$: *the path integral knows the ladder's spectrum* (88), rung for rung, zero-point included. Two routes, one kernel, one spectrum; the fork's two roads, walked to the same destination and checked against each other.

*Remark* 10.1 (The fork revisited)*.* [Fld. §10.3] could only gesture at what has just been done in full. The canonical route (Routes 1 here, Heisenberg/Schrödinger machinery throughout) slices time: unitarity is manifest, but a frame has been chosen — the same trade-off, at the same Legendre step, as in the mechanics and field notes. The path-integral route weights histories by the *scalar* $S[x]$: no slicing in the weight, no picture, no operator ordering — but unitarity is now a theorem to prove rather than a structure to see. At this document's level the choice is convenience; one level up, where $S$ is the Lorentz scalar of [Fld. §8.1], the path integral is what keeps covariance visible while the canonical formalism keeps probability visible — and quantum field theory, needing both, keeps both formalisms and translates freely. The translation's proof of concept is this section: (95) and (101) are the same object, as (106) verifies.

Part II is complete: evolution, pictures, spectra, and the two grand formalisms reconciled on the series' own mascot. Part III turns from time to *space* — rotations, and everything the word "angular momentum" is about to mean.

# Part III
# Symmetry

## 11 Rotations and Angular Momentum

Part III is the symmetry part, and it opens by finishing a table. The quantum Noether dictionary (Remark 5.1) has run twice — translations yielded $\hat{p}$, waiting yielded $\hat{H}$ — and its third row, rotations, is richer than both: three generators instead of one, and, for the first time, generators that *do not commute with each other.* That non-commutativity is inherited straight from the geometry of turning, it reproduces the classical bracket algebra of the mechanics notes on cue, and it is loose enough — no $\mathbf{1}$ on its right-hand side — to admit the finite-dimensional representations that Theorem 2.2 exempted. The standing laboratory is about to receive its derivation.

### 11.1 Symmetries act unitarily

What should "the system, rotated" mean on Hilbert space? A map $|\alpha\rangle \to |\alpha'\rangle$ that preserves all physical predictions — all transition probabilities, $|\langle\beta|\alpha\rangle|^2$. A foundational theorem pins down what such maps can be:

**Theorem 11.1** (Wigner)**.** *Any map of rays preserving all transition probabilities is implemented by an operator that is either unitary or antiunitary. [stated; the antiunitary option is realized in nature by time reversal, and is deferred with it.]*

Continuous symmetries connect to the identity, which is unitary, so each rotation $R$ acts by a unitary $U(R)$, and composition respects the group — up to a phase, since states are rays (§3.1): $U(R_2)U(R_1) = \mathrm{e}^{\mathrm{i}\gamma}\, U(R_2R_1)$. That innocent-looking phase has consequences: for most groups it can be removed by redefinitions; for the rotation group the irreducible ambiguity is exactly a *sign* — the sign Remark 6.2 watched a precessing spin acquire — and §12 is where it surfaces. [stated at this level; the classification of the phases is the theory of projective representations.]

For an infinitesimal rotation by $\mathrm{d}\varphi$ about the axis $\hat{n}$, the third run of the generator move (§5.2, §6.1):

$$U(\hat{n}, \mathrm{d}\varphi) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\,\mathrm{d}\varphi\ \hat{n}\cdot\hat{\vec{J}} + O(\mathrm{d}\varphi^2), \tag{107}$$

defining three Hermitian generators $\hat{J}_x, \hat{J}_y, \hat{J}_z$ — the *angular momentum* operators, $[\hat{J}] = $ [action] this time, no $\hbar$-repair needed — and completing the dictionary's table. What remains is their algebra.

### 11.2 The algebra by both roads

The document's standing pattern — one commutator, two independent derivations — applies verbatim.

*Road A: deform the classical bracket.* The mechanics notes recorded the closed bracket of the classical angular momenta, $\{L_x, L_y\} = L_z$ [Mech. §8.2]; the full family $\{L_i, L_j\} = \epsilon_{ijk}L_k$ is the same two-line computation from the fundamental brackets, run for each pair [verify once]. Dirac's rule (4) transcribes in one line:

$$\boxed{[\hat{J}_i, \hat{J}_j] = \mathrm{i}\hbar\,\epsilon_{ijk}\,\hat{J}_k.} \tag{108}$$

*Road B: read the geometry of turning.* Rotations about different axes do not commute, and the failure is quantitative. For the $3 \times 3$ rotation matrices, a direct expansion to second order gives the classic identity

$$R_x(\varepsilon)\, R_y(\varepsilon)\, R_x(-\varepsilon)\, R_y(-\varepsilon) = R_z(\varepsilon^2) + O(\varepsilon^3) \tag{109}$$

— go around the little $x$–$y$ commutator loop and you have turned, slightly, about $z$ (verify once with the matrices: the $O(\varepsilon)$ terms cancel by construction, and the surviving $\varepsilon^2$ entries sit exactly in the $xy$ block). Demand that the unitaries represent this, insert (107) on both sides, and match at order $\varepsilon^2$: the left side's surviving second-order term is $-[\hat{J}_x, \hat{J}_y]\,\varepsilon^2/\hbar^2$ (the $\hat{J}_x^2$ and $\hat{J}_y^2$ terms cancel pairwise between the $+\varepsilon$ and $-\varepsilon$ factors), the right side's is $-\mathrm{i}\hat{J}_z\,\varepsilon^2/\hbar$, whence $[\hat{J}_x, \hat{J}_y] = \mathrm{i}\hbar\,\hat{J}_z$ — (108) again, cyclically. The two roads' agreement is Remark 5.1's convergence, third instance: the bracket algebra of mechanics *was* the geometry of the symmetry group all along, and quantization preserves both because they are one.

*Remark* 11.2 (Why the laboratory was allowed). Look at the right-hand side of (108): operators, not $\mathrm{i}\hbar\,\mathbf{1}$. The trace obstruction of Theorem 2.2 evaporates — $\mathrm{tr}\ \epsilon_{ijk}\hat{J}_k = 0$ is perfectly satisfiable by traceless matrices — and finite-dimensional representations are *allowed*: the algebra itself does not fix the size of its stage, and §13 will find one stage per half-integer. The two-dimensional laboratory of §3.5 has operated on exactly this exemption since §2.3 first noted it; the exemption is now explained.

## 11.3 Orbital angular momentum: the classical entry

The oldest realization of (108) is the quantized classical angular momentum,

$$\hat{L}_i = \epsilon_{ijk}\, \hat{q}_j\, \hat{p}_k \tag{110}$$

(no ordering ambiguity: $\epsilon_{ijk}$ only pairs $j \neq k$, and $[\hat{q}_j, \hat{p}_k] = 0$ for $j \neq k$ — Dirac's rule applies without ambiguity). Verify the family with a single commutator:

$$\begin{aligned}[\hat{L}_x, \hat{L}_y] &= [\hat{q}_y\hat{p}_z - \hat{q}_z\hat{p}_y,\ \hat{q}_z\hat{p}_x - \hat{q}_x\hat{p}_z] \\ &= \hat{q}_y\hat{p}_x\,[\hat{p}_z, \hat{q}_z] + \hat{q}_x\hat{p}_y\,[\hat{q}_z, \hat{p}_z] = \mathrm{i}\hbar\left(\hat{q}_x\hat{p}_y - \hat{q}_y\hat{p}_x\right) = \mathrm{i}\hbar\,\hat{L}_z\end{aligned} \tag{111}$$

(only the terms sharing a $z$-pair survive; Leibniz (3) does the bookkeeping). And $\hat{L}$ is a generator in the concrete sense: in polar coordinates on the plane, the same two-line computation as §5.3 — drag the wavefunction, $\psi(\varphi - \mathrm{d}\varphi)$, match against the generator — gives

$$\langle \vec{x}|\,\hat{L}_z\,|\psi\rangle = -\,\mathrm{i}\hbar\,\frac{\partial}{\partial\varphi}\,\psi(\vec{x}) : \tag{112}$$

$\hat{L}_z$ is to the angle what $\hat{p}$ was to position — with the one crucial difference that the angle is *periodic*, which is about to quantize $\hat{L}_z$'s spectrum (§13) by the same fitting-waves mechanism as §8.3, and which is the constructive face of the bounded-coordinate warning in Remark 4.8.

## 11.4 Spin: internal rotation and the sign

Now the laboratory itself. Spin is angular momentum with no $\hat{q} \times \hat{p}$ inside — generators acting on an *internal* index of the state, touching the spatial argument not at all: the precise analogue of the internal–spacetime split that [Fld. §9.3] drew for field symmetries. For spin-$\frac{1}{2}$, $\hat{S}_i = \frac{\hbar}{2}\sigma_i$, and the Pauli identity (18)'s antisymmetric part *is* (108) — verified back in §3.5, now recognized as the rotation algebra in its smallest allowed representation (Remark 11.2).

The finite rotations come cheap, and they carry the section's punchline. Exponentiate the generator using the Pauli identity's symmetric part, $(\hat{n}\cdot\vec{\sigma})^2 = \mathbf{1}$: the power series splits into even and odd,

$$\boxed{U(\hat{n},\varphi) = \exp\Big[-\frac{\mathrm{i}\varphi}{2}\,\hat{n}\cdot\vec{\sigma}\Big] = \cos\frac{\varphi}{2}\,\mathbf{1} \;-\; \mathrm{i}\,\sin\frac{\varphi}{2}\,\hat{n}\cdot\vec{\sigma},} \tag{113}$$

— and set $\varphi = 2\pi$:

$$U(\hat{n},2\pi) = -\,\mathbf{1}. \tag{114}$$

A full turn is *minus* the identity — for any axis, any spin-$\frac{1}{2}$ state. Remark 6.2 watched one Hamiltonian produce this sign dynamically; (114) says the Hamiltonian was innocent: the sign belongs to the *representation*, hard-wired into (113)'s half-angles, which trace back through the exponential to the $\frac{\hbar}{2}$ in the generators — to what spin-$\frac{1}{2}$ *is*. It is also exactly the projective phase Theorem 11.1's fine print permitted: $U(2\pi)$ represents the same rotation as $U(0)$, differing by a sign that rays cannot see but interferometers can. What kind of mathematical object carries such a representation — and what experiment measures the sign — is the next section entire.

# 12 $SU(2)$, $SO(3)$, and the Double Cover

Section 11.4 left a sign in the room: $U(2\pi) = -\mathbf{1}$, hard-wired into the half-angles of (113). This section identifies the sign's origin — the states of a spin-$\frac{1}{2}$ do not carry the rotation group at all, but its *double cover* — explains why no redefinition can remove it, and then does what a physics document must: reports the experiment in which the sign was measured, interference fringes with a period of 720°.

## 12.1 Two groups for one geometry

The rotation group $SO(3)$ is an old acquaintance: the $\mathsf{G} = \mathbf{1}$, det $= +1$ member of the one-template family $\mathsf{M}^{\mathsf{T}}\mathsf{G}\,\mathsf{M} = \mathsf{G}$ [Fld. §3.1] — this subsection is that remark's quantum sequel. The new group is the one §11.4 was unknowingly computing in: $SU(2)$, the $2\times 2$ unitaries of unit determinant. Its general element is exactly (113),

$$U = a\,\mathbf{1} + \mathrm{i}\,\vec{b}\cdot\vec{\sigma}, \qquad a,\vec{b}\in\mathbb{R}, \qquad a^2 + |\vec{b}|^2 = 1 \tag{115}$$

(unitarity plus $\det U = 1$ force real $(a,\vec{b})$ on the unit sphere) — so as a space, $SU(2)$ *is the three-sphere* $S^3$. The bridge to rotations is conjugation. Package a vector as a matrix, $X = \vec{x}\cdot\vec{\sigma}$, and let $U$ act by

$$X \;\longrightarrow\; U\,X\,U^\dagger \;\equiv\; \vec{x}'\cdot\vec{\sigma}. \tag{116}$$

The map $\vec{x}\to\vec{x}'$ is linear, and it is a *rotation* by a two-line argument: $\det X = -|\vec{x}|^2$ (write it out from (17)), conjugation preserves determinants, so lengths are preserved; and $SU(2)$ being connected, the map connects to the identity — orientation preserved, $R(U)\in SO(3)$. Composition is automatic, $R(U_2U_1) = R(U_2)R(U_1)$. But the correspondence is not one-to-one:

$$\boxed{R(-U) = R(U)} \qquad \text{— two elements of } SU(2) \text{ per rotation,} \tag{117}$$

since the sign cancels between $U$ and $U^\dagger$ in (116) (and $\pm U$ are the *only* such pairs: the kernel is the center $\{\pm\mathbf{1}\}$). $SU(2)$ is a *double cover* of $SO(3)$: the sphere $S^3$, wrapped twice around the space of rotations, antipodal points $U, -U$ landing on the same $R$. The spin-$\frac{1}{2}$ sign is now identified rather than mysterious: the state transforms under $SU(2)$; "rotate by $2\pi$" is a path in $SO(3)$ that returns to the identity, but its *lift* to $SU(2)$ walks from $\mathbf{1}$ to the other preimage, $-\mathbf{1}$ — which re-derives Eq. (114) as a statement of geography.

*Remark* 12.1 (Why the sign cannot be redefined away). Could a cleverer assignment of phases make $U(2\pi) = +\mathbf{1}$? No, and the obstruction is topological. In $SO(3)$ the $2\pi$-rotation loop *cannot be continuously contracted to a point* — the group is not simply connected, $\pi_1(SO(3)) = \mathbb{Z}_2$ [stated; the "belt trick" is the tabletop proof, and the $4\pi$ loop *does* contract] — so a representation is free to assign the loop a sign, and no continuous redefinition can change a discrete assignment. $SU(2) = S^3$ *is* simply connected: it is the universal cover, its true (non-projective) representations are exactly the projective representations Theorem 11.1's fine print allowed for $SO(3)$, and the classification splits cleanly in two — representations blind to the sign (the true $SO(3)$ representations: integer spin) and representations that carry it (half-integer spin), a dichotomy the algebra of §13 is about to reproduce from the other end. [The general theory — covering groups, projective representations — is stated at exactly this depth and no further.]

## 12.2 Measuring the sign: neutrons and 720°

An overall sign on an isolated state is invisible (§3.1); the measurement therefore requires making the sign *relative*, as Remark 6.2's closing sentence anticipated. Split a beam in two, rotate the spin on *one* arm, recombine:

$$I \;\propto\; \big\| |\psi\rangle + U(\hat{n},\varphi)|\psi\rangle \big\|^2 = 2 + 2\,\mathrm{Re}\langle\psi|\,U(\hat{n},\varphi)\,|\psi\rangle \;\overset{(113)}{=}\; 2\Big(1+\cos\frac{\varphi}{2}\Big) \tag{118}$$

(for $|\psi\rangle$ an eigenstate of $\hat{n}\cdot\vec{\sigma}$; the general case only adds contrast factors). The fringe pattern is periodic in the rotation angle with period

$$\varphi_{\text{period}} = 4\pi = 720°, \tag{119}$$

and at $\varphi = 2\pi$ — one full rotation — the two arms interfere *destructively*: the state rotated by a full turn is minus itself, measurably.

**Example 12.2** (The experiment and its numbers)**.** The experiment was run in 1975 (Rauch and collaborators; Werner and collaborators, independently) with thermal neutrons in a single-crystal silicon interferometer — the beam split and recombined by Bragg diffraction, one arm threading a magnet. The rotation is Larmor precession (§6.3): a neutron of thermal speed $v \approx 2200\,\mathrm{m/s}$ crossing a field region of length $\ell \approx 2\,\mathrm{cm}$ precesses by $\varphi = \gamma_n B\,\ell/v$, and with $|\gamma_n| = 1.83\times10^8\,\mathrm{rad/(s\,T)}$ [experimental input] the field needed for one *fringe* period ($\varphi = 4\pi$) is

$$B_{4\pi} = \frac{4\pi\,v}{|\gamma_n|\,\ell} \approx 7.5\,\mathrm{mT} \tag{120}$$

— laboratory-bench scale, and sweepable. Sweeping it, the measured intensity oscillated with a period of $\approx 716° \pm 4°$ [experimental input; consistent with 720°, the small offset understood from field inhomogeneity], and emphatically not 360°: the neutron count itself, plotted against the precession angle, is a graph of Eq. (118) — a direct measurement of the minus sign in (114). A rotation by $2\pi$ is not nothing; it is $-1$, and a silicon crystal can tell.

## 12.3 Finite rotations tabulated: the $\mathscr{D}$-matrices

For bookkeeping that §16.3 will reference — the finite form of the tensor-operator definition — name the matrix elements of a finite rotation in a given representation. Parametrize rotations by Euler angles, $U(\alpha,\beta,\gamma) = \mathrm{e}^{-\mathrm{i}\alpha J_z/\hbar}\,\mathrm{e}^{-\mathrm{i}\beta J_y/\hbar}\,\mathrm{e}^{-\mathrm{i}\gamma J_z/\hbar}$, and define

$$\mathscr{D}^{(j)}_{m'm}(\alpha,\beta,\gamma) \;\equiv\; \langle j,m'|\,U(\alpha,\beta,\gamma)\,|j,m\rangle \tag{121}$$

— the *Wigner $\mathscr{D}$-functions*, one unitary matrix per rotation per representation (the labels $(j,m)$ are §13's to construct; the definition is parked here with the group, where it belongs). For the one representation already in hand, everything is explicit: with $J_z = \frac{\hbar}{2}\sigma_z$, $J_y = \frac{\hbar}{2}\sigma_y$ and the half-angle formula (113),

$$\mathscr{D}^{(1/2)}(\alpha,\beta,\gamma) = \begin{pmatrix} \mathrm{e}^{-\mathrm{i}(\alpha+\gamma)/2}\cos\frac{\beta}{2} & -\,\mathrm{e}^{-\mathrm{i}(\alpha-\gamma)/2}\sin\frac{\beta}{2} \\ \mathrm{e}^{\mathrm{i}(\alpha-\gamma)/2}\sin\frac{\beta}{2} & \mathrm{e}^{\mathrm{i}(\alpha+\gamma)/2}\cos\frac{\beta}{2} \end{pmatrix}, \tag{122}$$

half-angles throughout — the double cover legible in every entry ($\alpha \to \alpha + 2\pi$ multiplies the whole matrix by $-1$, Eq. (114) again). The higher-$j$ families follow the same definition once §13 builds their stages, and they are the rotation group's complete answer to "how does anything transform"; §16.3 consumes them as the finite form of the tensor-operator definition.

The group theory is settled: two groups for one geometry, and the sign has been identified, measured, and tabulated. What has *not* been done is find the representations themselves — which pairs $(j,m)$ exist, and why $j$ comes in halves. That is pure ladder algebra, the oscillator's argument reused, and it is next.

# 13 The Angular Momentum Spectrum from the Algebra

Sections 11 and 12 settled which group acts and what its topology permits; this section finds the stages — every Hilbert space the algebra (108) can act on irreducibly — using nothing but the algebra itself. The engine is a ladder bounded by norms, the oscillator's argument (§9.2) reused; the output is the celebrated $(j,m)$ catalogue, half-integers included; and the closing subsection explains why *orbital* angular momentum, for all its seniority, realizes only half the catalogue.

## 13.1 The commuting set and the ladder

The three generators do not commute with each other, so pick the largest compatible set (§4.2). The rotation-invariant combination $\hat{\vec{J}}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2$ commutes with all three components —

$$[\hat{\vec{J}}^2, \hat{J}_z] = \sum_i \big(\hat{J}_i\,[\hat{J}_i,\hat{J}_z] + [\hat{J}_i,\hat{J}_z]\,\hat{J}_i\big) = \mathrm{i}\hbar \sum_{i,k} \epsilon_{izk}\big(\hat{J}_i\hat{J}_k + \hat{J}_k\hat{J}_i\big) = 0 \tag{123}$$

(Leibniz, then the sym-against-antisym lemma of [Fld. §3.7]: the antisymmetric $\epsilon_{izk}$ contracts a product symmetrized in $i \leftrightarrow k$ to zero — a one-line lemma the series uses repeatedly) — so $\{\hat{\vec{J}}^2, \hat{J}_z\}$ is the working pair. Label the joint eigenstates

$$\hat{\vec{J}}^2|\lambda,m\rangle = \hbar^2\lambda\,|\lambda,m\rangle, \qquad \hat{J}_z|\lambda,m\rangle = \hbar m\,|\lambda,m\rangle, \tag{124}$$

and build the ladder from the transverse pair:

$$\hat{J}_\pm \equiv \hat{J}_x \pm \mathrm{i}\,\hat{J}_y, \qquad [\hat{J}_z,\hat{J}_\pm] = \pm\hbar\,\hat{J}_\pm, \qquad [\hat{J}_+,\hat{J}_-] = 2\hbar\,\hat{J}_z, \qquad [\hat{\vec{J}}^2,\hat{J}_\pm] = 0 \tag{125}$$

(each a one-line consequence of (108)). The first relation is the oscillator's $[\hat{N},\hat{a}^\dagger] = \hat{a}^\dagger$ wearing a $z$: $\hat{J}_\pm$ shifts $m$ by $\pm 1$; the last says it does so *within* a fixed $\lambda$ — the ladder climbs a finite tower, if the norms allow.

### 13.2 The norm constraint and the catalogue

They allow only so much. Multiply out (using (125)):

$$\hat{J}_{\mp}\hat{J}_{\pm} = \hat{J}_x^2 + \hat{J}_y^2 \pm \mathrm{i}\,[\hat{J}_x, \hat{J}_y] = \hat{\vec{J}}^2 - \hat{J}_z^2 \mp \hbar\,\hat{J}_z, \tag{126}$$

so the squared norms of the rungs above and below are

$$\big\|\hat{J}_{\pm}|\lambda, m\rangle\big\|^2 = \hbar^2\,\big[\lambda - m(m \pm 1)\big] \;\geq\; 0. \tag{127}$$

Unlike the oscillator's one-sided constraint (86), this one squeezes from *both* ends: large $|m|$ of either sign violates it. So the ladder must terminate both ways — there is a top rung killed by $\hat{J}_+$ and a bottom rung killed by $\hat{J}_-$ — and (127) fixes them:

$$\lambda = m_{\max}(m_{\max}+1) \quad \text{and} \quad \lambda = m_{\min}(m_{\min}-1) \qquad \Longrightarrow \qquad m_{\min} = -\,m_{\max} \equiv -j \tag{128}$$

(the quadratic's other root, $m_{\min} = m_{\max}+1$, contradicts $m_{\min} \leq m_{\max}$). The ladder runs from $-j$ to $+j$ in unit steps, so $2j$ is a nonnegative *integer* — and that is the entire derivation:

$$\boxed{\begin{aligned} &\hat{\vec{J}}^2 :\ \hbar^2\, j(j+1), \qquad j \in \big\{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, 2, \dots\big\}, \\ &\hat{J}_z :\ \hbar m, \qquad m \in \{-j, -j{+}1, \dots, +j\} \quad — 2j+1 \text{ states}, \\ &\hat{J}_{\pm}|j,m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\;|j, m{\pm}1\rangle. \end{aligned}} \tag{129}$$

[The derivation fixes the coefficient's modulus; taking it real and positive is a phase convention — Condon–Shortley, the one App. B uses.]

Three readings of the box:

- **The halves are half the answer.** Nothing was assumed but the commutators, and the algebra delivered integer *and* half-integer $j$ on equal footing — the half-integers are not an exotic appendix but fifty percent of the catalogue, and §12 has already identified them: they are precisely the representations that ride the double cover and carry the sign (114). Setting $j = \frac{1}{2}$ in (129) reproduces the standing laboratory from nothing — two states, $\hat{J}_z$ eigenvalues $\pm\hbar/2$, and the $\hat{J}_{\pm}$ matrix elements assemble exactly $\frac{\hbar}{2}\sigma_{x,y}$ — a derivation of the Pauli matrices that §3.5 could only posit.
- **Same engine, different termination.** The oscillator's norm constraint (86) bounded the ladder below only — one termination, infinite tower, one representation. Here (127) bounds both ends — two terminations that must be an integer number of rungs apart, *finite* towers, one per $j$: the finite-dimensionality that Remark 11.2 allowed is here explicitly constructed. One lemma of bookkeeping distinguishes a particle count from a spin.
- **The catalogue is complete.** Every irreducible representation of the algebra is one of (129)'s towers [stated; the standard induction], so the catalogue is exhaustive: this *is* what angular momentum can be.

### 13.3 Orbital angular momentum realizes only the integers

Now the realization of §11.3. In the position representation $\hat{L}_z = -\mathrm{i}\hbar\,\partial_\varphi$ (112), so its eigenfunctions are $\mathrm{e}^{\mathrm{i}m\varphi}$ — and the coordinate $\varphi$ is an *angle*: the harvest of the periodicity planted in §11.3 and flagged in Remark 4.8. A wavefunction on physical space takes one value per point, $\psi(\varphi + 2\pi) = \psi(\varphi)$, and the fitting-waves mechanism of §8.3 — walls there, closure here — quantizes

$$m \in \mathbb{Z} : \tag{130}$$

half-integer $m$ would make $\mathrm{e}^{\mathrm{i}m\varphi}$ double-valued, not a function on the circle at all. Orbital angular momentum therefore realizes only the integer-$j$ towers of (129). [Caveat: as stated this is the quick argument, and its edge — *why* must $\psi$ be single-valued when phases are cheap? — deserves acknowledgment. The airtight version: $\hat{\vec{L}} = \hat{\vec{q}} \times \hat{\vec{p}}$ acts on $L^2(\mathbb{R}^3)$, and that space decomposes into integer representations only, the spherical harmonics below being a complete angular basis; the half-integer towers simply do not occur in it. Half-integer angular momentum is not a funny wavefunction — it requires an *internal* index, which is what spin is. Stated, with the decomposition as the citation-level fact.]

The integer towers, realized as functions on the sphere, are the *spherical harmonics* $Y_{\ell m}(\theta, \varphi) = \langle \theta, \varphi | \ell, m \rangle$: joint eigenfunctions of $\hat{\vec{L}}^2$ (eigenvalue $\hbar^2 \ell(\ell+1)$) and $\hat{L}_z$ ($\hbar m$), constructible by the ladder itself — kill the top, $\hat{L}_+ Y_{\ell\ell} = 0$, a first-order equation giving $Y_{\ell\ell} \propto \sin^\ell\theta\, \mathrm{e}^{\mathrm{i}\ell\varphi}$, and descend with $\hat{L}_-$ — the oscillator's $(\hat{a}^\dagger)^n|0\rangle$ construction, sideways. The low family explicitly:

$$Y_{00} = \frac{1}{\sqrt{4\pi}}, \qquad Y_{10} = \sqrt{\frac{3}{4\pi}} \cos\theta, \qquad Y_{1,\pm1} = \mp\sqrt{\frac{3}{8\pi}} \sin\theta\, \mathrm{e}^{\pm\mathrm{i}\varphi} \tag{131}$$

[normalizations stated; completeness on the sphere stated] — $Y_{10} \propto \cos\theta = z/r$ is chemistry's $p_z$ orbital, and Fig. 6 draws the angular shapes through $\ell = 2$: $\ell$ counts total angular nodes, $|m|$ tilts them between polar and equatorial, and the $\varphi$-independence of every $|Y_{\ell m}|^2$ is $\hat{L}_z$-eigenstate stationarity made visible (§6.2: sharp $m$, uniformly distributed conjugate angle).

The kinematics of rotation is now complete: the group (§11), its cover (§12), and every stage it can play on (129), with the spherical harmonics as the integer towers' concrete realization. What is missing is a *dynamics* to put on the stage — a Hamiltonian that respects the rotations — and the payoff for choosing one is the hydrogen atom, the document's capstone.

# 14 Central Potentials and Hydrogen

The document's threads converge here. The stage is §13's; the boundary-condition discipline and the one-dimensional intuition are §8's; the naming scheme is §7.1's CSCO; and the open puzzle is §2.1's — an atom that should not exist, emitting discrete lines with no classical origin. This section builds the hydrogen atom from the Coulomb potential and answers that puzzle in nanometers.

## 14.1 Central potentials: three dimensions become one

Let $V = V(r)$: rotationally invariant, so $[\hat{H}, \hat{\vec{L}}] = 0$ — angular momentum is conserved (§7.1), and $\{\hat{H}, \hat{\vec{L}}^2, \hat{L}_z\}$ is a commuting set: the states can carry the labels $(E, \ell, m)$ jointly. Separate accordingly. In the position representation the kinetic operator splits into radial and angular parts,

$$-\hbar^2 \nabla^2 = -\hbar^2 \frac{1}{r} \frac{\partial^2}{\partial r^2} r \; + \; \frac{\hat{\vec{L}}^2}{r^2} \tag{132}$$

[the identity is a coordinate computation; stated, with the angular part recognizable as $\hat{\vec{L}}^2$ from (112) and its siblings], so the ansatz $\psi = R(r)\, Y_{\ell m}(\theta, \varphi)$ absorbs the angular dependence entirely

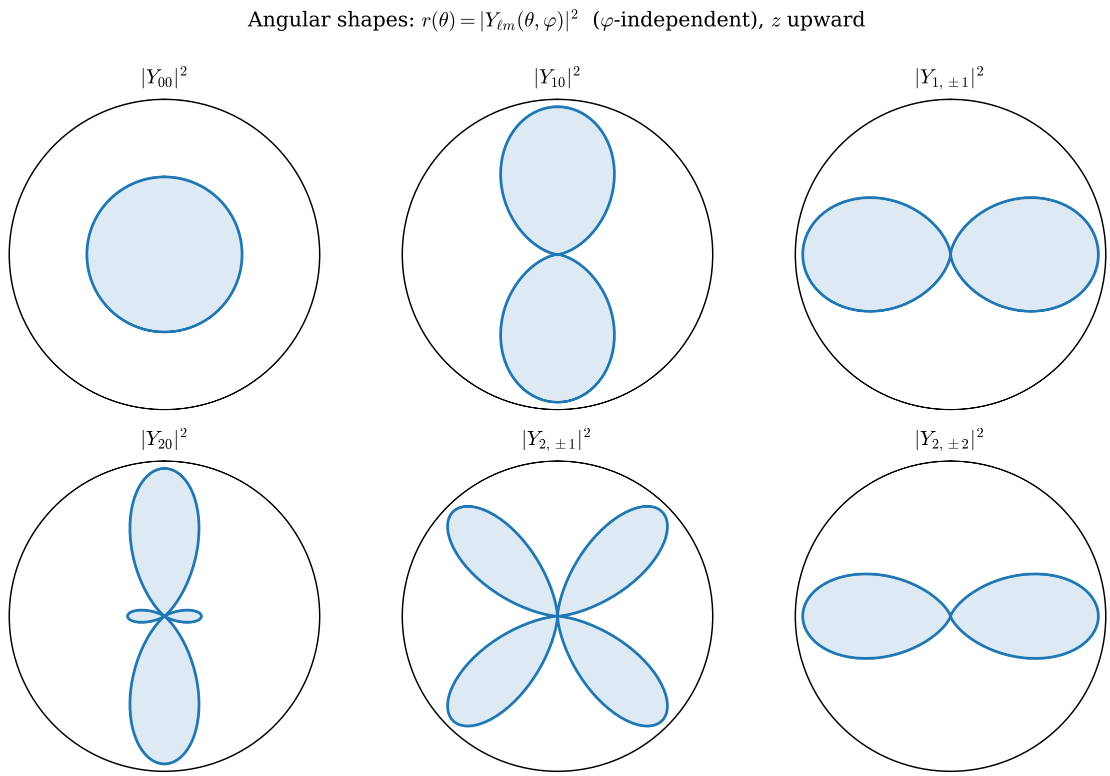


Figure 6: The integer catalogue on the sphere: angular probability profiles $|Y_{\ell m}(\theta)|^2$ through $\ell = 2$, exact closed forms, $z$ upward ($|Y_{\ell m}|^2$ is $\varphi$-independent, so the planar cut is the whole story). $\ell$ sets the angular structure; $m$ distributes it between pole and equator — $|m| = \ell$ hugs the equator, $m = 0$ the axis.

into §13's eigenvalue $\hbar^2\ell(\ell+1)$, and the substitution $u(r) = r\,R(r)$ lands on

$$\boxed{-\frac{\hbar^2}{2m}\,u'' + \Big[V(r) + \frac{\hbar^2\,\ell(\ell+1)}{2m\,r^2}\Big]u = E\,u, \qquad u(0) = 0} \tag{133}$$

— a *one-dimensional* Schrödinger problem on the half-line: the whole of §8 applies wholesale, with two new fixtures. The boundary condition $u(0) = 0$ is a hard wall at the origin ($R = u/r$ must stay finite — the quick reason; the airtight one: $u(0) \neq 0$ gives $R \sim 1/r$, and $\nabla^2(1/r) = -4\pi\,\delta^3(\vec{x})$ plants a delta source the equation cannot balance [stated]), so the origin is §8.3's wall relocated; and the potential has acquired the *centrifugal barrier* — the quantum version of the classical $\ell^2/2mr^2$ of the central-force reduction [Mech. §3.3], with $\ell(\ell+1)$ where $\ell^2$ might have been guessed (the ladder's $j(j+1)$, (129), insists). States of higher $\ell$ are pushed off the origin; only $\ell = 0$ states touch the nucleus.

### 14.2 Hydrogen: scales first, then the spectrum

The Coulomb problem is $V(r) = -e^2/r$, with $e^2 \equiv q_e^2/4\pi\epsilon_0$ as shorthand. Before solving anything, do what the house style demands: ask what the parameters $(\hbar, m, e^2)$ can build. Exactly one length and one energy can be built,

$$a_0 = \frac{\hbar^2}{me^2} \approx 0.529\,\text{Å}, \qquad \mathrm{Ry} = \frac{me^4}{2\hbar^2} = \frac{e^2}{2a_0} \approx 13.61\,\mathrm{eV} \tag{134}$$

— and these are Example 5.2's balance point made exact: the radius where localization cost ($\sim \hbar^2/2ma_0^2$) and Coulomb gain ($\sim e^2/a_0$) compete on equal terms. The atom's size was fixed by dimensional analysis before a single equation; the equation's job is the numerical coefficients and the *pattern* of levels.

Solve (133) in units $\rho = r/a_0$, $\epsilon = E/\mathrm{Ry}$. The two ends dictate asymptotics — decay $u \sim \mathrm{e}^{-\sqrt{-\epsilon}\,\rho}$ at infinity (bound state), $u \sim \rho^{\ell+1}$ at the origin (imposed by the centrifugal wall) — so peel both off, $u = \rho^{\ell+1}\,\mathrm{e}^{-\sqrt{-\epsilon}\,\rho}\,w(\rho)$, and feed $w$ a power series. The recursion that falls out relates consecutive coefficients by

$$\frac{c_{k+1}}{c_k} = \frac{2\big[\sqrt{-\epsilon}\,(k+\ell+1) - 1\big]}{(k+1)(k+2\ell+2)} \xrightarrow[k\to\infty]{} \frac{2\sqrt{-\epsilon}}{k}, \tag{135}$$

and the large-$k$ tail is the series of $\mathrm{e}^{+2\sqrt{-\epsilon}\,\rho}$: a non-terminating series overwhelms the decay and the state fails to normalize — the same normalizability constraint that bounded §9.2's ladder. Survival requires *termination*: some numerator must vanish, $\sqrt{-\epsilon}\,(n_r + \ell + 1) = 1$ for a nonnegative integer $n_r$, i.e., with $n \equiv n_r + \ell + 1$:

$$\boxed{E_n = -\frac{\mathrm{Ry}}{n^2}, \qquad n = 1, 2, 3, \ldots, \qquad \ell \in \{0, 1, \ldots, n-1\}, \qquad \text{degeneracy } \sum_{\ell=0}^{n-1}(2\ell+1) = n^2.} \tag{136}$$

**Example 14.1** (Hydrogen's spectral lines in nanometers)**.** Section 2.1 opened with spectral lines the classical atom could not have; §6.2 showed lines live at Bohr frequencies between levels; here are the levels, so here are the lines. Between $n = 2$ and $n = 1$:

$$\Delta E = \mathrm{Ry}\Big(1 - \tfrac{1}{4}\Big) = 10.20\,\mathrm{eV} \qquad \Longrightarrow \qquad \lambda = \frac{hc}{\Delta E} = 121.5\,\mathrm{nm} \tag{137}$$

— Lyman-$\alpha$, the ultraviolet workhorse of astronomy. Between 3 and 2: $\Delta E = 1.89\,\mathrm{eV}$, $\lambda = 656\,\mathrm{nm}$ — H$\alpha$, *the* red line of emission nebulae, now a two-line computation. The whole zoo compresses to the Rydberg formula $1/\lambda = R_\infty(1/n_1^2 - 1/n_2^2)$ with $R_\infty = \mathrm{Ry}/hc = 1.097 \times 10^7\,\mathrm{m}^{-1}$ — which derives the empirical rule of 1888, instantiates the Rydberg–Ritz combination principle (§6.2), and answers the document's opening puzzle in full, in the observed decimal places. [Scope: nonrelativistic, spinless electron, infinitely heavy proton — the reduced-mass correction is a factor $(1 + m_e/m_p)^{-1} \approx 1 - 1/1836$, already visible in precision data, and fine structure needs the tools of §§15–16 plus relativity; stated.]

### 14.3 The radial catalogue

Termination at $n_r = n - \ell - 1$ makes $w$ a polynomial of that degree (an associated Laguerre polynomial [normalization stated]), so the radial function $R_{n\ell}$ has exactly $n - \ell - 1$ nodes — the oscillation theorem of §8.3 passing its check in the radial column. The low family, explicitly ($\rho = r/a_0$, in units $a_0^{-3/2}$):

$$R_{10} = 2\,\mathrm{e}^{-\rho}, \qquad R_{20} = \frac{2-\rho}{2\sqrt{2}}\,\mathrm{e}^{-\rho/2}, \qquad R_{21} = \frac{\rho}{2\sqrt{6}}\,\mathrm{e}^{-\rho/2}, \tag{138}$$

and Fig. 7 draws the radial probability densities $P_{n\ell} = r^2|R_{n\ell}|^2$ through $n = 3$. Three features to read off: the ground state peaks at exactly $r = a_0$ (differentiate $\rho^2\mathrm{e}^{-2\rho}$); size grows quadratically, $\langle r\rangle \sim n^2 a_0$ — Rydberg atoms are *large*; and at fixed $n$, higher $\ell$ means fewer radial nodes but a stronger centrifugal push off the origin — the $(3,0)$ state touches the nucleus, the $(3,2)$ state cannot, a distinction that controls everything from hyperfine structure to which orbitals chemistry builds bonds with.

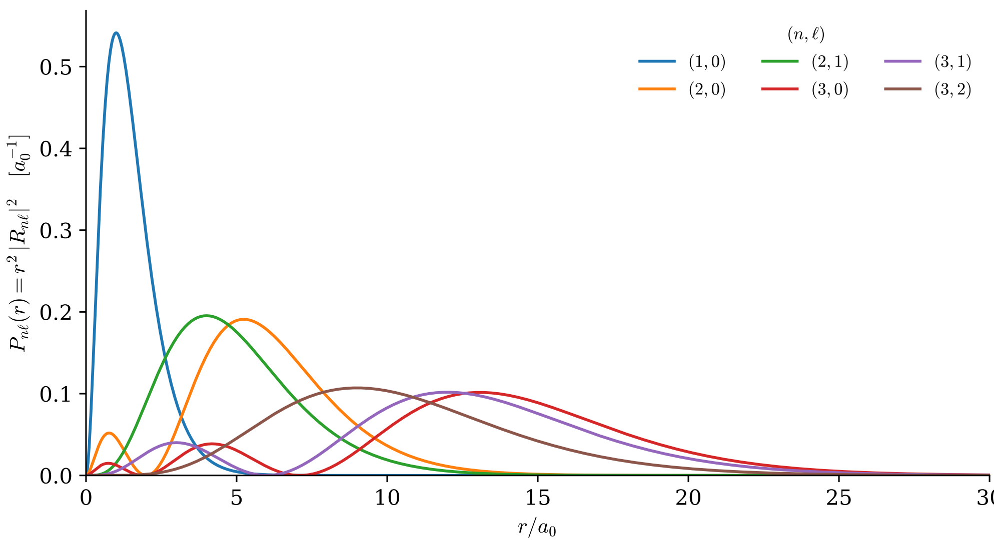


Figure 7: Hydrogen's radial probability densities $P_{n\ell}(r) = r^2|R_{n\ell}|^2$ through $n = 3$, exact closed forms; each curve numerically verified to unit norm by the figure script. Node count $n - \ell - 1$; ground-state peak at exactly $a_0$; the $n^2$ growth of size; the centrifugal exclusion of high-$\ell$ states from the origin.

*Remark* 14.2 (The degeneracy is not an accident — it is a bigger symmetry). Rotational symmetry explains the $(2\ell + 1)$-fold degeneracy in $m$ — states of one $\ell$-tower are rotations of each other —

but it does *not* explain why $\ell = 0, \ldots, n-1$ all land on the same $E_n$: a generic central potential splits them, and (136)'s $n^2$ is special to $1/r$. The classical fingerprint of the same specialness is that Coulomb orbits *close* (ellipses that do not precess — a property $1/r$ shares only with $r^2$ [stated; Bertrand's theorem]), and closed orbits mean an extra conserved quantity: the Runge–Lenz vector, pointing along the ellipse's fixed axis. Quantum mechanically the corresponding operator commutes with $\hat{H}$, and together with $\hat{\vec{L}}$ it closes a *larger* algebra — $SO(4)$, rotations of a four-dimensional space — whose representation theory delivers exactly the $n^2$-dimensional multiplets [stated; one of the subject's classic algebraic solutions, Pauli 1926]. The moral is the document's in miniature: where there is an unexplained degeneracy, look for the symmetry that explains it.

One atom is built. But the electron has spin the capstone ignored, and real atoms have more than one angular momentum to their name — orbital and spin, electron and nucleus. Composing them is not optional bookkeeping; it is where the tensor product of §3.6 finally does physics, and it is next.

# 15 Addition of Angular Momenta

The hydrogen capstone ended on a confession: the electron carries a spin the calculation ignored, so even one hydrogen atom holds *two* angular momenta — $\hat{\vec{L}}$ of the orbit, $\hat{\vec{S}}$ of the electron — and nature's Hamiltonians couple them. This section works out what "the total angular momentum of a composite" means and what values it can take. The material has a reputation for abstraction, so the treatment is deliberately slow: the *problem* is stated before the machinery, the commutator computation that creates the two-bases problem is displayed in full rather than left at [stated] level, and the simplest nontrivial case is then worked out completely.

## 15.1 A careful statement of the problem

Two subsystems with angular momenta $j_1$ and $j_2$ live, by §3.6, on the tensor product

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2, \qquad \dim \mathcal{H} = (2j_1 + 1)(2j_2 + 1), \tag{139}$$

with the obvious product basis $|j_1 m_1\rangle \otimes |j_2 m_2\rangle \equiv |m_1, m_2\rangle$ (the $j$'s fixed throughout, suppressed when unambiguous). Each factor carries its own generators, acting only on its own slot — $\hat{\vec{J}}_1 \otimes \mathbf{1}$ and $\mathbf{1} \otimes \hat{\vec{J}}_2$, written $\hat{\vec{J}}_1$, $\hat{\vec{J}}_2$ for short — and because they touch different slots they commute with each other: $[\hat{J}_{1i}, \hat{J}_{2k}] = 0$, always.

Now rotate the *whole* system. A rotation turns both parts at once, $U(R) = U_1(R) \otimes U_2(R)$, and expanding each factor to first order ((107), twice) shows the composite's generator is the sum:

$$\hat{\vec{J}} \;\equiv\; \hat{\vec{J}}_1 + \hat{\vec{J}}_2 \qquad (\text{meaning } \hat{\vec{J}}_1 \otimes \mathbf{1} + \mathbf{1} \otimes \hat{\vec{J}}_2). \tag{140}$$

This is what "adding angular momenta" means — not a numerical addition but an operator sum on a tensor product — and the first sanity check is that the sum still *is* an angular momentum:

$$[\hat{J}_i, \hat{J}_k] = [\hat{J}_{1i}, \hat{J}_{1k}] + [\hat{J}_{2i}, \hat{J}_{2k}] = \mathrm{i}\hbar\, \epsilon_{ikl}\big(\hat{J}_{1l} + \hat{J}_{2l}\big) = \mathrm{i}\hbar\, \epsilon_{ikl}\, \hat{J}_l \tag{141}$$

(the cross terms vanish — different slots), so the entire catalogue (129) applies to $\hat{\vec{J}}$: its representation content is some collection of $(j, m)$ towers. The question of the section: *which* towers, and how they sit inside the product basis.

Here is why the question has content. There are now two natural complete sets of commuting observables:

| *uncoupled* set | *coupled* set |
|---|---|
| $\hat{\vec{J}}_1^2,\ \hat{\vec{J}}_2^2,\ \hat{J}_{1z},\ \hat{J}_{2z}$ | $\hat{\vec{J}}_1^2,\ \hat{\vec{J}}_2^2,\ \hat{\vec{J}}^2,\ \hat{J}_z$ |
| basis $\lvert m_1, m_2\rangle$ | basis $\lvert j, m\rangle$ |
| answers: each part's $z$-component | answers: the total's magnitude and $z$-component |

Both are legitimate; they answer different experimental questions; and they are *different bases*, because the coupled set's $\hat{\vec{J}}^2$ refuses to commute with the uncoupled set's $\hat{J}_{1z}$. That refusal is the crux of the whole subject, so it deserves its computation. Expand

$$\hat{\vec{J}}^2 = \hat{\vec{J}}_1^2 + \hat{\vec{J}}_2^2 + 2\,\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2, \tag{142}$$

note the first two terms commute with everything in sight, and hit the cross term:

$$\big[\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2,\ \hat{J}_{1z}\big] = [\hat{J}_{1x}, \hat{J}_{1z}]\,\hat{J}_{2x} + [\hat{J}_{1y}, \hat{J}_{1z}]\,\hat{J}_{2y} = \mathrm{i}\hbar\big(\hat{J}_{1x}\hat{J}_{2y} - \hat{J}_{1y}\hat{J}_{2x}\big)\ \neq\ 0 \tag{143}$$

(only slot-1 operators feel the commutator; the result is the $z$-component of $\hat{\vec{J}}_1\times\hat{\vec{J}}_2$). Physically: the total magnitude involves the *relative orientation* of the two spins, and pinning down one spin's $z$-component disturbs exactly that. So sharp $(m_1, m_2)$ and sharp $(j, m)$ are incompatible bookkeeping — two adapted coordinate systems (Remark 3.5, once more) for one space, and the change of basis between them is this section's deliverable.

Two members are shared: $\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$ commutes with both sets, and acting on a product state,

$$\hat{J}_z\lvert m_1, m_2\rangle = \hbar\,(m_1+m_2)\lvert m_1, m_2\rangle \qquad\Longrightarrow\qquad m = m_1 + m_2: \tag{144}$$

the $z$-components *do* add numerically. This one additive quantum number is the crack through which the whole structure can be pried open.

### 15.2 Which totals occur: counting by $m$

Since every $j$-tower contributes exactly one state at each $m \in \{-j, \dots, j\}$, the multiplicities $N(m)$ — how many product states carry a given total $m$ — determine the tower content completely: the number of towers whose top rung is at $m$ is $N(m) - N(m+1)$. And $N(m)$ is elementary counting on (144): the number of ways to write $m = m_1 + m_2$ with $|m_1| \le j_1$, $|m_2| \le j_2$.

Count it for a concrete case first, $j_1 = 1$, $j_2 = \frac{1}{2}$ (six product states):

| $m$ | product states $(m_1, m_2)$ | $N(m)$ |
|---|---|---|
| $+\frac{3}{2}$ | $(1, +\frac{1}{2})$ | 1 |
| $+\frac{1}{2}$ | $(1, -\frac{1}{2}),\ (0, +\frac{1}{2})$ | 2 |
| $-\frac{1}{2}$ | $(0, -\frac{1}{2}),\ (-1, +\frac{1}{2})$ | 2 |
| $-\frac{3}{2}$ | $(-1, -\frac{1}{2})$ | 1 |

Read the staircase top-down: one tower tops out at $m = \frac{3}{2}$ (so $j = \frac{3}{2}$ occurs, once), one more appears at $m = \frac{1}{2}$ ($N$ jumped $1 \to 2$: a $j = \frac{1}{2}$ tower), and then nothing new — $N$ never exceeds 2. Conclusion: $1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$, dimensions $6 = 4 + 2$. The general case is the same staircase: $N(m)$ starts at 1 for $m = j_1 + j_2$, climbs by one per step down, and plateaus once the shorter ladder is exhausted — at $m = |j_1 - j_2|$, after which no new towers appear. Hence the *triangle rule*:

$$\boxed{j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j, \qquad \text{each } j \text{ exactly once,}} \tag{145}$$

the possible totals being exactly the triangle-compatible values — the quantum shadow of the classical fact that two vectors of lengths $j_1, j_2$ compose to anything between $|j_1 - j_2|$ and $j_1 + j_2$, discretized in unit steps. The dimension check is a telescoping sum worth displaying once:

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (j_1+j_2+1)^2 - (j_1-j_2)^2 = (2j_1+1)(2j_2+1) \checkmark \tag{146}$$

(each $2j+1$ is a difference of consecutive squares, $(j+1)^2 - j^2$; the sum collapses to its endpoints).

### 15.3 Clebsch–Gordan coefficients: the change of adapted coordinates

The two bases are related by a unitary matrix whose entries have a name:

$$|j, m\rangle = \sum_{m_1+m_2=m} \underbrace{\langle m_1, m_2|j, m\rangle}_{\text{Clebsch–Gordan coefficient}} |m_1, m_2\rangle, \tag{147}$$

the sum restricted by (144) — the one selection rule every coefficient obeys for free. Nothing about them is new machinery: (147) is a change of basis exactly as in §3.3, computable in every case by the two tools already in hand — *ladders* ($\hat{J}_\pm = \hat{J}_{1\pm} + \hat{J}_{2\pm}$ walks down a tower once its top is known, and the top of the top tower is unmissable: $|j_1+j_2,\, j_1+j_2\rangle = |j_1, j_2\rangle$, the unique maximal state) — and *orthogonality* (each new tower's top is fixed, up to phase, by being orthogonal to everything already built at its $m$). Conventional phases (Condon–Shortley: all coefficients real, and the highest-$m_1$ coefficient in each tower positive) pin the remaining signs [stated]; beyond that there is nothing to memorize, only an algorithm — run explicitly in the next subsection — and a lookup habit: Appendix B tabulates the small products, every entry machine-generated and verified.

### 15.4 The complete worked case: $\frac{1}{2} \otimes \frac{1}{2}$

Two spin-$\frac{1}{2}$'s: four product states $|{+}{+}\rangle, |{+}{-}\rangle, |{-}{+}\rangle, |{-}{-}\rangle$, and the triangle rule (145) promises $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$: a *triplet* and a *singlet*, $4 = 3 + 1$.

*Step 1: the top.* The maximal state is the unique $m = 1$ state:

$$|1, 1\rangle = |{+}{+}\rangle. \tag{148}$$

*Step 2: ladder down, factors and all.* Apply $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$. On the left, the catalogue's matrix element (129) with $j = 1$, $m = 1$: $\hat{J}_-|1,1\rangle = \hbar\sqrt{1\cdot 2 - 1\cdot 0}\,|1,0\rangle = \hbar\sqrt{2}\,|1,0\rangle$. On the right, each one-particle lowering carries $\hbar\sqrt{\frac{1}{2}\cdot\frac{3}{2} - \frac{1}{2}\cdot(-\frac{1}{2})} = \hbar$:

$$\big(\hat{J}_{1-} + \hat{J}_{2-}\big)|{+}{+}\rangle = \hbar\,|{-}{+}\rangle + \hbar\,|{+}{-}\rangle. \tag{149}$$

Equate and divide by $\hbar\sqrt{2}$:

$$|1, 0\rangle = \frac{1}{\sqrt{2}}\big(|{+}{-}\rangle + |{-}{+}\rangle\big). \tag{150}$$

Once more: $\hat{J}_-|1,0\rangle = \hbar\sqrt{2}\,|1,-1\rangle$, while the right side gives $\hbar(|{-}{-}\rangle + |{-}{-}\rangle)/\sqrt{2} = \hbar\sqrt{2}\,|{-}{-}\rangle$, so $|1,-1\rangle = |{-}{-}\rangle$ — the ladder closes exactly where the catalogue said it must.

*Step 3: the leftover.* Four states, three used. The $m = 0$ subspace is two-dimensional and the triplet took one direction; the orthogonal complement, normalized and with the conventional phase, is

$$|0, 0\rangle = \frac{1}{\sqrt{2}}\big(|{+}{-}\rangle - |{-}{+}\rangle\big) \qquad \text{— the } \textit{singlet}. \tag{151}$$

*Step 4: verify the labels.* That $|0,0\rangle$ really has $j=0$ should be *checked*, not trusted, and the check exercises the machinery. First rewrite the cross term of (142) in ladder form — one line,

$$\hat{J}_{1+}\hat{J}_{2-}+\hat{J}_{1-}\hat{J}_{2+}=2\big(\hat{J}_{1x}\hat{J}_{2x}+\hat{J}_{1y}\hat{J}_{2y}\big)\implies \hat{\vec{J}}_1\cdot\hat{\vec{J}}_2=\hat{J}_{1z}\hat{J}_{2z}+\tfrac{1}{2}\big(\hat{J}_{1+}\hat{J}_{2-}+\hat{J}_{1-}\hat{J}_{2+}\big),\tag{152}$$

then evaluate on the singlet, term by term. The $zz$ term gives $(\frac{\hbar}{2})(-\frac{\hbar}{2})=-\frac{\hbar^2}{4}$ on both $|+-\rangle$ and $|-+\rangle$. The ladder terms swap the spins with a factor $\hbar^2$ each ($\hat{J}_{1-}\hat{J}_{2+}|+-\rangle=\hbar^2|-+\rangle$, $\hat{J}_{1+}\hat{J}_{2-}|-+\rangle=\hbar^2|+-\rangle$, the other orderings annihilate), so on the antisymmetric combination the swap costs a sign:

$$\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2\,|0,0\rangle=\Big(-\frac{\hbar^2}{4}-\frac{\hbar^2}{2}\Big)|0,0\rangle=-\frac{3\hbar^2}{4}\,|0,0\rangle,\tag{153}$$

and assembling (142) with $\hat{\vec{J}}{}^2_{1,2}=\frac{3}{4}\hbar^2$:

$$\hat{\vec{J}}{}^2|0,0\rangle=\Big(\tfrac{3}{4}+\tfrac{3}{4}-\tfrac{3}{2}\Big)\hbar^2\,|0,0\rangle=0\ \checkmark\tag{154}$$

— and the same evaluation on $|1,0\rangle$ (swap costs no sign) gives $+\frac{\hbar^2}{4}+\frac{\hbar^2}{2}$, hence $\hat{\vec{J}}{}^2=2\hbar^2=1(1{+}1)\hbar^2$ ✓. The summary table, with every entry now derived:

| state | $(j,m)$ | $\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2/\hbar^2$ |
|---|---|---|
| $\|{+}{+}\rangle$ | $(1,+1)$ | $+\frac{1}{4}$ |
| $\frac{1}{\sqrt{2}}(\|{+}{-}\rangle+\|{-}{+}\rangle)$ | $(1,0)$ | $+\frac{1}{4}$ |
| $\|{-}{-}\rangle$ | $(1,-1)$ | $+\frac{1}{4}$ |
| $\frac{1}{\sqrt{2}}(\|{+}{-}\rangle-\|{-}{+}\rangle)$ | $(0,0)$ | $-\frac{3}{4}$ |

The last column is the section's most-used practical output: any Hamiltonian containing $\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2$ — and nature's spin–spin couplings do — is *diagonal* in the coupled basis, constant on each multiplet, splitting singlet from triplet by $\hbar^2$.

**Example 15.1** (The algorithm scales: $\frac{3}{2}\otimes\frac{1}{2}$)**.** Re-run the machine, compactly, on the next case up — eight product states $|m_1,m_2\rangle$, $m_1\in\{\pm\frac{3}{2},\pm\frac{1}{2}\}$, $m_2=\pm\frac{1}{2}$. First the staircase:

| $m$ | product states | $N(m)$ |
|---|---|---|
| $+2$ | $(\frac{3}{2},+\frac{1}{2})$ | 1 |
| $+1$ | $(\frac{3}{2},-\frac{1}{2})$, $(\frac{1}{2},+\frac{1}{2})$ | 2 |
| $0$ | $(\frac{1}{2},-\frac{1}{2})$, $(-\frac{1}{2},+\frac{1}{2})$ | 2 |
| $-1$ | $(-\frac{1}{2},-\frac{1}{2})$, $(-\frac{3}{2},+\frac{1}{2})$ | 2 |
| $-2$ | $(-\frac{3}{2},-\frac{1}{2})$ | 1 |

One tower tops at $m=2$, one new one at $m=1$, none after: $\frac{3}{2}\otimes\frac{1}{2}=2\oplus 1$, dimensions $8=5+3$, as (145) promised. Now two rows of coefficients, by the same two tools. *Ladder*: $|2,2\rangle=|\frac{3}{2},+\frac{1}{2}\rangle$, and applying $\hat{J}_-$ — left side $\hbar\sqrt{2\cdot 3-2\cdot 1}\,|2,1\rangle=2\hbar\,|2,1\rangle$, right side $\hbar\sqrt{3}\,|\frac{1}{2},+\frac{1}{2}\rangle+\hbar\,|\frac{3}{2},-\frac{1}{2}\rangle$ (the $\sqrt{3}$ from (129) at $j_1=\frac{3}{2}$, $m_1=\frac{3}{2}$; the $\hbar$ from the spin-$\frac{1}{2}$ slot) — gives

$$|2,1\rangle=\frac{\sqrt{3}}{2}\,|\tfrac{1}{2},+\tfrac{1}{2}\rangle+\frac{1}{2}\,|\tfrac{3}{2},-\tfrac{1}{2}\rangle.\tag{155}$$

*Orthogonality*: the $m = 1$ subspace is two-dimensional and the $j = 2$ tower has claimed one direction; the $j = 1$ tower's top is the complement, with the overall sign fixed by the Condon–Shortley convention — *the coefficient of the highest-$m_1$ state is positive*:

$$|1, 1\rangle = \frac{\sqrt{3}}{2} \, |\tfrac{3}{2}, -\tfrac{1}{2}\rangle - \frac{1}{2} \, |\tfrac{1}{2}, +\tfrac{1}{2}\rangle, \tag{156}$$

and continued lowering fills in the rest — nothing new happens, only more square roots. The case is not academic: a nucleus of spin $\frac{3}{2}$ coupled to an electron spin is exactly this decomposition, and the two hyperfine levels $F = 2, 1$ of $^{87}$Rb's ground state — the pair between which laser cooling and atomic-clock physics runs — are these two towers [stated].

*Remark* 15.2 (The entanglement flag delivered). Section 3.6 flagged that most states of a composite are entangled and promised a physical specimen. The singlet is *the* specimen: $|0, 0\rangle$ is manifestly not a product (try $(a|+\rangle + b|-\rangle) \otimes (c|+\rangle + d|-\rangle)$: matching coefficients forces $ac = bd = 0$ against $ad = -bc \neq 0$), and its correlations are total — measure either spin along *any* axis and the other is fixed opposite, since $j = 0$ makes the state invariant under the simultaneous rotation $U \otimes U$ (no direction is special: a rotationally invariant state has no Bloch vector to point). This is the two-particle state of the Einstein–Podolsky–Rosen discussion and of Bell's inequalities [stated; the flag planted in §3.6 is delivered to exactly this depth and the literature does the rest], and it was built here by nothing more exotic than a ladder and an orthogonality.

**Example 15.3** (The 21-centimeter line)**.** Hydrogen's ground state, $(n, \ell) = (1, 0)$, carries two idle spins: the electron's and the proton's. Their magnetic coupling is proportional to $\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p$ — so the table above *is* the level diagram: a triplet and a singlet, split by the hyperfine interaction, singlet lower. The measured splitting is tiny,

$$\Delta E = 5.87 \, \mu\text{eV}, \qquad f = \Delta E/h = 1420.4 \, \text{MHz}, \qquad \lambda = 21.1 \, \text{cm} \tag{157}$$

[experimental input; predicting $\Delta E$'s magnitude needs the magnetic moments and the contact interaction — note it involves $|\psi(0)|^2$, nonzero only for the $\ell = 0$ states of Fig. 7 — stated], corresponding to $\Delta E/k_B \approx 68 \, \text{mK}$: any astrophysical environment populates the triplet, and the slow magnetic-dipole decay back to the singlet emits the *21-centimeter line* — the radio signature by which neutral hydrogen is mapped across galaxies, including the first maps of the Milky Way's spiral arms. The structure of that line — why there are exactly two levels with statistical weights $3 : 1$ — is precisely the four-row table derived above; radio astronomy runs on a Clebsch–Gordan decomposition.

The composition of *states* is settled: totals by the triangle rule, bases by ladders and orthogonality. What remains for the symmetry story is the composition of states with *operators* — how observables themselves transform under rotation, and the theorem that squeezes every rotational matrix element through one reduced number. That theorem is next.

# 16 Tensor Operators and the Wigner–Eckart Theorem

One asymmetry remains in the symmetry story. States have been sorted into rotation multiplets — the $(j, m)$ towers — but *operators* have not, and physical predictions end in matrix elements $\langle\beta|\hat{A}|\alpha\rangle$ with an operator in the middle. This closing technical section repairs the asymmetry, carefully: it defines what it means for an operator to transform, discovers that operators fall into the *same* towers as states, and proves — to the depth of one displayed recursion — the theorem that then squeezes every rotational matrix element in atomic physics through a single number. The abstraction is front-loaded; the payoff, at the end, is as concrete as which spectral lines exist.

## 16.1 How operators transform

Rotate the system: states go $|\psi\rangle \to U|\psi\rangle$. For an operator, "transform" means whatever keeps the bookkeeping consistent, and expectation values dictate it: the rotated experiment measures $\langle\psi|U^\dagger \hat{A}\, U|\psi\rangle$, so the operator's transform is $\hat{A} \to U^\dagger \hat{A}\, U$. Now impose physics on specific operators. The position operator is a *vector*: rotating the apparatus must rotate its readings,

$$U^\dagger(R)\, \hat{q}_i\, U(R) = R_{ik}\, \hat{q}_k. \tag{158}$$

Feed in an infinitesimal rotation (107) about $\hat{n}$: the left side is $\hat{q}_i + \frac{\mathrm{i}}{\hbar}\, \mathrm{d}\varphi\, [\hat{n}\cdot\hat{\vec{J}}, \hat{q}_i]$, the right side is $\hat{q}_i + \mathrm{d}\varphi\, (\hat{n}\times\hat{\vec{q}})_i$, and matching — for every axis — gives the commutator form of "vectorhood":

$$\boxed{[\hat{J}_i, \hat{V}_k] = \mathrm{i}\hbar\, \epsilon_{ikl}\, \hat{V}_l} \qquad \text{(vector operator)}, \qquad [\hat{J}_i, \hat{S}\,] = 0 \quad \text{(scalar operator)}. \tag{159}$$

This is a *definition by commutators*, and it already has instances: $\hat{\vec{q}}$, $\hat{\vec{p}}$, $\hat{\vec{L}}$, $\hat{\vec{S}}$ all satisfy it (for $\hat{\vec{L}}$ it is (108) itself), while $\hat{\vec{q}}\cdot\hat{\vec{p}}$, $\hat{\vec{q}}^{\,2}$, $\hat{H}_{\text{central}}$ are scalars. Notice what (159) formally *is*: the statement that the components $\hat{V}_k$, under the adjoint action of $\hat{\vec{J}}$ (action by commutator), shuffle among themselves exactly the way a $j = 1$ triple of states would. That observation — operators organize into the same multiplets as states — is the entire section in embryo, and making it precise is next.

## 16.2 Cartesian tensors: the obvious generalization and its redundancy

Given (159), the natural next move is more indices. Define a *Cartesian tensor* of rank $n$ as an object with $n$ vector indices, transforming with one rotation matrix per index — constructively, take products of vectors: the *dyadic*

$$\hat{T}_{ik} = \hat{U}_i\hat{V}_k, \qquad U^\dagger\, \hat{T}_{ik}\, U = R_{ii'}\, R_{kk'}\, \hat{T}_{i'k'}, \tag{160}$$

nine components, and the pattern continues to any rank. This is the tensor grammar of the field-theory notes [Fld. §3.7], imported to operator components, and it is perfectly *correct*. It is also, as an organizing scheme, *redundant* — and watching the redundancy is the fastest route to the right definition. Split the dyadic identically into three pieces:

$$\hat{U}_i\hat{V}_k = \underbrace{\frac{\hat{\vec{U}}\cdot\hat{\vec{V}}}{3}\,\delta_{ik}}_{\text{trace: 1}} + \underbrace{\frac{\hat{U}_i\hat{V}_k - \hat{U}_k\hat{V}_i}{2}}_{\text{antisymmetric: 3}} + \underbrace{\left(\frac{\hat{U}_i\hat{V}_k + \hat{U}_k\hat{V}_i}{2} - \frac{\hat{\vec{U}}\cdot\hat{\vec{V}}}{3}\,\delta_{ik}\right)}_{\text{symmetric traceless: 5}} \tag{161}$$

(an identity — add the three terms and cancel). Now rotate. Each block stays strictly within itself, for a transparent reason each: the trace is a scalar because $R\,\delta\,R^{\mathsf{T}} = \delta$ — the *defining equation* of the rotation group, [Fld. §3.1]'s template once more; the antisymmetric part packages as a vector, $\frac{1}{2}\epsilon_{ikl}(\hat{U}_k\hat{V}_l - \hat{U}_l\hat{V}_k) = (\hat{\vec{U}}\times\hat{\vec{V}})_i$, and vectors rotate among themselves; and symmetry-plus-tracelessness survives conjugation by any $R$. So the nine components never mix as nine: rotations act on (161) block-diagonally, in pieces of size

$$3\times 3 = 1 + 3 + 5. \tag{162}$$

The Cartesian packaging is *reducible*: it lumps together subspaces that never speak to one another, like insisting on one $9\times 9$ matrix where three small blocks act. The counts $1, 3, 5$ are a broad hint — they are $2k+1$ for $k = 0, 1, 2$, the dimensions of the state catalogue (129) — and they pose the section's real question: what are the *minimal* rotation-closed operator families, the irreducible pieces the Cartesian language keeps bundling?

## 16.3 Spherical tensors: the irreducible organization

The answer's shape is already in hand. Irreducible multiplets of the rotation algebra are the $(2k+1)$-component towers of §13, laddered by $\hat{J}_\pm$ and graded by $\hat{J}_z$; demand operator families organized the same way. Define a *spherical tensor of rank $k$* as any $2k+1$ operators $\hat{T}_q^{(k)}$, $q = -k, \ldots, k$, obeying

$$\boxed{[\hat{J}_z, \hat{T}_q^{(k)}] = \hbar\, q\, \hat{T}_q^{(k)}, \qquad [\hat{J}_\pm, \hat{T}_q^{(k)}] = \hbar\sqrt{k(k+1) - q(q\pm 1)}\, \hat{T}_{q\pm 1}^{(k)}} \tag{163}$$

— the catalogue's matrix elements (129), transplanted verbatim from action-on-states to action-by-commutator: *an operator multiplet is a tower that commutators ladder.* (Equivalently, in finite form, the components mix by the $\mathscr{D}^{(k)}$-matrices of §12.3; the infinitesimal version (163) is taken as the definition because it is convention-proof and computation-ready. That the adjoint action is a legitimate "action" at all — that acting by nested commutators obeys the same algebra — is the Jacobi identity of §2, invoked one last time.)

A scalar is the $k = 0$ case. A vector fits with the spherical repackaging

$$\hat{V}_0^{(1)} = \hat{V}_z, \qquad \hat{V}_{\pm 1}^{(1)} = \mp \frac{\hat{V}_x \pm \mathrm{i}\, \hat{V}_y}{\sqrt{2}}, \tag{164}$$

and one check shows the definition engaging: using (159),

$$[\hat{J}_z,\ \hat{V}_{+1}^{(1)}] = -\frac{[\hat{J}_z, \hat{V}_x] + \mathrm{i}\, [\hat{J}_z, \hat{V}_y]}{\sqrt{2}} = -\frac{\mathrm{i}\hbar\, \hat{V}_y + \hbar\, \hat{V}_x}{\sqrt{2}} = \hbar\, \hat{V}_{+1}^{(1)} \quad \checkmark \tag{165}$$

— eigenvalue $q = +1$, as (163) demands (the $\hat{J}_\pm$ relations check the same way). The paradigm rank-$k$ example: the spherical harmonics themselves, read as multiplication operators, $Y_{km}(\hat{q})$ — they were *built* as $(k, m)$ towers in §13.3.

Now return to the dyadic and name its blocks in the new language. The counts (162) identify the three pieces of (161) as spherical tensors of ranks $0, 1, 2$ — dot product, cross product, and *quadrupole* — so the decomposition every linear-algebra student knows (trace, antisymmetric, symmetric-traceless) is a Clebsch–Gordan statement about operators,

$$1 \otimes 1 = 0 \oplus 1 \oplus 2, \tag{166}$$

with the triangle rule (145) presiding, exactly as it did for composite states. Two morals. First, *Cartesian tensors are reducible; spherical tensors are their irreducible pieces* — the spherical language is not a rival convention but the finer sieve, and the route of §16.2 was deliberate: the redundancy was found first, the irreducible organization built to remove it. Second, a warning about the overloaded word "rank": a Cartesian tensor of $n$ indices contains spherical ranks up to $k = n$ *and below* (the dyadic holds $k = 0, 1, 2$ at once); when a physicist says "the quadrupole operator," only the five-component $k = 2$ piece is meant, the trace and antisymmetric pieces already stripped. [Sakurai's tensor-operator discussion is the standard reference for this decomposition.]

## 16.4 The key observation and the theorem

The section converges on one line. Act with a generator on the state $\hat{T}_q^{(k)}|j, m\rangle$ and let Leibniz split the product:

$$\hat{J}_i \left( \hat{T}_q^{(k)} |j, m\rangle \right) = \underbrace{[\hat{J}_i,\ \hat{T}_q^{(k)}]}_{\text{rotates the operator: rep } k} |j, m\rangle \ + \ \hat{T}_q^{(k)} \underbrace{\hat{J}_i |j, m\rangle}_{\text{rotates the state: rep } j} . \tag{167}$$

Compare with §15.1: this *is* the two-slot structure of $\hat{\vec{J}}_1 + \hat{\vec{J}}_2$, with the adjoint action in slot one and the state in slot two — Leibniz has reproduced the addition of angular momenta, verbatim. Therefore $\hat{T}_q^{(k)}|j,m\rangle$ transforms exactly like the product state $|k,q\rangle \otimes |j,m\rangle$, and all of §15 applies for free: its total-$j'$ content obeys the triangle rule, its $\hat{J}_z$-eigenvalue is $\hbar(q+m)$, and its overlap with $\langle j',m'|$ must be proportional to the corresponding Clebsch–Gordan coefficient. That is already the theorem, seen from above; here is its statement and proof mechanism.

**Theorem 16.1** (Wigner–Eckart)**.**

$$\boxed{\langle j',m'|\,\hat{T}_q^{(k)}\,|j,m\rangle \;=\; \langle k,q;\,j,m|j',m'\rangle \;\times\; \langle j'\|\,\hat{T}^{(k)}\|\,j\rangle} \tag{168}$$

— *the Clebsch–Gordan coefficient carries* all *dependence on* $(m',q,m)$*, and the second factor, the* reduced matrix element*, is one single number for the whole* $(j',k,j)$ *family, independent of every orientation label.*

*Proof mechanism, one recursion displayed.* Sandwich the defining relation $[\hat{J}_\pm, \hat{T}_q^{(k)}] = \hbar\sqrt{k(k+1)-q(q\pm 1)}\,\hat{T}_{q\pm1}^{(k)}$ between $\langle j',m'|$ and $|j,m\rangle$, letting $\hat{J}_\pm$ act left and right by the catalogue (129):

$$\begin{aligned}\sqrt{j'(j'{+}1)-m'(m'{\mp}1)}\;\langle j',m'{\mp}1|\hat{T}_q^{(k)}|j,m\rangle \;-\; \sqrt{j(j{+}1)-m(m{\pm}1)}\;\langle j',m'|\hat{T}_q^{(k)}|j,m{\pm}1\rangle \\ = \sqrt{k(k{+}1)-q(q{\pm}1)}\;\langle j',m'|\hat{T}_{q\pm1}^{(k)}|j,m\rangle.\end{aligned} \tag{169}$$

Regard the numbers $\langle j',m'|\hat{T}_q^{(k)}|j,m\rangle$, at fixed $(j',k,j)$, as one unknown per label triple $(m',q,m)$: (169) is a homogeneous linear recursion linking them — and it is, coefficient for coefficient, *the same recursion the Clebsch–Gordan coefficients satisfy* (obtained by the identical sandwich of $\hat{J}_\pm = \hat{J}_{1\pm} + \hat{J}_{2\pm}$ on product states in §15.3). A linear system does not care what its unknowns mean: its solution space here is one-dimensional [the induction walking the recursion through all labels from the maximal one is stated], so any solution is a multiple of the CG solution — and the multiple is (168)'s reduced matrix element. Geometry determines everything but one number; the one number is where the physics lives. □

Two immediate harvests, one structural, one practical:

- **Selection rules for free.** The matrix element vanishes unless the CG coefficient survives:

$$m' = q+m, \qquad |j-k| \;\le\; j' \;\le\; j+k \tag{170}$$

  — with no dynamical computation required.

- **Ratios are universal.** Within a multiplet family, ratios of matrix elements are ratios of CG coefficients — pure geometry, identical for every rank-$k$ operator and every physical system. Dynamics enters only through the one reduced element: the theorem *factorizes* every rotational matrix element into (universal geometry) $\times$ (one number of physics), the same division of labor the document has met as kinematics-versus-dynamics throughout, here made into an equation.

### 16.5 The payoff in lines: dipole selection rules

Aim the theorem at Fig. 7. Atoms radiate, in leading approximation, through the *dipole* operator $\hat{\vec{q}}$ — a vector, hence via (164) a rank-1 spherical tensor. The selection rules (170) with $k=1$, applied to the orbital labels of hydrogen:

$$\Delta m \in \{-1,0,+1\}, \qquad \Delta\ell \in \{-1,0,+1\}, \tag{171}$$

and one further cut: under the parity operator of Remark 8.1, now on $L^2(\mathbb{R}^3)$ ($\vec{x} \to -\vec{x}$), the spherical harmonics carry $(-1)^\ell$ [stated; visible in (131)], while the dipole operator is odd — so the matrix element survives only if the parities differ, killing $\Delta\ell = 0$:

$$\boxed{\Delta\ell = \pm 1.} \tag{172}$$

Now read hydrogen's low levels through this rule. The $2p \to 1s$ transition is allowed — that *is* Lyman-$\alpha$ (Example 14.1), lifetime 1.6 ns [experimental]. But $2s \to 1s$ has $\Delta\ell = 0$: dipole-*forbidden*, and with no lower level to reach otherwise, the $2s$ state is stranded — *metastable*, decaying only by simultaneous two-photon emission with a lifetime of about an eighth of a second [experimental; a factor $\sim 10^8$ beyond its neighbor]. A hundred-million-fold lifetime ratio between two states one figure apart, traced to a single vanishing Clebsch–Gordan coefficient: that is the Wigner–Eckart theorem doing what it was built for — and it is why astrophysicists can find hydrogen's $2s$ population glowing in a continuum while $2p$ blinks in a line.

The symmetry story is complete, and it closes symmetric: states in $(j, m)$ towers, operators in $(k, q)$ towers, matrix elements factored through the geometry that both share. What remains is not another theorem but the outlook, and the handoff the series has been building toward.

# 17 Outlook

The document closes as the others in this series did: a list of what was left out, each item with its pointer; one last piece of physics that the tools in hand can almost reach, and the handoff. The last piece is the largest — the series' second closing of a circle — and it is why the whole five-document arc was walked in this order.

## 17.1 Deferred material, with pointers

What a first graduate course contains and this document deferred, each with its pointer: *time-dependent perturbation theory* lives in the interaction picture flagged at §7.4 — transition rates, the golden rule, and the actual strength of the spectral lines whose existence §16.5 settled; *scattering theory* continues §8.5's transmission problem into three dimensions, with the propagator of §10 as its natural engine; *density matrices* (Remark 3.6) carry the statistical and open-system story — decoherence, the modern descendant of the gap Remark 4.3 flagged; *time reversal*, with its antiunitary implementation, is the deferral Theorem 11.1 announced; *stationary perturbation theory* and the *variational method as an approximation scheme* are the standard machinery for Hamiltonians near ones solved here (the exact variational computations of §9 and Example 5.2 are its seeds); *charged particles in electromagnetic fields* — minimal coupling, $\hat{H} = (\hat{\vec{p}} - q\vec{A})^2/2m + q\phi$, and the gauge structure it carries — is deliberately left untouched here: it belongs with the field theory; *relativistic* quantum mechanics is not a patch but a rebuild, and it is where this outlook is headed. One omission, however, is too structural to defer without a paragraph, because without it quantum mechanics does not even predict that matter is stable in bulk.

## 17.2 Identical particles

Two electrons are not two similar objects; they are *indistinguishable in principle* — no measurement labels which is which. On the tensor product $\mathcal{H} \otimes \mathcal{H}$ of §3.6, swapping the slots is an operator $P_{12}$ with $P_{12}^2 = \mathbf{1}$, so its eigenvalues are $\pm 1$ — and indistinguishability demands physical states be eigenstates (anything else would let the swap, which changes nothing physical, change the state by more than a phase). Nature's rule, an empirical input at this level: *every species commits globally to one sign* —

$$\psi(2,1) = +\,\psi(1,2) \quad \text{(bosons)} \qquad \text{or} \qquad \psi(2,1) = -\,\psi(1,2) \quad \text{(fermions)}, \tag{173}$$

and the assignment is by spin: integer-spin species are bosons, half-integer fermions [stated — the *spin–statistics theorem*, unprovable inside nonrelativistic quantum mechanics and a celebrated theorem of relativistic QFT: the sign of §12, which a $2\pi$ rotation writes on a spinor, grows up to become the minus sign of exchange]. The fermionic minus sign has an immediate, world-building corollary: put two fermions in the same single-particle state and antisymmetry forces $\psi = -\psi = 0$ —

$$\boxed{\text{at most one fermion per state}} \tag{174}$$

— the *Pauli exclusion principle*. Electrons therefore cannot all collapse into hydrogen-like ground states: they stack, filling the $(n, \ell, m, m_s)$ catalogue of §14 floor by floor — which is the periodic table — the organizing structure of chemistry — read off the labels this document constructed. [The many-electron problem in earnest — Slater determinants, exchange energies — is deferred with the rest; the principle is the point.]

## 17.3 The second closing: quantizing the field

Now the closing the series was built for. Recall where the field-theory notes ended [Fld. §14.1–14.2]: a free field, expanded in its modes, *is* a countable family of independent harmonic oscillators — one complex amplitude $a_{\vec{k}}$ per wavevector, each obeying the classical oscillator equations — and the notes could only *state* what quantization would mean. This document's §9 closed with the missing sentence's grammar: the pair $(\hat{a}, \hat{a}^\dagger)$, $[\hat{a}, \hat{a}^\dagger] = 1$, is the entire quantum content of one oscillator. Compose the two documents:

$$\boxed{a_{\vec{k}} \;\longrightarrow\; \hat{a}_{\vec{k}}, \qquad [\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = \delta_{\vec{k}\vec{k}'}, \qquad \hat{H} = \sum_{\vec{k}} \hbar\omega_{\vec{k}}\Big(\hat{a}^\dagger_{\vec{k}}\hat{a}_{\vec{k}} + \tfrac{1}{2}\Big)} \tag{175}$$

— one ladder per mode, nothing else. And now read the spectrum with this document's eyes. The ground state $|0\rangle$ (every mode on its floor) is the *vacuum.* The state $\hat{a}^\dagger_{\vec{k}}|0\rangle$ costs energy $\hbar\omega_{\vec{k}}$ and carries momentum $\hbar\vec{k}$ (the mode's Noether charges, [Fld. §9.4], now operator eigenvalues) — a discrete, countable lump of field excitation with definite energy and momentum. There is a word for that:

*a particle.*

$\hat{a}^\dagger_{\vec{k}}$ *creates* one; $\hat{N}_{\vec{k}}$ counts them; multi-particle states are multi-rung states; and the identical-particle symmetrization of §17.2 is automatic — the ladder operators carry no particle labels to antisymmetrize, only mode labels, so "which quantum is which" is not even a question the formalism can pose. The photoelectric lumps of §2.1, the first experimental puzzle on the first page, are the rungs of (175) applied to the electromagnetic field: *photons*, each $E = \hbar\omega$ — Einstein's 1905 heuristic, now a two-line consequence of composing two documents. The coherent states of §9.4 become laser light; the zero-point halves, summed over infinitely many modes, diverge — the subject's first renormalization, flagged and left for the sequel [stated].

This is the series' second closed circle. The first [Fld. §14.4] ran *mechanics* → *chains* → *fields*: the continuum limit. This one runs *oscillator* → *quantum oscillator* → *quantum field*: the quantization. The two arcs compose into the diagram the five documents have been drawing —

$$\text{classical mechanics} \xrightarrow{\text{[Mech.], [Osc.]}} \text{classical fields} \xrightarrow{\text{[Fld.]}} ?$$
$$\text{classical mechanics} \xrightarrow{\text{this document}} \text{quantum mechanics} \xrightarrow{?} ?$$

— and both question marks are the same subject: quantum field theory, where the Heisenberg-picture operators carry spacetime labels (Remark 7.2's closing prediction), the action of [Mech.] weighs histories of *fields* in the path integral of §10, and the spin–statistics sign of §17.2 finally gets its proof. The reader holding a QFT course — Schwartz, or the lecture notes this series was built to accompany — now has every prerequisite for its first three chapters.

## 17.4 Coda

The document promised, in §1, to keep $\hbar$ visible so its arrival and departure could be watched; here is the departure. Every classical limit met on the way was the same limit: $S \gg \hbar$ collapsed the path sum onto Hamilton's principle (§10.4); $\hbar\omega \ll k_B T$ unfroze the ladders into equipartition (Example 9.3); $|\alpha|^2 \gg 1$ turned rungs into amplitudes (§9.4); $n \gg 1$ laid quantum ripples on the classical dwell time (Fig. 4). One deformation parameter, dialed down, and the whole classical series reassembles — which is the final sense in which these five documents are one: the mechanics notes are this document's $\hbar \to 0$ shadow, and the field notes are its $N \to \infty$ stage. The brackets became commutators; the commutators built amplitudes; the amplitudes, summed over histories, explained why the brackets' own variational principle was ever true. The circle does not just close; it explains why it was a circle.

# A Gaussian Integrals

This appendix keeps the Gaussian reference material where reference material belongs: out of the narrative, available on demand. Everything is elementary; the point is to have the formulas stated once, with their conditions, so the text can cite rather than rederive.

**The basic integral.** For $a > 0$,

$$I(a) = \int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{-ax^2} = \sqrt{\frac{\pi}{a}}, \tag{176}$$

by the classic squaring trick: $I^2$ is a two-dimensional integral, polar coordinates make it elementary, $I^2 = \int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} \mathrm{d}r\, r\, \mathrm{e}^{-ar^2} = \pi/a$.

**Linear term: complete the square.** For $a > 0$ and any complex $b$,

$$\int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{-ax^2+bx} = \sqrt{\frac{\pi}{a}}\, \mathrm{e}^{b^2/4a} \tag{177}$$

— write $-ax^2 + bx = -a(x - b/2a)^2 + b^2/4a$ and shift the contour (legitimate for entire integrands with Gaussian decay; [stated] at this document's level of rigor). The imaginary-$b$ case is the Fourier transform of the Gaussian:

$$\int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{-x^2/4\sigma^2}\, \mathrm{e}^{-\mathrm{i}kx} = 2\sigma\sqrt{\pi}\, \mathrm{e}^{-\sigma^2 k^2} \qquad \text{— a Gaussian again, width } \tfrac{1}{2\sigma} \text{ in } k, \tag{178}$$

the reciprocal-width fact that §5.5 turns into physics.

**Complex Gaussian.** Equation (176) extends to complex $a$ with $\operatorname{Re} a > 0$ (same formula, principal branch of the root), and — with more care — to purely imaginary $a$ as an improper (Fresnel) limit:

$$\int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{\mathrm{i}cx^2} = \sqrt{\frac{\pi}{|c|}}\, \mathrm{e}^{\mathrm{i}\frac{\pi}{4}\operatorname{sgn} c}, \qquad c \in \mathbb{R} \setminus \{0\} \tag{179}$$

[stated; the contour rotation is standard]. This is the version the path integral of §10 consumes in bulk: the weights there are $\mathrm{e}^{\mathrm{i}S/\hbar}$, pure phases, and (179) is how pure phases nonetheless integrate to finite answers.

**Moments.** Differentiating (176) under the integral sign with respect to $a$:

$$\int_{-\infty}^{\infty} \mathrm{d}x\, x^2\, \mathrm{e}^{-ax^2} = \frac{1}{2a}\sqrt{\frac{\pi}{a}}, \qquad \text{hence} \qquad \langle x^2 \rangle = \frac{\int x^2\, \mathrm{e}^{-x^2/2\sigma^2} \mathrm{d}x}{\int \mathrm{e}^{-x^2/2\sigma^2} \mathrm{d}x} = \sigma^2 : \tag{180}$$

the parameter in $\mathrm{e}^{-x^2/2\sigma^2}$ *is* the variance — the reading-off used at Eq. (48). Odd moments vanish by symmetry; higher even moments follow by further differentiation, $\langle x^{2n} \rangle = (2n-1)!!\, \sigma^{2n}$.

## B Computed Clebsch–Gordan Tables

This appendix tabulates the Clebsch–Gordan coefficients $\langle m_1, m_2 | J, M\rangle$ for the smallest products, in the Condon–Shortley convention of §15.3 (all coefficients real; the highest-$m_1$ coefficient in each tower positive). Every entry below was *machine-generated* by symbolic computation from the ladder-and-orthogonality algorithm of §15.4 and checked: each $M$-block of each table was verified to form an orthonormal matrix, and spot values were checked against the standard particle-data-group table. Signs sit outside the roots. Rows are grouped by $M = m_1 + m_2$ (Eq. (144)); the columns are the towers $J$ that the triangle rule (145) admits; a dot marks a combination the $m$ selection rule forbids, while an explicit 0 marks an allowed combination whose coefficient the symmetries force to vanish.

Two symmetry relations extend the tables beyond what is printed [stated; both follow from the ladder construction and the Condon–Shortley phases]:

$$\langle j_1\, j_2\, m_1\, m_2 | JM\rangle = (-1)^{J-j_1-j_2}\,\langle j_2\, j_1\, m_2\, m_1 | JM\rangle, \tag{181}$$

so a product in the opposite order ($\frac{1}{2} \otimes 1$, say) is read from the table printed here ($1 \otimes \frac{1}{2}$) at the cost of the sign $(-1)^{J-j_1-j_2}$; and

$$\langle j_1\, j_2\, m_1\, m_2 | JM\rangle = (-1)^{J-j_1-j_2}\,\langle j_1\, j_2, -m_1, -m_2 | J, -M\rangle, \tag{182}$$

which is why each table's lower half mirrors its upper half up to that same sign. Equation (182) also explains the explicit zeros below: at $m_1 = m_2 = M = 0$ it reads $\langle 0\,0 | J\,0\rangle = (-1)^{J-j_1-j_2}\langle 0\,0 | J\,0\rangle$, forcing the coefficient to vanish whenever $J - j_1 - j_2$ is odd — the $J = 1$ column of $1 \otimes 1$ and the $J = 2$ column of $2 \otimes 1$. Larger products ($\frac{3}{2} \otimes \frac{3}{2}$, $2 \otimes \frac{3}{2}$, $2 \otimes 2$) are in the particle-data-group table.

$\frac{1}{2} \otimes \frac{1}{2}$

| $m_1$ | $m_2$ | $J = 1$ | $J = 0$ |
|---|---|---|---|
| $+\frac{1}{2}$ | $+\frac{1}{2}$ | 1 | · |
| $+\frac{1}{2}$ | $-\frac{1}{2}$ | $\sqrt{1/2}$ | $\sqrt{1/2}$ |
| $-\frac{1}{2}$ | $+\frac{1}{2}$ | $\sqrt{1/2}$ | $-\sqrt{1/2}$ |
| $-\frac{1}{2}$ | $-\frac{1}{2}$ | 1 | · |

$1 \otimes \frac{1}{2}$

| $m_1$ | $m_2$ | $J = \frac{3}{2}$ | $J = \frac{1}{2}$ |
|---|---|---|---|
| $+1$ | $+\frac{1}{2}$ | 1 | · |
| $+1$ | $-\frac{1}{2}$ | $\sqrt{1/3}$ | $\sqrt{2/3}$ |
| $+0$ | $+\frac{1}{2}$ | $\sqrt{2/3}$ | $-\sqrt{1/3}$ |
| $+0$ | $-\frac{1}{2}$ | $\sqrt{2/3}$ | $\sqrt{1/3}$ |
| $-1$ | $+\frac{1}{2}$ | $\sqrt{1/3}$ | $-\sqrt{2/3}$ |
| $-1$ | $-\frac{1}{2}$ | 1 | · |

$\frac{3}{2} \otimes \frac{1}{2}$

| $m_1$ | $m_2$ | $J=2$ | $J=1$ |
|---|---|---|---|
| $+\frac{3}{2}$ | $+\frac{1}{2}$ | $1$ | $\cdot$ |
| $+\frac{3}{2}$ | $-\frac{1}{2}$ | $\sqrt{1/4}$ | $\sqrt{3/4}$ |
| $+\frac{1}{2}$ | $+\frac{1}{2}$ | $\sqrt{3/4}$ | $-\sqrt{1/4}$ |
| $+\frac{1}{2}$ | $-\frac{1}{2}$ | $\sqrt{1/2}$ | $\sqrt{1/2}$ |
| $-\frac{1}{2}$ | $+\frac{1}{2}$ | $\sqrt{1/2}$ | $-\sqrt{1/2}$ |
| $-\frac{1}{2}$ | $-\frac{1}{2}$ | $\sqrt{3/4}$ | $\sqrt{1/4}$ |
| $-\frac{3}{2}$ | $+\frac{1}{2}$ | $\sqrt{1/4}$ | $-\sqrt{3/4}$ |
| $-\frac{3}{2}$ | $-\frac{1}{2}$ | $1$ | $\cdot$ |

$1 \otimes 1$

| $m_1$ | $m_2$ | $J=2$ | $J=1$ | $J=0$ |
|---|---|---|---|---|
| $+1$ | $+1$ | $1$ | $\cdot$ | $\cdot$ |
| $+1$ | $+0$ | $\sqrt{1/2}$ | $\sqrt{1/2}$ | $\cdot$ |
| $+0$ | $+1$ | $\sqrt{1/2}$ | $-\sqrt{1/2}$ | $\cdot$ |
| $+1$ | $-1$ | $\sqrt{1/6}$ | $\sqrt{1/2}$ | $\sqrt{1/3}$ |
| $+0$ | $+0$ | $\sqrt{2/3}$ | $0$ | $-\sqrt{1/3}$ |
| $-1$ | $+1$ | $\sqrt{1/6}$ | $-\sqrt{1/2}$ | $\sqrt{1/3}$ |
| $+0$ | $-1$ | $\sqrt{1/2}$ | $\sqrt{1/2}$ | $\cdot$ |
| $-1$ | $+0$ | $\sqrt{1/2}$ | $-\sqrt{1/2}$ | $\cdot$ |
| $-1$ | $-1$ | $1$ | $\cdot$ | $\cdot$ |

$2 \otimes \frac{1}{2}$

| $m_1$ | $m_2$ | $J=\frac{5}{2}$ | $J=\frac{3}{2}$ |
|---|---|---|---|
| $+2$ | $+\frac{1}{2}$ | $1$ | $\cdot$ |
| $+2$ | $-\frac{1}{2}$ | $\sqrt{1/5}$ | $\sqrt{4/5}$ |
| $+1$ | $+\frac{1}{2}$ | $\sqrt{4/5}$ | $-\sqrt{1/5}$ |
| $+1$ | $-\frac{1}{2}$ | $\sqrt{2/5}$ | $\sqrt{3/5}$ |
| $+0$ | $+\frac{1}{2}$ | $\sqrt{3/5}$ | $-\sqrt{2/5}$ |
| $+0$ | $-\frac{1}{2}$ | $\sqrt{3/5}$ | $\sqrt{2/5}$ |
| $-1$ | $+\frac{1}{2}$ | $\sqrt{2/5}$ | $-\sqrt{3/5}$ |
| $-1$ | $-\frac{1}{2}$ | $\sqrt{4/5}$ | $\sqrt{1/5}$ |
| $-2$ | $+\frac{1}{2}$ | $\sqrt{1/5}$ | $-\sqrt{4/5}$ |
| $-2$ | $-\frac{1}{2}$ | $1$ | $\cdot$ |

$\frac{3}{2} \otimes 1$

| $m_1$ | $m_2$ | $J = \frac{5}{2}$ | $J = \frac{3}{2}$ | $J = \frac{1}{2}$ |
|---|---|---|---|---|
| $+\frac{3}{2}$ | $+1$ | $1$ | $\cdot$ | $\cdot$ |
| $+\frac{3}{2}$ | $+0$ | $\sqrt{2/5}$ | $\sqrt{3/5}$ | $\cdot$ |
| $+\frac{1}{2}$ | $+1$ | $\sqrt{3/5}$ | $-\sqrt{2/5}$ | $\cdot$ |
| $+\frac{3}{2}$ | $-1$ | $\sqrt{1/10}$ | $\sqrt{2/5}$ | $\sqrt{1/2}$ |
| $+\frac{1}{2}$ | $+0$ | $\sqrt{3/5}$ | $\sqrt{1/15}$ | $-\sqrt{1/3}$ |
| $-\frac{1}{2}$ | $+1$ | $\sqrt{3/10}$ | $-\sqrt{8/15}$ | $\sqrt{1/6}$ |
| $+\frac{1}{2}$ | $-1$ | $\sqrt{3/10}$ | $\sqrt{8/15}$ | $\sqrt{1/6}$ |
| $-\frac{1}{2}$ | $+0$ | $\sqrt{3/5}$ | $-\sqrt{1/15}$ | $-\sqrt{1/3}$ |
| $-\frac{3}{2}$ | $+1$ | $\sqrt{1/10}$ | $-\sqrt{2/5}$ | $\sqrt{1/2}$ |
| $-\frac{1}{2}$ | $-1$ | $\sqrt{3/5}$ | $\sqrt{2/5}$ | $\cdot$ |
| $-\frac{3}{2}$ | $+0$ | $\sqrt{2/5}$ | $-\sqrt{3/5}$ | $\cdot$ |
| $-\frac{3}{2}$ | $-1$ | $1$ | $\cdot$ | $\cdot$ |

$2 \otimes 1$

| $m_1$ | $m_2$ | $J = 3$ | $J = 2$ | $J = 1$ |
|---|---|---|---|---|
| $+2$ | $+1$ | $1$ | $\cdot$ | $\cdot$ |
| $+2$ | $+0$ | $\sqrt{1/3}$ | $\sqrt{2/3}$ | $\cdot$ |
| $+1$ | $+1$ | $\sqrt{2/3}$ | $-\sqrt{1/3}$ | $\cdot$ |
| $+2$ | $-1$ | $\sqrt{1/15}$ | $\sqrt{1/3}$ | $\sqrt{3/5}$ |
| $+1$ | $+0$ | $\sqrt{8/15}$ | $\sqrt{1/6}$ | $-\sqrt{3/10}$ |
| $+0$ | $+1$ | $\sqrt{2/5}$ | $-\sqrt{1/2}$ | $\sqrt{1/10}$ |
| $+1$ | $-1$ | $\sqrt{1/5}$ | $\sqrt{1/2}$ | $\sqrt{3/10}$ |
| $+0$ | $+0$ | $\sqrt{3/5}$ | $0$ | $-\sqrt{2/5}$ |
| $-1$ | $+1$ | $\sqrt{1/5}$ | $-\sqrt{1/2}$ | $\sqrt{3/10}$ |
| $+0$ | $-1$ | $\sqrt{2/5}$ | $\sqrt{1/2}$ | $\sqrt{1/10}$ |
| $-1$ | $+0$ | $\sqrt{8/15}$ | $-\sqrt{1/6}$ | $-\sqrt{3/10}$ |
| $-2$ | $+1$ | $\sqrt{1/15}$ | $-\sqrt{1/3}$ | $\sqrt{3/5}$ |
| $-1$ | $-1$ | $\sqrt{2/3}$ | $\sqrt{1/3}$ | $\cdot$ |
| $-2$ | $+0$ | $\sqrt{1/3}$ | $-\sqrt{2/3}$ | $\cdot$ |
| $-2$ | $-1$ | $1$ | $\cdot$ | $\cdot$ |